DynamiX Device Reference

Checked against the live device registry

October 27, 2026 · kernel figures drawn by `make_figures.py` and `make_wavelet_figures.py` from the app's own kernel code

How to read this document

A *device* is one processing tool in a layer's chain. **Transforms** compute (they head chains and produce a result bundle); **filters** select from a computed result (list indexes and masks — scrubbing a filter knob redraws without recomputing). Chains are always transforms-then-filters; interleaving is rejected at construction.

Each tool has a display name and a device id. The device browser lists tools by scientific family and, like the chain strip, shows a tool's display name with its id in gray; this reference names a tool the same way: WTMM 2D · `wtmm2d`. Saved chains, rack files, layer names, cache keys and provenance record the device id. Each tool declares the data it consumes and produces, and one rule derived from those declarations says whether a step applies (§2). The tables of families, contracts, applicability and output rows are written by `make_tables.py` from the tools' own declarations, as the kernel figures are drawn from the kernel code.

Conventions that hold everywhere:

- **Scales are stored in pixels**, converted only for display.
- **The shown level is view state**: the Transport sets it for each layer (§1), and no chain step holds it.
- **The wavelet family belongs to the method**: each tool's menu lists only the kernel class its method admits, under the method's own names.
- **Inert knobs**: a parameter that binds only under one choice of another parameter (e.g. q only for q-family wavelets) is ignored otherwise; where the device declares it, the knob is *grayed out* in the device box while it does not apply (its value is kept).
- **View-only knobs** (marked *view* in the tables) choose what is shown from a result that is already computed and cached — Show, Component, Group, Orientation, Show PC. Switching one redraws at once; it never recomputes.
- **Orientations are** (**angle**, **frame**), the frame being the pixel grid or grid, true or magnetic north (§6.1, Gradient arrows and anisotropy).
- **No-support masking**: pixels with no honest signal support (nodata flats, FFT dust) are NaN in every exponent map.
- Each table lists: parameter (UI label), default, hard range, and meaning. Soft ranges (knob-drag limits) are narrower than hard ranges.
- **A step that does not apply** (a V-chain filter after an analysis without V-chains) stays in its chain with its knobs, drawn inactive with the reason, and the engine skips it. Bypassing a step is the user's own act.

Wavelet normalization: L^1

Every wavelet in the app is normalized in the L^1 (pure-dilation) convention, $\psi_a(x) = a^{-d}\,\psi(x/a)$ in $d = 2$ dimensions, so that at a Hölder singularity the transform scales as $|T_\psi[f](x,a)| \sim a^{h(x)}$ with no offset. The L^2 convention ($a^{-d/2}$, unit energy $\int \psi^2 = 1$ — e.g. the normalization constants A and A_q of Borges et al. 2004,

whose transform carries $a^{-1/2}$) shifts every exponent by $d/2$. Turiel's microcanonical formalism uses the same L^1 dilation $\Psi_r(x) = r^{-d}\Psi(x/r)$ (Turiel et al. 2008, eqs. 6–8). Borges et al.'s own WTMM example agrees in practice: their eq. 41 reads $T \sim a^{\alpha}$ (their fig. 5 recovers $\alpha = 0.5$), which holds for L^1 scaling, not for the $a^{-1/2}$ of their eq. 10.

- `wtmm2d` **normalizes in the Fourier domain.** Each filter is built as $\hat{\psi}(a\mathbf{k})$ from a smoothing function with $\hat{\theta}(0) = 1$ (unit mass at every scale), times the derivative factor $i\,a\mathbf{k}$: scale enters only through $a\mathbf{k}$, which *is* the L^1 dilation.
- **The Hölder tools normalize in real space** by default, on the pixel grid: positive kernels (`holder_measure`) to unit discrete mass; Mexican hats (`holder_multiaffine`) to exact zero mean, then unit L^1 norm ($\sum|\psi_r| = 1$), at every scale. With *Wavelet space* set to Fourier (§9) they are built as $\hat{\psi}(r\mathbf{k})$ instead, like `wtmm2d`'s filters.
- A constant prefactor that does not depend on the scale (such as A or A_q) cannot change any exponent — it only shifts $\log|T|$ — and the app's own normalization replaces it anyway. Only the *scale dependence* of the normalization matters.
- **Validated on a synthetic cusp** $|x - x_0|^h$: both routes recover the slope h (tests pin it; an L^2 slip would read $h + 1$ in 2-D). The smallest scales are biased by the discretization of both kernel and signal: at a sampled cusp the slope settles above $\sigma \approx 5\,\text{px}$ (`wtmm2d`) and $r \approx 10\,\text{px}$ (the Mexican hats), later for sharp-cutoff kernels ($q_T < 1$).
- **Two different q's.** q_T (the knob *Tsallis q* of the Hölder tools) is the Tsallis/Borges shape parameter of a kernel; q in $Z(q, a)$, $\tau(q)$ and $D(h)$ is the moment order of the partition functions. They are unrelated.

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1 The workspace: datasets, layers, ROIs, derivatives

Datasets. Every opened file is a *dataset*: one row in the layer tree, and that row *is* its raw layer (its H hides the raster; right-click → Remove dataset takes the dataset and everything on it). A plain image (PNG/JPEG) opens as one luminance band; a multiband GeoTIFF / netCDF / HDF stack opens as (n_y, n_x, n_c).

Children. A tool dropped on a dataset never replaces the data: it spawns a *child* layer holding that tool, nested under the dataset. A child re-runs its chain on its parent's data whenever that changes. Right-click on a result row: *Delete layer* (its children move up to its parent) or *Delete layer and children*; shift-click or Cmd-click (Ctrl on Windows) selects several rows for one Delete. **H never cascades**: each row hides only its own drawing (the dataset's H hides the dataset raster, not the results over it).

ROIs. A saved region of interest is stored in whole *file* pixels and shown as a row `dataset @A` under its dataset. Selecting the ROI row makes it the region the next tool runs on; results (`tool @A`) nest under it. The ROI row's H hides only its outline. An ROI cannot be deleted while results use it (delete them first). Analyzers read their own declared margin around the ROI; decompositions (`pca`, `tucker_HOOI_HOSVD`, `ssa2d`) analyze the ROI itself (margin 0: a margin would mix outside data into the statistic).

Output rows. A result whose tool produces several things carries a collapsed *Outputs* group under its row: one row per output, each with its own H (the output-rows table below lists every tool's rows). Raster outputs share the canvas one at a time: showing one hides the others, and hiding the one shown puts back what the tool's `show` shows by default (the data, or an MFA tool's h map). They follow the tool's `show` knob, and switching between them re-uses what was computed. The *edges* row hides the maxima dots and gradient arrows (and the H-lines in the Vector and Geo views); the result keeps its maxima, so filters, the spectrum and the anisotropy window still read them. The layer's own H hides everything the result draws. An output that needs its own computation (the `mz_edges` reconstruction, an MFA tool's band, one level of a per-scale output, §1) runs the first time it is shown, with progress and Stop, and stays cached; Freeze (the row's F) keeps it.

Output rows: the rows each tool carries under its result.

Tool	Output	Row	Kind	Computed	Per level
Interpolating wavelets (DD / DAF) · `interpolating_wavelets`	`low`	low	raster	when shown	
	`horizontal`	horizontal	raster	when shown	yes
	`vertical`	vertical	raster	when shown	yes
	`diagonal`	diagonal	raster	when shown	yes
WTMM 2D · `wtmm2d`	`modulus`	modulus	raster	when shown	yes
	`argument`	argument	raster	when shown	yes
WTMM 2D, ROI · `wtmm2d_roi`	`modulus`	modulus	raster	when shown	yes
	`argument`	argument	raster	when shown	yes
Mallat–Zhong edges · `mz_edges`	`edges`	edges	vector	with the analysis	
	`coarse`	coarse	raster	when shown	
	`thumbnail`	thumbnail	raster	when shown	
	`recon`	recon (edges + coarse)	raster	when shown	
	`recon_edges_only`	recon (edges only)	raster	when shown	
	`residual`	residual	raster	when shown	
	`modulus`	modulus	raster	when shown	yes
	`argument`	argument	raster	when shown	yes
CDF edges · `cdf_edges`	`edges`	edges	vector	with the analysis	
	`filtered`	u (filtered)	raster	with the analysis	
	`edge_channel`	v (edge detector)	raster	with the analysis	
	`re_scale`	u at σ	raster	when shown	yes
	`im_scale`	v at σ	raster	when shown	yes
Perona–Malik edges · `pm_edges`	`edges`	edges	vector	with the analysis	
	`filtered`	filtered	raster	with the analysis	
	`filtered_scale`	filtered at σ	raster	when shown	yes
Microcanonical MFA, measure · `holder_measure`	`band`	band reconstruction	raster	when shown	
	`band_mask`	band mask	raster	when shown	
Microcanonical MFA, multi-affine · `holder_multiaffine`	`band`	band reconstruction	raster	when shown	
	`band_mask`	band mask	raster	when shown	
DWT · `dwt2d`	`approximation`	approximation	raster	when shown	yes
	`horizontal`	horizontal	raster	when shown	yes

Tool	Output	Row	Kind	Computed	Per level
	`vertical`	vertical	raster	when shown	yes
	`diagonal`	diagonal	raster	when shown	yes
Inpaint · `inpaint`	`residual`	residual	raster	when shown	
	`heldout`	held-out error	raster	when shown	
Wavelet ridge (Passot–Newell) · `wavelet_ridge`	`modulus`	—T— over orientations	raster	when shown	yes
	`orientation`	orientation of max —T—	raster	when shown	yes
	`phase`	phase at max —T—	raster	when shown	yes
Wavelet leaders · `wavelet_leaders`	`leader`	log2 leader	raster	when shown	yes
	`p_leader`	log2 p-leader	raster	when shown	yes
	`wse`	log2 WSE leader	raster	when shown	yes
	`oscillation`	oscillation ratio	raster	when shown	yes
CWT leaders · `cwt_leaders`	`leader`	log2 leader	raster	when shown	yes
	`p_leader`	log2 p-leader	raster	when shown	yes

Progress is live: every compute reports one increasing fraction from inside its own loops.

The right panel shows what applies to the selected row: the colors, widths and arrows of the WTMM overlays appear with a result that has maxima or chains, *Reconstruct...* with an h map, colormap, stretch and hillshade with a single-band display (a multiband display uses the Composite section), and the view-specific sections in the Vector and Geo views.

Derivative datasets (Fork)

A *child* keeps re-processing its parent. A **derivative** is forked once from what a result shows and from then on is an *original raw dataset*: nothing links it back, and a tool dropped on it spawns a child that analyses the derivative's own pixels.

Where	result row → right-click → *Fork derivative dataset...*; or the *Fork derivative...* button of the Components window
Raster	a checklist of every raster the result holds: the one on screen first (after its view knobs — e.g. Group "1", Show residual = data − C1), then the components / PCs, recon and residual per input band of a multiband field, filtered, h(x)...; every checked raster becomes a band (n checked → an n-band dataset)
Vector	the extrema and maxima lines the result shows (after its filters)
Both	raster and vector go in one file
Placement	a dataset row of its own, or nested inside the dataset it came from (removing that dataset lifts the derivative to the top)
Keep	*Save to a file...* (a Save-As dialog; permanent), or *Temporary*: a file in this session's scratch folder, left out of a saved project (a notice names it) until *Save derivative as...* on its row writes it where you choose

Grid	the result's own grid, never resampled: the whole field, or the ROI window for an ROI result
Georeference	the parent's CRS rides over (with the grid's axes); the parent file's layout (its source path, window, full size) does not
File	a RasterField `.npz` (values, axes, frame, units, provenance) — it opens like any `.npz`; vectors ride under `vec_*` keys
History	`provenance["derived"]`: parent path and SHA-256, the layer, its whole chain with every parameter (view knobs included), the ROI window, the date; `provenance["bands"]` names the bands

A derivative that carries vectors gets `derived_vectors` (a loader the shell places, in the browser's Dev group) as its first step: its frozen extrema and maxima lines draw like any result's, and the extrema and chain filters can be dropped after it.

The Components window

Right panel → Decomposition → *Components…*, on a `ssa2d` or `tucker_HOOI_HOSVD` result:

- **thumbnails** — the eigenarrays (`ssa2d`: the $L_r \times L_c$ window patterns) or the component images, each labeled with its share;
- **the shares** on a log scale (group members in amber) — a slowly decaying floor is noise;
- **the w-correlation matrix** (`ssa2d`; black 0 to white 1) — separable components are nearly w-orthogonal, so groups show as blocks on the diagonal; hovering reads $w(C_i, C_j)$.

Clicking thumbnails picks the **group**: click one, Cmd-click (Ctrl) to add or remove, shift-click for a range, *All* for every one. The pick is written into the tool's Group knob (and the window follows the knob), so the canvas redraws from the cache.

Multidimensional rasters

Import. Opening a multi-grid container (netCDF, HDF5, HDF4) shows an import dialog: the image grids grouped by sensor and resolution with a checkbox per band, and the remaining grids (geolocation lattices, calibration tables, supplements) in a collapsed *Ancillary grids* group, each importable on its own. Each ticked group becomes one multiband dataset (n_y, n_x, n_c). Grids of different resolution are separate datasets: ASTER's VNIR (15 m), SWIR (30 m) and TIR (90 m) arrive as three. HDF4 grids are named by their swath group (`VNIR_Band3N/ImageData`), since ASTER names every band's array `ImageData`. A dataset is its file together with its band list, so two groups imported from one file are two datasets, each with its own row, cache and saved state.

Band rows. A multiband dataset lists its bands under a collapsed *Bands (n)* row. Each band row is a child layer whose chain starts with `band_select`: selecting it shows that band, and a tool dropped on it runs on that band alone. *Remove band from project* (right-click, or Delete) takes the band out of the dataset together with its row and any results under it; the file is untouched, and the last band stays. The removal changes the dataset's identity, so a result computed on the fuller stack is recomputed.

Composite. A multiband dataset draws as an RGB composite. The *Composite* section of the right panel lists the bands with a channel (R, G, B or none; each channel takes one band) and Solo / Mute buttons: mute drops a band's channel; one solo shows that band alone in gray; several solos keep each in its channel. A picture whose file stores bytes (a color PNG, JPEG or plain TIFF opens as a stack named R, G, B) draws at linear 0–255, so it reads as the file shows it, and a derivative forked with every band keeps the mix. Its *Stretch* applies one type to all channels, each computed from its own band's values:

Stretch	Mapping to $[0, 1]$
linear (min–max)	minimum to maximum (normalization)
percent clip	the p-th to $(100 - p)$-th percentiles, clipped ($p = 2$ is ENVI's linear 2 %)
std-dev	mean $\pm k\sigma$, clipped
log	$\log(1 + v - \min v)$, then minimum to maximum
histogram equalize	each value's rank among the finite values
bipolar	$-m$ to $+m$, m the $(100 - p)$-th percentile of $\lvert v \rvert$: zero maps to mid-scale, for signed data (principal components, residuals)

The single-band Display stretch offers the same six types. Stretches are display only; analysis reads the band values themselves.

What a field stage shows. A chain that ends on a field stage shows the field it produces: `noise` alone shows the noised field (a stack through the composite), a band row its band, a bus its stack. Bypassing the stage shows the raw data again, and *Fork derivative dataset…* takes the produced field, one band per checkbox.

Buses: routing bands into one stack

Multi-band tools (`pca`, `tucker_HOOI_HOSVD`) take a stack (n_y, n_x, n). A **bus** assembles one from the bands of any datasets on the same grid. It heads a child layer's rack (`bus`), and a tool dropped after it analyses the bus's stack. Right-click a dataset row → *Route bands to a new bus…*:

- **Sends** are picked per band from every loaded dataset; a dataset on another grid is listed grayed out with the reason. A send references the band in its file (the file path and the band's grid id or index); the bus reads the band when it computes, over the window it needs.
- **Channel order** (drag to reorder) is the band axis the tool receives. `pca` and `tucker_HOOI_HOSVD` give the same decomposition in any order, with the per-band loadings in that order; band math and the composite's default channels depend on it.
- **Same grid**: every send has the bus's shape, pixel step, georeference anchor and CRS. Combining grids of different resolution requires resampling, which the bus does not do.
- **Windows**: on an ROI the bus reads each band over the ROI's processing window, reflect-padded at the file edges exactly as the dataset itself is.
- **Returns**: the results of the tool after the bus are ordinary result rows; forking one makes it a dataset of its own.
- *Edit bus sends…* (right-click the bus row) re-opens the routing; a routing already computed is answered from the cache.

Live band stacks. Select two or more layers (shift- or Cmd-click), right-click → *Build band stack from N selected layers*: a bus whose sends are the rasters those layers show (a reconstruction, a noised band, an h map), referenced live. Each send carries a fingerprint of its layer's chain; editing a referenced layer changes the fingerprint, and the stack recomputes when next shown. Referenced layers are computed first when needed, and a deleted one is reported by name. The layers share the stack's grid (an ROI result has its own grid and is refused). Layer sends and file sends mix in one bus.

The shown level: the Transport

The Transport under the canvas (a slider, play, and a reading of the scale in pixels and in the dataset's units) sets the level the active layer shows. The level is view state: each layer keeps its own, saved with the project, and scrubbing it leaves the chain as it is. After the last Transform that applies, when the analysis holds an extrema stack of more than one level, the engine takes the shown level out of the stack and then runs the filters, so a

filter knob redraws one level. The V-chains stay whole: the 3-D scale-space view lifts them through every scale, and the anisotropy window reads every level. The Transport works on every multi-scale layer, the *WTMM + Hölder* rack's included, and the vector (3-D) view draws a layer's extrema at the level that layer shows. Bands stay rows of their dataset; the Transport ranges over scales only.

Each inspector window has a Transport of its own. With its *M* button on (the default) it follows the main Transport; with it off the inspector holds a level of its own. Two views of one layer at two levels resolve the same cached analysis, each at its level.

`scale_select` is superseded. It stays registered, in the browser's Dev group. Of the built-in racks only *WTMM standard*, the default chain, holds it. Opening a project whose chain holds it sets the layer's level from its `scale_idx` once; the step stays in the chain, drawn inactive with the reason "the scale is set by the Transport".

Per-scale outputs and levels marked for figures

The multi-scale analyzers carry one raster row per scale-space output, off by default (*per level* in the output-rows table):

- WTMM 2D · `wtmm2d` and WTMM 2D, ROI · `wtmm2d_roi`: `modulus` and `argument`, the wavelet transform's modulus and argument at one scale;
- Mallat–Zhong edges · `mz_edges`: `modulus` and `argument` at one dyadic level;
- CDF edges · `cdf_edges`: `re_scale` and `im_scale`, the Re and Im snapshots at σ_k;
- Perona–Malik edges · `pm_edges`: `filtered_scale`, the diffused raster at σ_k.

A row that is on shows the level the Transport is on and reads that level's scale in pixels; a row that is off computes nothing, and the levels it computed while on stay cached until the bounds below evict them. Turning a row on is a choice for that layer alone. Each level is its own cache entry. WTMM 2D and the Mallat–Zhong edges compute the one level asked for (one FFT pair, one filter cascade); the CDF and Perona–Malik evolutions are sequential, so the first request runs the evolution once and keeps every snapshot as one entry. Each output runs the tool's own per-scale routine with the analysis's knobs (WTMM 2D: its dither and a^η lift as the analysis applied them); with the `nms` detector the maxima of WTMM 2D's level-k modulus are the stored level-k extrema. A WTMM 2D, ROI · `wtmm2d_roi` layer made by *Save + run WTMM* draws its rows on the ROI's own grid; a result run on an ROI row (`tool @A`) computes no lazy outputs.

Marking a level for figures. Right-click a per-level row → *Mark level n for figures* (n counts from 1 at the finest level; *Unmark* on a marked level). The row reads *marked*, and the marks are saved with the layer. A marked level is pinned in the cache while its row is on, so eviction passes it by; unmarking it, or turning the row off, releases it. *File* → *Export Derived Raster (.npz)…* writes the level on show; its provenance names the output (`<device id>/<output>`), the level and the marked levels (counted from 0), and the status line lists the marked levels.

Memory. The RAM cache holds up to 64 unpinned entries and eight unpinned analysis results. Its automatic byte budget is one quarter of physical RAM; a positive `cache_ram_bytes` setting overrides it. The stage store has its own 2 GB bound. Large evictions spill asynchronously into the process's scratch folder and reload on demand. One write can be pending; further victims reuse a disk copy or are dropped with a logged reason, so a slow writer cannot retain an unbounded queue in RAM. The disk budget is the smaller of 32 GB and half the initially available disk space. A system-memory reserve of 20% triggers release of evictable entries, stages and GPU buffers. Shown results, frozen entries and required upstream entries remain protected; their memory and a running computation can exceed the cache budget. Scratch cache files are removed at exit. View → Cache Inspector opens a tool window with RAM, disk, stage and GPU memory, entry ownership and a bounded event history. It can drop or spill selected entries, reload disk entries, clear the unpinned tiers, drop intermediate stages and trim GPU buffers. Frozen or marked entries must be released from their owning controls before they can be dropped. Shown-analysis deletion and bulk clearing require confirmation.

2 Tool families and data contracts

Families. The device browser lists tools by scientific family (the families table), and a tool in several families is listed under each. The *Dev* group, collapsed, holds the parameter-schema stubs, the superseded tools (kept registered so that saved projects open and resolve) and the steps the shell places itself. A tool that declares no family is listed by its id under *Other*. The last group, *Racks*, lists the built-in racks and the user's own (§3).

Families: the tools the device browser lists under each family.

Family	Tools (display name · device id)
Data preparation	Noise / dither · `noise`; Fill · `fill`; Periodic extension (DAF) · `periodic_extension`; Fractional integration (Fourier) · `fractional_integration`; Dyadic interpolation (DD4) · `dyadic_interpolation`; DAF low-pass · `daf_smoothing`; Interpolating wavelet filter (DD / DAF) · `interpolating_wavelet_filter`; Band select · `band_select`
Edge detection	WTMM 2D · `wtmm2d`; WTMM 2D, ROI · `wtmm2d_roi`; Mallat–Zhong edges · `mz_edges`; CDF edges · `cdf_edges`; Perona–Malik edges · `pm_edges`
Regularity	WTMM 2D · `wtmm2d`; WTMM 2D, ROI · `wtmm2d_roi`; Mallat–Zhong edges · `mz_edges`; Microcanonical MFA, measure · `holder_measure`; Microcanonical MFA, multiaffine · `holder_multiaffine`; Wavelet leaders · `wavelet_leaders`; Directional scaling functions · `directional_scaling`; CWT leaders · `cwt_leaders`
Skeletons	Wavelet skeleton · `wavelet_skeleton`
Decomposition	Interpolating wavelets (DD / DAF) · `interpolating_wavelets`; PCA · `pca`; Tucker (HOOI / HOSVD) · `tucker_HOOI_HOSVD`; 2-D SSA · `ssa2d`; DWT · `dwt2d`
Reconstruction	Mallat–Zhong edges · `mz_edges`; Microcanonical MFA, measure · `holder_measure`; Microcanonical MFA, multiaffine · `holder_multiaffine`; Tucker (HOOI / HOSVD) · `tucker_HOOI_HOSVD`; 2-D SSA · `ssa2d`; Inpaint · `inpaint`
Topology	Chain topology · `chain_topology`
Selection: extrema	Orientation wedge · `orientation_wedge`; Modulus threshold · `modulus_threshold`; H-chain length · `hline_length`; H-chain modulus · `hline_modulus`; H-chain Hölder · `hline_holder`
Selection: V-chains	Min V-chains per H-chain · `min_vchains`; Chain Hölder · `chain_holder`; Chain modulus · `chain_modulus`; Chain length · `chain_length`; Seam classifier · `chain_classify`; Group filter · `group_filter`
Points	Back-project points · `backproject`
Measures	Disorder function · `disorder_function`; Hessian measure (Nathan–Gunaratne) · `hessian_measure`
Patterns	Wavelet ridge (Passot–Newell) · `wavelet_ridge`
Microstructure	Lineal path · `lineal_path`
Dev (collapsed)	`bus`; `holder_map`; `band_recon`; Band reconstruction, measure · `band_recon_measure`; Band reconstruction, multiaffine · `band_recon_multiaffine`; `tucker_havok`; Scale · `scale_select`; `group_paint`; `derived_vectors`; `stub_wavelet`; `stub_holder`; `stub_wedge`

Contracts. Devices pass named kinds of data, the contracts, and every tool declares the ones it consumes and the ones it produces (the declarations table). In a declaration "or" separates alternatives (Tucker takes a Field or a Stack) and "+" joins contracts needed together. *What it received* passes the step's input through: `noise` keeps a field a field and a stack a stack, and `chain_topology` adds the topology to the analysis it reads. A declaration marked † depends on the parameters; the table gives it at the defaults, then each choice that changes it. A `bus` produces a Field from one send and a Stack from several.

Contracts: the named kinds of data and the result keys each one covers.

Contract	Meaning	Result keys
Field	a single-band field	the `RasterField` itself
Stack	a band stack	the `RasterField` itself
Points	a point set	the `PointSet` itself
ExtremaStack	an extrema stack	`extrema`, `scales`, `_shape`
PointAlpha	a per-point α	`extrema[i]["alpha"]`
VChains	V-chains	`chains`
Topology	the H/V topology	`topology`, `_topology_vids`
HMap	an h-map	`h_map`, `r2_map`
Components	decomposition components	`raster_out`
Curve	a curve (r, L(r))	`r`, `L`
Samples	per-level sample sets	`samples`

Declarations: what each tool consumes and produces (T: Transform, F: Filter; †: depends on the parameters).

Tool		Consumes	Produces
Noise / dither · `noise`	T	Field or Stack	what it received
Fill · `fill`	T	Field or Stack	what it received
Periodic extension (DAF) · `periodic_extension`	T	Field	what it received
Fractional integration (Fourier) · `fractional_integration`	T	Field	what it received
Dyadic interpolation (DD4) · `dyadic_interpolation`	T	Field or Stack	what it received
DAF low-pass · `daf_smoothing`	T	Field or Stack	what it received
Interpolating wavelets (DD / DAF) · `interpolating_wavelets`	T	Field	Components
Interpolating wavelet filter (DD / DAF) · `interpolating_wavelet_filter`	T	Field	what it received
Band select · `band_select`	T	Stack or Field	Field
`bus`	T	Field or Stack	Stack†
WTMM 2D · `wtmm2d`	T	Field	ExtremaStack + VChains
WTMM 2D, ROI · `wtmm2d_roi`	T	Field or Stack	ExtremaStack + VChains
Mallat–Zhong edges · `mz_edges`	T	Field	ExtremaStack + PointAlpha†; `algorithm` = `printed`: ExtremaStack; `wavelet` = `frac_bspline`: ExtremaStack
Wavelet skeleton · `wavelet_skeleton`	T	Field	ExtremaStack
CDF edges · `cdf_edges`	T	Field	ExtremaStack
Perona–Malik edges · `pm_edges`	T	Field	ExtremaStack
`holder_map`	T	Field	HMap
Microcanonical MFA, measure · `holder_measure`	T	Field	HMap
Microcanonical MFA, multiaffine · `holder_multiaffine`	T	Field	HMap
`band_recon`	T	Field	HMap
Band reconstruction, measure · `band_recon_measure`	T	Field	HMap

Tool		Consumes	Produces
Band reconstruction, multiaffine · `band_recon_multiaffine`	T	Field	HMap
PCA · `pca`	T	Stack	Components
`tucker_havok`	T	Field or Stack	Components
Tucker (HOOI / HOSVD) · `tucker_HOOI_HOSVD`	T	Field or Stack	Components
2-D SSA · `ssa2d`	T	Field	Components
DWT · `dwt2d`	T	Field	Components
Inpaint · `inpaint`	T	Field or Stack	Components
Chain topology · `chain_topology`	T	ExtremaStack + VChains	what it received + Topology
Min V-chains per H-chain · `min_vchains`	F	Topology	—
Scale · `scale_select`	F	ExtremaStack	—
Orientation wedge · `orientation_wedge`	F	ExtremaStack	—
Modulus threshold · `modulus_threshold`	F	ExtremaStack	—
H-chain length · `hline_length`	F	ExtremaStack	—
H-chain modulus · `hline_modulus`	F	ExtremaStack†; `mode` = `sup_all_scales`: ExtremaStack + VChains	—
H-chain Hölder · `hline_holder`	F	ExtremaStack + PointAlpha	—
Chain Hölder · `chain_holder`	F	VChains	—
Chain modulus · `chain_modulus`	F	VChains	—
Chain length · `chain_length`	F	VChains	—
Seam classifier · `chain_classify`	F	VChains	—
`group_paint`	F	VChains	—
Group filter · `group_filter`	F	VChains	—
Back-project points · `backproject`	T	Points	Points
`derived_vectors`	T	Field or Stack	ExtremaStack + VChains
Lineal path · `lineal_path`	T	Field	Curve
Disorder function · `disorder_function`	T	Field	Curve
Hessian measure (Nathan–Gunaratne) · `hessian_measure`	T	Field	Curve
Wavelet ridge (Passot–Newell) · `wavelet_ridge`	T	Field	Components
Wavelet leaders · `wavelet_leaders`	T	Field	HMap + Samples
Directional scaling functions · `directional_scaling`	T	Field	Curve
CWT leaders · `cwt_leaders`	T	Field	HMap + Samples

Every tool carries its origin, the paper behind its method; the browser shows it in the tool's tooltip.

Origins: the method each tool implements; code ported from another program is named in `NOTICE`.

Tool	Citation
Noise / dither · `noise`	—
Fill · `fill`	Pottier, Turiel & Garçon (2008), Remote Sens. Environ.: missing data from the turbulent cascade

Tool	Citation
Periodic extension (DAF) · `periodic_extension`	Frishman, Hoffman, Rakauskas & Kouri (1996); Hoffman, Gunaratne, Zhang & Kouri (2000): the Hermite DAF and its periodic extension
Fractional integration (Fourier) · `fractional_integration`	Bessel potential, Dumeur et al. (2025), Eq. 12; explicit reference length and mean-preserving Riesz option
Dyadic interpolation (DD4) · `dyadic_interpolation`	Deslauriers & Dubuc, Interpolation dyadique, EPM/RT-87/22 (1987), classical four-point scheme
DAF low-pass · `daf_smoothing`	Hermite DAF gain; Hoffman et al. (2000), Eq. 5; US6847737
Interpolating wavelets (DD / DAF) · `interpolating_wavelets`	DD predictor; US6847737 Eqs. 47–48 GLDAF shell; local half-transpose lifting update and DC normalization
Interpolating wavelet filter (DD / DAF) · `interpolating_wavelet_filter`	DD predictor; US6847737 Eqs. 47–48 GLDAF shell; local half-transpose lifting update and DC normalization
Band select · `band_select`	—
WTMM 2D · `wtmm2d`	Arneodo, Decoster & Roux (2000), Eur. Phys. J. B 15, 567; chaining and follow ported from xsmurf, Centre de Recherche Paul Pascal, Bordeaux (1999)
WTMM 2D, ROI · `wtmm2d_roi`	Arneodo, Decoster & Roux (2000), Eur. Phys. J. B 15, 567; chaining and follow ported from xsmurf, Centre de Recherche Paul Pascal, Bordeaux (1999)
Mallat–Zhong edges · `mz_edges`	Mallat & Zhong (1992), IEEE Trans. Pattern Anal. Mach. Intell. 14(7), 710; ported from LastWave's dwtrans2d 2.1 (1998–2002)
Wavelet skeleton · `wavelet_skeleton`	Mallat & Hwang (1992), IEEE Trans. Inf. Theory 38(2), 617
CDF edges · `cdf_edges`	Araújo, Barbeiro, Cuesta & Durán (2017), Cross-diffusion systems for image processing I and II; Barbeiro (2023)
Perona–Malik edges · `pm_edges`	Perona & Malik (1990), IEEE Trans. Pattern Anal. Mach. Intell. 12(7), 629
Microcanonical MFA, measure · `holder_measure`	Turiel, Yahia & Pérez-Vicente (2008), J. Phys. A 41, 015501; Pont, Turiel & Pérez-Vicente (2006), Phys. Rev. E 74, 061110
Microcanonical MFA, multiaffine · `holder_multiaffine`	Turiel, Yahia & Pérez-Vicente (2008), J. Phys. A 41, 015501; Pont, Turiel & Pérez-Vicente (2006), Phys. Rev. E 74, 061110; Turiel (2009)
PCA · `pca`	Pearson (1901), Phil. Mag. 2, 559; Hotelling (1933), J. Educ. Psychol. 24, 417
Tucker (HOOI / HOSVD) · `tucker_HOOI_HOSVD`	De Lathauwer, De Moor & Vandewalle (2000), SIAM J. Matrix Anal. Appl. 21, 1253 and 1324; Broomhead & King (1986), Physica D 20, 217
2-D SSA · `ssa2d`	Golyandina & Usevich (2010); Golyandina, Korobeynikov, Shlemov & Usevich (2015), J. Stat. Softw. 67(2)
DWT · `dwt2d`	Mallat (1989), IEEE Trans. PAMI 11, 674; the PyWavelets banks (Lee et al. 2019, J. Open Source Softw. 4, 1237); bl3 after Pottier, Turiel & Garçon (2008)
Inpaint · `inpaint`	Pottier, Turiel & Garçon (2008), Remote Sens. Environ.: missing data from the turbulent cascade
Chain topology · `chain_topology`	—
Min V-chains per H-chain · `min_vchains`	—
Orientation wedge · `orientation_wedge`	—
Modulus threshold · `modulus_threshold`	—
H-chain length · `hline_length`	—
H-chain modulus · `hline_modulus`	—
H-chain Hölder · `hline_holder`	—
Chain Hölder · `chain_holder`	—

Tool	Citation
Chain modulus · `chain_modulus`	—
Chain length · `chain_length`	—
Seam classifier · `chain_classify`	—
Group filter · `group_filter`	—
Back-project points · `backproject`	—
Lineal path · `lineal_path`	Torquato (2002), Random Heterogeneous Materials, Springer, §2.4
Disorder function · `disorder_function`	Gunaratne, Jones, Ouyang & Swinney (1995), Phys. Rev. Lett. 75, 3281
Hessian measure (Nathan–Gunaratne) · `hessian_measure`	Nathan & Gunaratne (2005), Phys. Rev. E 71, 035101
Wavelet ridge (Passot–Newell) · `wavelet_ridge`	Passot & Newell (1994), Physica D 74, 301; the q-Morlet of Borges, Tsallis, Miranda & Andrade (2004), J. Phys. A 37, 9125
Wavelet leaders · `wavelet_leaders`	Jaffard, Lashermes & Abry (2006), Wavelet Analysis and Applications, Birkhäuser, 201–246; p-leaders: Jaffard et al. (2016), Leonarduzzi et al. (2017); (θ, ω)-leaders: Dumeur et al. (2025); GMWLP histogram: Cardo & Corvalán (2009), Fractals
Directional scaling functions · `directional_scaling`	Ben Slimane, Ben Abid, Ben Omrane & Halouani (2019), Symmetry 11, 825: the directional thermodynamic formalism on the hyperbolic wavelet basis
CWT leaders · `cwt_leaders`	Jaffard 1994 Lemma 1; Jaffard, Lashermes & Abry 2006; the cylinder supremum on the WTMM's transform (spec 30); GMWLP histogram: Cardo & Corvalán (2009), Fractals
`bus`	—
`holder_map`	Turiel, Yahia & Pérez-Vicente (2008), J. Phys. A 41, 015501; Pont, Turiel & Pérez-Vicente (2006), Phys. Rev. E 74, 061110
`band_recon`	Turiel & del Pozo (2002), IEEE Trans. Image Process.: reconstruction from the most singular manifold
Band reconstruction, measure · `band_recon_measure`	Turiel & del Pozo (2002), IEEE Trans. Image Process.: reconstruction from the most singular manifold
Band reconstruction, multiaffine · `band_recon_multiaffine`	Turiel & del Pozo (2002), IEEE Trans. Image Process.: reconstruction from the most singular manifold
`tucker_havok`	Brunton, Brunton, Proctor, Kaiser & Kutz (2017), Nat. Commun. 8, 19
Scale · `scale_select`	—
`group_paint`	—
`derived_vectors`	—
`stub_wavelet`	—
`stub_holder`	—
`stub_wedge`	—

The rule. Before the first step the layer's input is available: a Field for a single-band raster, a Stack for an (n_y, n_x, n_c) raster with $n_c > 1$, Points for a point set. A band row's `band_select` turns its dataset's Stack into a Field. Then, step by step:

- a Transform applies when one of its alternatives is available, and its products then replace the available set;
- a Filter applies when one of its alternatives is available, and leaves the set as it is;
- a step that does not apply leaves the set as it is.

The engine skips a step that does not apply, so the step is absent from the dependency set and from the cache keys, and the last Transform that applies is the analysis the layer's output rows come from. The chain strip draws the step inactive with the reason, which names the missing contract, the tools that produce it and the

current analysis: "needs V-chains: WTMM 2D and WTMM 2D, ROI produce it; the current analysis (CDF edges) does not". The step keeps its place and its knobs, and it runs again as soon as its input is there. A tool that declares no consumes always applies, and a Transform that declares no products counts as producing every contract, so a tool from outside this build keeps working.

Where the rule is used.

- **Browser.** A tool that cannot apply as the next step on the active layer is grayed, with the reason in its tooltip. A grayed row still drags, and the step lands drawn inactive. A tool that reads a Field or a Stack stays available on an analyzed layer, because dropping it there forks a sibling.
- **Chain strip.** Each step that does not apply is drawn inactive with its reason.
- **Fork.** A primary analyzer reads a Field or a Stack and produces neither: the edge, regularity, skeleton and decomposition tools. Dropped onto a layer headed by a different one, it forks a sibling layer; dropped onto a dataset row, it spawns a child. The sibling begins with the parent's leading routing steps (a band row's `band_select`, a `bus` with its sends), so it analyses the same field.
- **Shipping.** A filter on the parent moves to the sibling, with its knobs, when it applies after the sibling's analysis and does not apply after the parent's.
- **Racks.** A rack is grayed where its first step cannot apply, and the rack editor runs the rule along the rack (§3).

Which tool applies after which. The applicability table is the rule run for every selection tool after every analysis. The extrema filters apply after each tool that produces an extrema stack. The V-chain filters, Chain topology · `chain_topology` and `hline_modulus` in mode `sup_all_scales` apply after the WTMM tools, which produce V-chains. Min V-chains per H-line · `min_vchains` applies after Chain topology, and H-line Hölder · `hline_holder` after the Mallat–Zhong edges of the authors' algorithm, whose analysis stamps a per-point α. A derivative dataset's frozen vectors (`derived_vectors`) carry an extrema stack and V-chains, the contracts of a WTMM 2D analysis, so the same tools apply after them.

Applicability: • where a selection tool applies after an analysis (the contracts rule, §2).

applies after →	Interpolating wavelets (DD / DAF)	WTMM 2D	WTMM 2D, ROI	Mallat–Zhong edges	Mallat–Zhong edges, printed	Mallat–Zhong edges, frac_bspline	Wavelet skeleton	CDF edges	Perona–Malik edges	Microcanonical MFA, measure	Microcanonical MFA, multiaffine	PCA	Tucker (HOOI / HOSVD)	2-D SSA	DWT	Inpaint	Lineal path	Disorder function
Chain topology · `chain_topology`		•	•															
Min V-chains per H-chain · `min_vchains`																		
Orientation wedge · `orientation_wedge`		•	•	•	•	•	•	•	•									
Modulus threshold · `modulus_threshold`		•	•	•	•	•	•	•	•									
H-chain length · `hline_length`		•	•	•	•	•	•	•	•									
H-chain modulus · `hline_modulus`		•	•	•	•	•	•	•	•									

applies after →	Interpolating wavelets (DD / DAF)	WTMM 2D	WTMM 2D, ROI	Mallat–Zhong edges	Mallat–Zhong edges, printed	Mallat–Zhong edges, frac_bspline	Wavelet skeleton	CDF edges	Perona–Malik edges	Microcanonical MFA, measure	Microcanonical MFA, multiaffine	PCA	Tucker (HOOI / HOSVD)	2-D SSA	DWT	Inpaint	Lineal path	Disorder function
H-chain modulus · `hline_modulus`, `mode` = `sup_all_scales`		•	•															
H-chain Hölder · `hline_holder`				•														
Chain Hölder · `chain_holder`		•	•															
Chain modulus · `chain_modulus`		•	•															
Chain length · `chain_length`		•	•															
Seam classifier · `chain_classify`		•	•															
Group filter · `group_filter`		•	•															

3 Racks

A *rack* is a named chain to drop onto a layer. The browser's *Racks* group lists the built-in racks and the user's own; double-clicking one opens it in the rack editor.

Built-in	read-only: *WTMM standard* (the default chain), *WTMM + Hölder*, *M–Z edges*, *CDF edges*, *PM edges* and *Wavelet skeleton*; *Duplicate to edit* in the editor makes a user copy
Save	*Save chain as rack…* (right-click a result or band row) or *Save as rack…* (the chain strip) records every step with all its knobs, view-only ones included, and its bypass state, under the name you give. A band row's own `band_select` and a `bus` with its sends belong to their layer and stay out
Store	one JSON file per rack, named after the rack, in the `racks` folder beside the settings file: `~/Library/Application Support/GeoDynamix/racks/` on macOS, `%APPDATA%\GeoDynamix\racks\` on Windows; with `DYNAMIX_SETTINGS_PATH` set, beside the settings file it names
File	`{"format": "geodynamix-rack", "schema": 1, "name", "description", "steps": [{"device", "params", "bypassed"}], "created", "app_version"}`, with device ids and parameter names as a saved chain stores them
Editor	*Racks* → *Rack Editor…*: the family tree on the left and the rack's steps on the right; drag a tool in to add it and drag a step to reorder; the selected step's knobs; *Save*, *Save as…*, *Delete* (user racks) and *Duplicate to edit*
Validation	live while editing: the order must be transforms-then-filters, and the contracts rule (§2) runs along the rack from the input chosen under *Validate from a*: a single-band field, a band stack or a point set
Grayed	a rack whose first step cannot apply on the active layer is grayed with the reason. A rack file that names a device or a parameter this build lacks, or that cannot be read, is listed grayed with the reason, and the other racks load

Drop	split and merge: the rack's Transforms go after the chain's Transforms and its Filters after the chain's Filters, so the order stays legal. A rack Transform whose tool already runs in the chain is left out. A rack headed by a different primary analyzer forks a sibling, as dropping that analyzer does

4 Field preparation

4.1 Noise / dither · `noise` — perturbation

Adds controlled noise to the field before analysis. Its main use is *dithering* quantized data ($\pm\frac{1}{2}$ LSB) so that quantization plateaus do not read as zero-gradient flats.

Parameter	Default	Range	Meaning
`dist` (Distribution)	uniform	uniform / gaussian	noise law
`amplitude`	0	$[0, 10^9]$	0 = auto $\frac{1}{2}$ LSB; uniform: half-width; gaussian: σ
`seed`	0	$[0, 2^{31}]$	RNG seed (deterministic reruns)
`sign` (Apply)	add	add / subtract	subtract undoes a previous add exactly

4.2 Periodic extension (DAF) · `periodic_extension` — a periodic field

Makes a non-periodic field periodic, so that a scale-space tool after it can transform it with periodic boundaries and no padding. A gap after the last row and column is filled with the self-tuned Hermite DAF (Frishman, Hoffman, Rakauskas & Kouri 1996; Hoffman, Gunaratne, Zhang & Kouri 2000): the band limit f_b holds the chosen fraction of the power, or is a given wavelength $\lambda_b = 1/f_b$; the gap is $2/f_b$, and σ gives a gain of 0.99 at f_b under order M. The output is the field plus the gap, its real block stamped, so WTMM keeps no maxima in the gap, takes its threshold from real pixels and resolves its Edge `auto` to `periodic`. Below 6 points per wavelength at f_b the fill is rough: the tool extends anyway and the warning under the band knob says so (lower the power fraction, or give a longer λ_b); WTMM's automatic edge handling reflects instead at that point. Only a stencil that cannot fit the field with its gap passes the field through unchanged. The reading under the knob gives λ_b, the gap and σ for the field at hand. On an ROI it extends the ROI's own pixels, and the extended field is the row's result, placed at the ROI; a tool after it on that row works on the extended field, or fork the output to analyze it as its own dataset.

Parameter	Default	Range	Meaning
`band` (Band limit)	auto	auto, given	the band limit from the data (a fraction of the power) or as a given wavelength
`band_frac` (Power fraction)	0.99	$[0.5, 0.9999]$	fraction of the power below f_b, under *auto*
`lambda_b` (λ_b)	20 px	$[4, 10^5]$	the wavelength at the band limit, under *given*
`M` (DAF order)	12	$[2, 40]$	Hermite order of the DAF
`lam` (Regularization)	10^{-3}	$[0, 1]$	λ of the gap solve
`layout` (Layout)	centred	origin, centred	where the data sit in the extended raster: at the origin with the gap after the last row and column, or in the middle with the gap split around them (the same periodic field, rolled; the real-block stamp carries the offset and WTMM's real block follows it)
`round_to` (Size multiple)	none	none, 32, 64, 128, 256	the extended size rounded up to a multiple by widening the gap, so a dyadic tool after it (wavelet leaders, dwt2d) runs exactly periodic: its pyramid needs 2^J to divide the size, else the wrap is a seam

4.3 Band select · `band_select` — one band of a stack

Slices one band out of a multiband field as the working field; grid, frame and CRS are unchanged. Band rows start with it and name their band, so removing another band from the dataset leaves each row on its own band.

Parameter	Default	Range	Meaning
`band` (Band)	1	$[1, 256]$	the band, 1-based (a band row selects by name)

4.4 Fill · `fill` — the inpainting engine as a field stage

Fills the declared regions of the host band (or of `target_band` on a stack) before the analyzer, with `inpaint`'s engine and knobs (below) but no held-out blocks and no Show. The clone carries `filled_fraction`, `fill_route`, `fill_kind` and the run's stamps in its provenance, and every analyzer result downstream carries them as `_fill`, so the strip and the Band dialog say that analysis inside the region reads a reconstruction. With `mask_source` = region and no region drawn it passes the field through unchanged (the one no-op decidable from the parameters alone); a data-decided empty mask runs and records `filled_fraction` = 0.

4.5 Fractional integration (Fourier) · `fractional_integration` — an integrated or differentiated field

Applies a Fourier operator to the field before analysis. The Bessel potential $(1+(L|k|)^2)^{-s/2}$ (Dumeur et al. 2025, Eq. 12) is the default; the Riesz operator $(L|k|)^{-s}$ acts at nonzero frequencies only. Both preserve the mean; k is the angular frequency per pixel and L the reference length. Positive order smooths; negative order differentiates. The record carries the order, the operator, L and the boundary, and the convention $h_{\text{original}} = h_{\text{integrated}} - s$. WTMM's `integration_order` applies the same operator inside its own chain. WTMM's `fracint_alpha`, labeled coefficient lift η, instead multiplies the wavelet coefficients by a^{η}. The fitter is seeded with the sum of input integration and coefficient lift, and subtracts that order when reporting exponents in the original frame. Bessel's low-frequency crossover and lacunary signals require care: a shift is not a universal inverse theorem. A global input integration precedes a fork before ROI analysis.

Parameter	Default	Range	Meaning
`integration_order` (Input integration s)	0	$[-6, 6]$	the order; 0 passes the field through
`integration_operator`	bessel	bessel, riesz	advanced: the operator
`integration_length` (Reference length)	0 px	$[0, 10^9]$	advanced: L in pixels; 0 takes the domain size over 2π
`edge` (Boundary)	mirror	mirror, periodic	how the transform sees the field's edges

4.6 Dyadic interpolation (DD4) · `dyadic_interpolation` — a finer grid

Refines the grid by the four-point Deslauriers–Dubuc predictor (Deslauriers & Dubuc 1987, the classical stencil $(-1, 9, 9, -1)/16$), band by band: L refinements give $(n-1)\,2^L + 1$ samples along each axis, the original nodes retained (rows and columns whose indices are multiples of 2^L) and the spacing divided by 2^L. The source axes must be uniformly spaced, and the native field is interpolated before any Periodic extension. The new samples are derived, so the result is marked to be forked: fork the derivative dataset and analyze it on its own grid. The reading under the knob gives the output size and the factor. Figure 11 draws the predictor's scaling function.

Parameter	Default	Range	Meaning
`levels` (Refinements)	1	$[1, 8]$	L: each refinement halves the spacing
`edge` (Boundary)	reflect	reflect, periodic	the predictor's stencil beyond the edges

4.7 DAF low-pass · `daf_smoothing` — an explicit Hermite DAF filter

Smooths the field with the Hermite distributed approximating functional, the taps

$$\delta_M(j\,|\,\sigma) = \frac{e^{-z^2}}{\sqrt{2\pi}\,\sigma} \sum_{n\le M/2} \frac{(-\frac{1}{4})^n}{n!}\, H_{2n}(z), \qquad z = \frac{j}{\sigma\sqrt{2}}$$

(Frishman, Hoffman, Rakauskas & Kouri 1996, Eq. 9), whose gain $e^{-x}\sum_{j\le M/2} x^j/j!$, $x = (k\sigma)^2/2$ (Hoffman, Gunaratne, Zhang & Kouri 2000, Eq. 5), is flat to order M in the passband. Both settings are given by hand; nothing is tuned from the data (the self-tuned DAF is the Periodic extension's, §4.2). It runs as a field stage, on an ROI with the filter's reach as its margin, and band by band on a stack. The tool publishes the gain against frequency as a Curve and records the residual RMS. Figure 11 draws the taps and the gain for four orders.

Parameter	Default	Range	Meaning
`sigma` (σ)	3 px	$[0.25, 100]$	the Gaussian width
`order` (Hermite order M)	12	$[0, 40]$, even	flatness of the passband; $M = 0$ is the Gaussian
`edge` (Boundary)	reflect	reflect, periodic	how the filter sees the field's edges

4.8 Interpolating wavelet filter (DD / DAF) · `interpolating_wavelet_filter` — threshold and reconstruct

Decomposes the field with the interpolating wavelets (§7.5, same pyramid knobs), soft-thresholds every detail coefficient by `threshold` and reconstructs; at threshold 0 the reconstruction is the field. The coefficients are cached, so moving the threshold (an inspector action) reconstructs without decomposing again. The record carries the settings, the residual RMS and that the transform is a biorthogonal lifting construction. The leaders read the DWT pyramid, so this reconstruction stays out of them.

Parameter	Default	Range	Meaning
`bank`, `order`, `sigma`, `levels`, `edge`			as `interpolating_wavelets`
`threshold` (Soft threshold)	0	$[0, 10^9]$	advanced: in field units

5 Routing

5.1 `bus` — sends into one stack

The head of a bus layer's rack (§1, Buses). It has no knobs: the routing dialog writes its send list. Its output is the stack (n_y, n_x, n) in channel order, or one band for one send, on the incoming field's grid (the whole dataset, or an ROI's processing window). The shell places it, and the browser lists it in the Dev group.

6 Primary analyzers

Each primary analyzer is its own representation of the field: it reads a single-band field (or a band stack) and produces an analysis. Dropping a different primary analyzer onto an analyzed layer forks a sibling layer, and the filters that apply only after the new analyzer move with it (§2).

6.1 WTMM 2D · `wtmm2d` — canonical WTMM (Arnéodo)

The canonical multifractal machinery: a continuous wavelet transform over a geometric scale grid, modulus *maxima* detected per scale (along-gradient non-maxima suppression, or xsmurf's `follow`), maxima chained across scales into maxima lines, feeding partition functions $\to \tau(q) \to$ Legendre $D(h)$. The maxima are dense by design — they are the statistical support, not a lineament map (use the filters, or the other analyzers, for sparse features). Legendre spectra are concave hulls by construction; the microcanonical tools (§9) measure $D(h)$ directly — together they diagnose non-concavity.

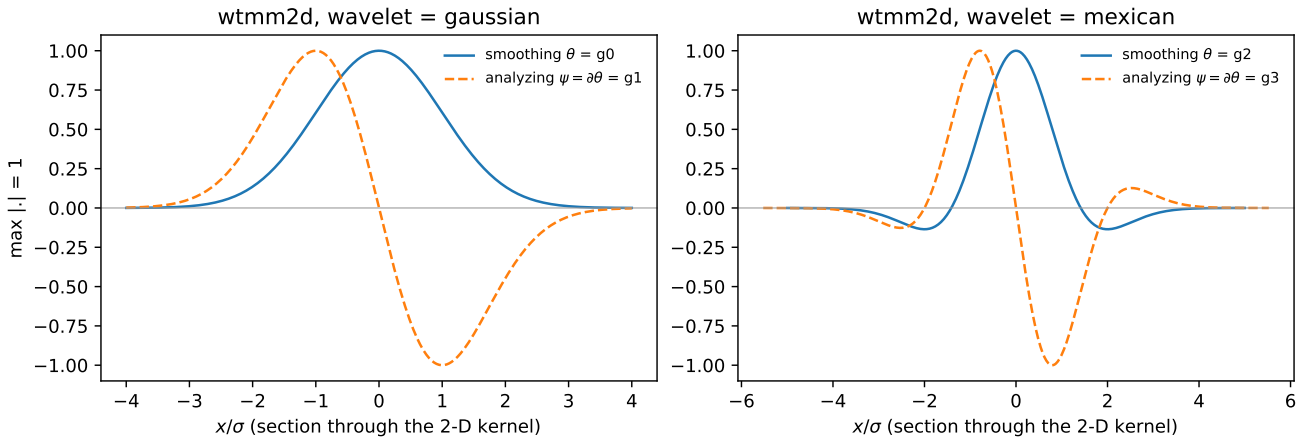


Figure 1: `wtmm2d`: the smoothing function θ and the analyzing wavelet $\psi = \partial\theta$ on the same axes (sections through the 2-D kernels, x in units of the smoothing σ). The wavelet is the derivative of the smoothing function, so the modulus maxima of ψ are the edges of θ-smoothed data. Figure 5 draws θ, ψ_1 and ψ_2 of every smoother in 3-D.

The smoothing function. `wavelet` names θ; the analyzing wavelet is always its gradient. Four smoothers share one scale convention (a nominal scale a is a real-space $\sigma = a/(\pi\sqrt{2})$) and one Fourier construction, every derivative filter being $(\mathrm{i}s_x)^p(\mathrm{i}s_y)^q\,\hat\theta(s)$ with $s = a\,k$: the *Gaussian* ($\psi = g_1$) and the *Mexican hat* $|k|^2\hat G$ ($\psi = g_3$), the two xsmurf kernels; the *2-D isotropic polyharmonic B-spline* of degree α (Van De Ville, Blu & Unser 2005: $\hat\theta = (V(\omega)/\|\omega\|^2)^{(\alpha+1)/2}$ with the localization $V = 4\sin^2\frac{\omega_1}{2} + 4\sin^2\frac{\omega_2}{2} - \frac{8}{3}\sin^2\frac{\omega_1}{2}\sin^2\frac{\omega_2}{2}$, isotropic to fourth order), its knot spacing chosen so its width matches the Gaussian's at the same a; it is a different function from the one-dimensional fractional B-spline of `mz_edges` (Unser & Blu's symmetric spline, §6.3), with which it shares only the order and the stored name `frac_bspline`; the *separable fractional B-spline* (`frac_bspline_sep`), that one-dimensional spline as a tensor product, $\hat\theta = |\mathrm{sinc}\,\frac{\omega_1}{2}|^{\alpha+1}|\mathrm{sinc}\,\frac{\omega_2}{2}|^{\alpha+1}$ with the same knot spacing, so both splines match the Gaussian to second order; and the *q-Gaussian* $e_q(-\rho^2/2\sigma^2)$ through its radial q-Fourier transform (Borges et al.), the same kernel the microcanonical tools offer, $q \to 1$ being the Gaussian exactly; the *q-Mexican hat*, $-\Delta$ of that q-Gaussian smoother (its analyzer the q-g_3, in closed form by the chain rule on e_q; Borges's own index is $q_B = 2 - 1/q$), which needs $q > 2/3$ for a continuous analyzer at the support's edge; and a *q-Meyer* smoother, the generalization Borges et al. leave open in their closing remark: Meyer's radial scaling-function taper with $\cos(\frac{\pi}{2}\nu)$ replaced by $\cos_q(x_0\nu)$, x_0 the q-cosine's first zero, so it is bandlimited and isotropic, its dilation fixed at $q = 1$ so that its band peak is the Gaussian's. In two dimensions the polyharmonic ψ is continuous for $\alpha > 2$ (the knob allows $\alpha \geq 1$ for exploration, default 3) and the q-Gaussian has finite mass for $q < 2$; its variance diverges from $q = 5/3$, so its scale is matched by the parameter σ, never by the variance. The wavelet scale bar and the $a_{\min}$ reading draw each smoother's own section and locate its band peak λ^* numerically (λ^* of the polyharmonic spline at $\alpha = 3$ is the Gaussian's within 3 %; the q-Gaussian's tails lengthen it above $q = 1$ and shorten it below). The polyharmonic spline's section is evaluated in the space domain, as the localized sum of Green's functions of $(-\Delta)^{(\alpha+1)/2}$ (`dynamix.core.polyharmonic_exact`), so it carries no truncation ringing; below $\alpha = 3$ it is singular at the knots and is drawn scaled by its main lobe and clipped there. The separable spline's sections come from Unser & Blu's exact series; its ψ is continuous for $\alpha > 1$.

How isotropic the two splines are. Neither is isotropic beyond second order, and both favor the diagonals. The polyharmonic localization V is isotropic to fourth order, but $\hat\theta = (V/\|\omega\|^2)^{(\alpha+1)/2}$ only to second: $\log\hat\theta$ carries $\gamma|\omega|^4\sin^2 2\varphi/1440$ ($\gamma = \alpha + 1$); the separable spline's carries $\gamma|\omega|^4\sin^2 2\varphi/5760$. At the analyzer's band peak $|\hat\theta|$ is larger on the diagonal than on the axis by 2.4 % (polyharmonic) and 0.6 % (separable) at $\alpha = 3$, by 3.8 % and 1.0 % at $\alpha = 1.5$. On the isotropic cusp $r^{1/2}$, half a scale from its center, the gradient argument's $\sin 4\varphi$ deviation is $-0.9°$ to $-1.2°$ (polyharmonic) and $-0.20°$ to $-0.26°$ (separable) at $a = 8$–32 px and $\alpha = 3$; the Gaussian's is zero, and a straight edge is unbiased for every smoother ($\nabla(\theta * f) \parallel n$ whenever $f = g(n\cdot x)$). The polyharmonic spline's own merits are its isotropic high-frequency decay $\|\omega\|^{-(\alpha+1)}$ and its $(-\Delta)^{(\alpha+1)/2}$ structure. Figure 7 draws both splines and these measurements. The two new smoothers run on the device (mlx) on Apple Silicon and on FFTW3 elsewhere, exactly as the pair does; the three engines agree to float32 tolerance.

Parameter	Default	Range	Meaning
`n_oct` (Octaves)	3	$[1, 12]$	scale range: a spans `n_oct` octaves above $a_{\min}$
`n_voice`	4	$[1, 64]$	scales per octave (grid density)
`q_min` (q min)	-3	$[-20, 20]$	the first moment order of the partition functions (MOMENTS group; out of the key at the defaults, so a saved chain keeps its results)
`q_max` (q max)	6	$[-20, 20]$	the last moment order, included when the step lands on it
`q_step` (Δq)	0.5	$[0.05, 5]$	the step between moment orders; the tables and the Partition functions window carry the list that ran
`a_min` ($a_{\min}$)	1.0	$[0.01, 64]$ px	finest analyzed scale
`wavelet` (Smoother θ)	mexican	mexican / gaussian / frac_bspline / frac_bspline_sep / q_gaussian / q_mexican / q_meyer	the smoothing function; ψ is its gradient: g_3, g_1, $\partial\theta_\alpha$, $\partial\theta_q$, q-g_3, $\partial\theta_q$
`alpha` (α)	3	$[1, 8]$	degree of the two fractional splines (shown with `frac_bspline` and `frac_bspline_sep`); $\alpha > 2$ for a continuous ψ in 2-D
`q_tsallis` (q)	1.5	$(0, 2)$	Tsallis q of the three q smoothers (shown with them); $q \to 1$ is the Gaussian, the Mexican hat, the Meyer; the q-Mexican hat needs $q > 2/3$
`dither`	off	bool	$\pm\frac{1}{2}$ LSB dither for quantized rasters
`fracint_alpha` (η)	0	$[-6, 6]$	per-scale a^η lift (pseudo fractional integration; shifts effective h by η; 0 leaves a chain's slope equal to its h)
`detector` (Detector)	nms	nms / follow	maxima detection: non-maxima suppression along the gradient, or xsmurf's `follow` (contour following of the zero-crossings of κ)
`interpolate`	off	bool	subpixel refinement of each maximum along its gradient (`follow` always refines)

`tie_rule` (Ties)	asymmetric	asymmetric, strict, keep_both	advanced, under `nms`: a pixel tied with its neighbor along the gradient. *asymmetric* is xsmurf's NMS (its `wtmm2d` command, `Remove_Gradient_NonMaxima_Slice_2D`): a pixel must beat the neighbor ahead of the gradient and at least equal the one behind, so one pixel of a tie survives, the one ahead; within about 1.8° of an axis ($|g|/\|g\| > 0.9995$) the neighbors are the nearest pixels. *strict* keeps neither, *keep_both* (the rule before, interpolated, $\geq$ on both sides) keeps both
`min_chain_len`	2	$[1, 1024]$	discard maxima lines spanning fewer scales
`smooth`	off	bool	fill modulus dips along maxima lines before the single-max test
`thresh`	0.001	$[0, 1]$	modulus floor for maxima detection (fraction)
`dist2_max`	50	$[0, 10^6]$ px^2	max squared jump when chaining maxima across scales
`box_ratio`	1.0	$[0.001, 64]$	chaining search-box anisotropy
`similitude`	0.8	$[0, 1]$	cross-scale similarity acceptance for chaining
`band_per`	voice	voice, octave	where the similitude band applies: per voice, between neighboring scales (xsmurf's vchain), or over a whole octave with each step taking its share, so the voices no longer change which h can chain; the reading beside the similitude knob gives the chainable range of h and the largest h the analyzing wavelet resolves, $h < n_\psi$
`chainer`	xsmurf_bench	the chainers	which cross-scale chainer links the maxima (`dynamix.core.chaining`); the default is the benchmark, the xsmurf `vchain` port, and a recipe at that default keeps the key it had
`angle_test`	off	on, off	spec 14, advanced: a candidate also needs $1 - \lvert\Delta \arg\rvert/\pi$ above the angle similarity (a gradient flipped by π accepted); off is xsmurf, whose `vchain` has the test commented out
`angle_threshold`	0.8	0–1	the angle similarity the angle test asks for
`radius_rule`	xsmurf	xsmurf, scale_proportional	the search radius: xsmurf's box and squared-distance cap, or $r = c\, a_{s+1}(a_{s+1}/a_s - 1)$ as both
`radius_c`	1	0.01–100	the constant c of the scale-proportional radius
`tie_break`	distance	distance, modulus	which claimant a coarser maximum keeps: the closest (xsmurf) or the one whose modulus is closest

`border_policy`	none	none, coi_trim, coi_drop	superseded by the Border cut and hidden on the whole-raster tool (a recipe naming it runs as before): chains near the image edge kept (xsmurf), cut at the first scale within k scales of the edge, or dropped
`coi_k`	2	0.1–10	under `border_policy`: the border zone in scales, $k\,a$ pixels from the edge
`edge` (Edge)	auto	auto, reflect, periodic, edge_pad, zero_pad, gaussian_taper	what lies beyond the raster for the FFT, whose convolution is circular: *auto* is periodic on a Periodic extension output (§4.2) and reflect otherwise, and the reading beside the knob says which; *reflect* the data mirrored without repeating the edge sample (JPEG2000's symmetric extension), *edge_pad* the edge values repeated and *zero_pad* zeros, each in a pad as wide as `pad_width` gives (xsmurf cv2d's MIRROR, PADDING and 0_PADDING); *periodic* no padding, the raster wrapped onto itself, with a warning where its opposite edges differ more than three times its neighboring pixels (the RMS of each); *gaussian_taper* the data rolled off to their mean in a band at the raster's edges, as wide as the field's dominant wavelength or the coarsest scale's reach, whichever is longer, with the band's maxima dropped (Jones and Gunaratne 1998), reflect and the reason on the knob where the band would take more than a quarter of the raster. Spec 22's *cut* and *daf_periodic* are hidden: a recipe naming them runs as it did. On an ROI layer the halo reads the parent: *auto* reflects past the parent's edges, and a chosen *periodic*, *edge_pad* or *zero_pad* does too, with a warning; the Border cut runs on whole rasters only
`extent` (Analyze)	real_data	real_data, whole_field	on a Periodic extension output (§4.2): *real_data* keeps the maxima to the real block and reads the threshold from it, the filled gap being padding; *whole_field* analyses the extended field, gap included, so the two spectra can be compared; on any other field the two are the same

`pad_width` (Padding)	measured_reach	measured_reach, fixed_px, border_pct	the pad under reflect, edge_pad and zero_pad, once for the whole transform at the widest scale's width: the coarsest scale's measured reach (`edge_eps`; a scale whose reach does not converge takes the widest converged one, and the reading beside the Edge knob says so), `pad_px` for every scale (the width before spec 25), or xsmurf's $\lfloor a \cdot$ `pad_pct` $\rfloor$
`pad_px` (Padding px)	32	$[0, 8192]$ px	advanced, under fixed_px: the pad width
`pad_pct` (Padding %)	0.72	$[0, 10]$	advanced, under border_pct: the pad as a fraction of the scale
`border_cut` (Border cut)	none	none, measured_reach, border_pct, k_scale, auto	off by default: the padding is cropped off before the maxima are found, so the mirror never carries one. Chosen, it is a step between the H-chains and the V-chains, in xsmurf's order (`rm_ext` before `vchain`): the maxima within the border of the raster's edges are dropped before chaining. *auto* is measured reach, and none on a Periodic extension output or where the measured-reach border would cover the raster (its maxima near the edges are then kept, with a warning saying why); the border is the measured reach, xsmurf's $\lfloor a \cdot$ `border_pct` $\rfloor$ or $k\,a$; the reading gives the coarsest scale's border in px, and a chosen border that covers the raster from some scale up keeps no maxima there and warns
`border_pct` (Border %)	0.72	$[0, 10]$	advanced, under border_pct: xsmurf's `border_percent`
`border_k` (Border k)	2	0.1–10	advanced, under k_scale: the border in scales, $k\,a$ pixels
`border_scope` (Border scope)	largest	largest, per_scale	advanced: every scale cut by the coarsest scale's border, as xsmurf's `rm_ext` does, or each scale by its own
`cut_scope` (Cut scope)	largest	largest, per_scale	hidden, under spec 22's `cut`: the coarsest scale's reach at every scale, or each scale's own, never wider than the coarsest's, which also stands in where a scale's reach does not converge

`edge_eps` (Edge ε)	0.001	$[10^{-6}, 0.1]$	advanced: the share of the analyzing wavelet's $\lvert\psi\rvert$ mass left outside its reach, measured on 1024 and 2048 grids and, where the run's own FFT grid is larger, on that grid too (the next power of two, at most 8192); where two radii differ by more than 25% the reach does not converge, and an ROI reads the whole parent at that scale (up to 64 Mpx) or warns
`show` (*view*)	edges	3 outputs	`edges` (the edge stack), or the `modulus` or `argument` at the shown level (the Outputs rows, §1)

The edge footprint. While an Edge, padding or Border cut knob moves, and for four seconds after, the Raster view draws what they do at the shown scale, from the numbers the run uses: the padded extent as a dashed outline (for *periodic*, as far as the coarsest scale reads round the wrap), the padding's content faintly outside the raster (the mirror, the repeated edge values, the zeros, the wrapped opposite edges, or the mirror of the tapered raster), and the band the Border cut or the taper removes hatched inside it. *View* ▷ *Edge Footprint* keeps it on, following the shown scale. After a Periodic extension in the same chain it reads the extension the last run cached, so the gap shows beyond the raster.

Gradient arrows and anisotropy

The argument $\mathcal{A} = \arg T_\psi$ of each maximum points uphill in the pixel frame: 0° toward increasing column, +90° toward increasing row (clockwise on screen). **Gradient arrows** (Display section: off / on WTMMM / on all maxima) draw one arrow per point at the current scale, half the scale long (at most 12 px) and scaled by modulus. The WTMMM are the maxima of the modulus along each maxima line (xsmurf's `ssm`).

Directional roughness (Spectrum section) plots the WTMMM statistics of Arnéodo, Decoster and Roux (2000, §6): the density $P_a(\mathcal{A})$ at several scales, flat for an isotropic field; the WTMMM of one scale in the gradient plane (T_{ψ_1}, T_{ψ_2}), round for an isotropic field; and the density of $\log_2 \mathcal{M}$ in four axial sectors of $\mathcal{A}$ (mod 180°, centered on 0°, 45°, 90°, 135°), whose shapes coincide when $\mathcal{M}$ and $\mathcal{A}$ are independent. Below them, Khalil, Joncas, Nekka, Kestener and Arnéodo's reading of the anisotropy as numbers (2006, ApJS 165:512, §4.2.4): the two components of the WTMMM scale on their own, $T_{\psi_1} \sim a^{H_x}$ and $T_{\psi_2} \sim a^{H_y}$, so $\ln P_a$ of each component across the scales of a fit window collapses onto one curve once rescaled by a^H; H_x and H_y are read by the slope against $\log_2 a$ of the component's median absolute value (the default) or standard deviation, each with the slope's standard error, or by the collapse itself, and the rescaling spins start at the estimate so that a drag is the paper's sensitivity sweep (their Fig. 14; $\pm\sigma$, $\pm 2\sigma$ companions on request). The readout names the component the modulus follows over the window (its global H is that component's, their §5.2.4), the directional power spectra of the one-dimensional cuts with $\beta_x = 2H_x + 1$, $\beta_y = 2H_y + 1$ and the radial β, and the moment orders q whose rescaled modulus pdfs $a^{-qH}\mathcal{M}^q P_a(\mathcal{M})$ keep their area within 10 % across the scales (their eq. 32: the q the partition functions can carry). The anisotropy against scale is the peak of $P_a(\mathcal{A})$ modulo 180°: its height over the isotropic level and its angle, which sharpens and turns toward fine scales as $\tan \mathcal{A} \sim a^{H_y - H_x}$ (their eq. 38). When a wavelet-leaders result run with its directional option is landed on any layer, its $\eta(2, e_i)/2$ stand beside H_x, H_y as the DWT's reading of the same exponents.

Frames. Angles are shown in the pixel frame or, for a georeferenced dataset, as azimuths: from grid north for a projected CRS (from the directions its axes run), from true north by adding the meridian convergence at each point, and from true north directly on a longitude/latitude grid. A dataset without a CRS (a sensor swath, a pixel grid) offers the pixel frame only.

6.2 WTMM 2D, ROI · `wtmm2d_roi` — WTMM over one ROI window

The parent's WTMM re-pointed at a native-resolution window: the knobs of `wtmm2d` (the smoother and its α / q included, declared once for both) plus the window itself. Used by the overview drill-down: draw an ROI on the decimated whole-extent, compute native.

Parameter	Default	Range	Meaning
`roi_row`, `roi_col`	0	px	window origin (native pixels)
`roi_h`, `roi_w`	512	≥ 8 px	window size
`boundary`	auto	auto / reflective	boundary handling at the window edge
`edge` (Edge)	reflect	reflect, cut, gaussian_taper, daf_periodic	the ROI tool keeps spec 22's Edge and the visible `border_policy` chain trim: the data mirrored, with the warning naming the scales whose halo runs past the raster and the false maxima mirror lines can carry; each scale's maxima dropped and its chains trimmed inside its cone of influence, the measured reach from the raster's edges; the Gaussian taper as on the whole raster; or the DAF extension of the parent (Hoffman et al. 2000; the band limit holding 99% of the power, a gap of two wavelengths at it, $M = 12$, σ for gain 0.99 there, Tikhonov 10^{-3}), reflect and the reason on the knob below six points per wavelength at the band limit

6.3 Mallat–Zhong edges · `mz_edges` — dyadic multiscale edges

Mallat and Zhong's multiscale edges ("Characterization of Signals from Multiscale Edges", 1992), computed with the authors' own algorithm: LastWave's `dwtrans2d` package (Bacry, Mallat, Zhong, Hwang et al.), ported to numba and matching its C bit for bit on square fields. The dyadic wavelet transform is computed à trous by direct FIR convolution with the paper's Table I filters ($h = \{\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}\}$ and a two-tap derivative g). The edges at scale 2^l px are `extrema2`'s gradient-modulus maxima: the gradient direction snapped to the row or column axis, a neighbor whose gradient points the opposite way counted lower, and a gap-filling pass at the finest scale; they are linked into per-scale H-lines (`line_id`). Stored moduli carry LastWave's normalization (factors close to $1/\lambda_l$). `Border` *mirror* runs the whole algorithm on the field mirrored to twice its size, for rasters whose edges do not wrap; *periodic* is LastWave's own mode. Two defects of the C are corrected: non-square fields and a single level ($J = 1$). `Algorithm` *printed* selects the paper's printed algorithm as the research port implements it (Fourier-domain filters on the mirrored torus), kept for comparison. Fields with nodata pixels are refused, and numba is required.

Where the edges are drawn. In the authors' transform a sample at index j describes position $j - \frac{1}{2}$, so maxima and the coarse channel from the second level up are drawn half a pixel west and north of their index. At the finest level the two gradient components sit half a pixel apart (W_x at $(j - \frac{1}{2}, i)$, W_y at $(j, i - \frac{1}{2})$), as in the authors' convention; `Co-locate level 1` averages each onto $(j - \frac{1}{2}, i - \frac{1}{2})$ for detection only, and the stored values the reconstruction uses are unchanged.

Reconstruction from the edges. The Outputs rows show *coarse* (the coarse channel S_{2^J} in the data's units), *thumbnail* (S_{2^J} sampled every 2^J pixels, each block centered on its sample), *recon (edges + coarse)*, *recon (edges only)* (with $S = 0$) and *residual* (the data minus the reconstruction). The reconstruction is LastWave's `e2recons`: an initial pass, then alternating projections. Each iteration analyses the current image, restores the stored values

at the maxima and spreads each correction between neighboring maxima along rows (W_x) and columns (W_y) as a sinh interpolation with decay $e^{-\kappa/2^l}$ per sample (row and column ends pinned to zero, as in the C), optionally clips the modulus to decrease away from each maximum, and synthesizes with the coarse channel held fixed. $\kappa = 1$ is the paper's constant; LastWave uses $\kappa = 2\ln 5.8 \approx 3.516$, which reconstructs markedly better when the coarse channel is pinned and slightly worse for edges only.

The reconstruction knobs live in the right panel's *Reconstruction* section, shown when `mz_edges` is in the layer's rack. By default (`Live` off) no reconstruction row computes on its own: each shows a reconstruction already cached for the current knobs (the recon row also its cached preview), else the data, and reads "press Run" until Run computes the row on show. *Run* computes the reconstruction either to convergence (the relative change falls under the tolerance, the iteration cap is reached, or the constraint residual rises three iterations in a row, which returns the best iterate) or for a fixed count, with progress and Stop. With `Live` on, showing the recon row computes a one-iteration preview, Decay κ, Clip and Coarse recompute it as they change, and Run continues from it. Each recon row reads its iterations, why the run stopped and its SNR against the data (mean removed; for edges only, against the data minus the synthesized coarse part). ROI results do not reconstruct yet (their rows say so), and the thumbnail shows in the 2-D view. Under *printed* the reconstruction is the research port's alternating projections with its own `Mode`, run for the iteration count when a row is shown.

Reconstructing from selected maxima (LastWave). The rack's filters reach the reconstruction: at every level the engine runs the chain's filters that apply on that level alone, and the Reconstruction section's level table decides what each level contributes. *all*: every maximum; *own* (the default): the maxima the filters keep there; *near s*: the level's own maxima within `near_radius` px of a maximum kept at level s (with the α check, only those whose modulus fits the source chain's decay $2^{\alpha(l-s)}$ within the tolerance, in $\log_2$); *coder s*: the kept positions of level s carrying this level's own transform values (Mallat & Zhong's coding mode, §IX); *predict s*: the kept positions and arguments of s with the modulus $M_s\,2^{\alpha(l-s)}$; *open*: no constraints, so the alternating projections fill the level from the others. The *§IX preset* selects at level 2 and codes levels 1 and 3 from it (the paper codes with $J = 3$). A source must have a selection of its own (*all*, *own* or *near*). With Per-level filters on, the filter knobs edit the level the Transport shows and each level keeps its own settings (turning it on starts every level from the current ones). The canvas draws, at the level on show, the maxima the reconstruction reads there, and the section reads how many of the level's maxima those are. α comes from the analysis: each maximum propagates up the levels to the nearest maximum of the same sign within 2^{l-1} px, α is the slope of $\log_2$ of the stored modulus over the linked levels from 2 up (the coarsest level takes the α of the chains reaching it), and a line takes the median of its points'. From level 2 up the stored modulus follows $2^{l\alpha}$ (a step 0, a line -1, a point -2); level 1's staggered two-tap gradient departs from it by a factor that depends on the full direction of the edge's normal (0.86 to 1.37 on straight binary steps), so a prediction at level 1 carries that factor. In mirror mode a selection is applied as removals to all four images of the mirrored field. Every reconstruction row is keyed on the filters and the table, so each selection caches on its own.

Parameter	Default	Range	Meaning
`n_levels` (Levels J)	4	$[1, 10]$	dyadic levels; scales $2, 4, \ldots, 2^J$ px
`algorithm`	lastwave	2 choices	the authors' algorithm, or the paper's printed algorithm (research port)
`border`	mirror	mirror / periodic	the algorithm on the mirrored field, or LastWave's periodic mode
`colocate_l1` (Co-locate level 1)	off	bool	detect the finest level on co-located gradient components
`dither` (printed)	off	bool	$\pm\frac{1}{2}$ LSB dither before analysis
`interpolate` (printed)	off	bool	subpixel refinement of each maximum along its gradient
`show` (*view*)	edges	8 outputs	the raster shown in place of the data (the Outputs rows)
`recon_live` (Live, *view*)	off	bool	off: reconstruction rows wait for Run; on: they compute on their own

`kappa` (Decay κ, *view*)	1.0	$[0.1, 10]$	decay of the correction between maxima, $e^{-\kappa/2^l}$ per sample; 1 = the paper, 3.516 = LastWave
`clip` (*view*)	off	bool	clip the modulus to decrease away from each maximum (LastWave's clipping)
`run_mode` (*view*)	converge	2 choices	run to convergence, or a fixed number of iterations
`iterations` (*view*)	20	$[1, 500]$	the total count (fixed) or the cap (converge), the preview's iteration included
`tolerance` (*view*)	10^{-3}	$[10^{-6}, 10^{-1}]$	relative change of the image that counts as converged
`coarse` (*view*)	full	full / thumbnail	the coarse channel a reconstruction pins: S_{2^J}, or the one decoded from its thumbnail
`mode` (printed, *view*)	separable	3 modes	how the research port's projection interpolates between the maxima
`recon_levels` (Levels, *view*)	(every level own)	table	each level's state and source, and with Per-level filters each level's filter settings
`per_level` (Per-level filters, *view*)	off	bool	the filter knobs edit the level the Transport shows
`near_radius` (*view*)	1	$[0, 8]$ px	how far a *near* maximum may sit from a kept source maximum
`alpha_check` (α check, *view*)	off	bool	a *near* maximum must also fit its source chain's decay
`alpha_tol` (α tolerance, *view*)	0.5	$[0.05, 4]$	the $\log_2$ misfit the α check allows
`alpha_fallback` (*view*)	0	$[-3, 2]$	the α of a chain too short to fit

Fractional B-spline. `Wavelet` *frac_bspline* replaces the quadratic spline by Unser & Blu's symmetric fractional B-spline of order $\alpha \geq 1$ as the smoothing function, the wavelet staying its derivative (one vanishing moment, maxima are edges). Its refinement filter is an infinite impulse response — the fractional order has no compact support — so the analysis runs the Fourier engine on the mirrored torus whatever `Algorithm` says, and its reconstruction is the fractional bank's closed-form synthesis pair, never LastWave's FIR `e2recons`; the live preview refuses fractional bundles with a pointed message. The wavelet scale bar draws ψ_α at the level on show from the exact series, and λ^* from the fractional cascade's own level response (its continuum constant solves $\tan(\omega/4) = (\alpha+1)\,\omega/4\alpha$; $\alpha = 3$ gives the quadratic spline's 1.8595).

6.4 Wavelet leaders · `wavelet_leaders` — leaders, p-leaders and (θ, ω)-leaders on the DWT pyramid

Three multiresolution quantities of the Jaffard school on the critically sampled orthogonal pyramid of `dynamix.core.dwt` (PyWavelets in periodization on the mirror-extended field, each level half the resolution of the last), computed in one pass over levels $j = 1 \dots J$ (cell side 2^j px): the **leader** $\ell(\lambda) = \sup|d|$ over the three bands, the 3×3 block 3λ and every descendant (Jaffard; Dumeur et al. 2025, Def. 1.2), the **p-leader** $\ell^{(p)} = (\sum |d|^p \, 2^{-2(j-j')})^{1/p}$ over the same set with Leonarduzzi et al. 2017's finite-resolution factor when $\eta(p) > 0$, and the (θ, ω)**-leader** ℓ^w, the sup over a window of finer levels and neighboring cells (Dumeur et al. 2025, Def. 2.2), Meyer's weak scaling exponent. Coefficients are L^1 (2^{-j} times the orthonormal ones), every band rolled onto the field's dyadic grid by its atom's centroid, and cells within the bank's border of an edge or of a non-finite pixel are left out (`wavelet`: sym4 5 cells, sym7 6, db4 5, bl3 13). `edge` continues the field to the pyramid's working image: *auto* runs periodic on a Periodic extension's output (its stamp; no border is lost, the real block alone counts) and mirrors otherwise; *mirror*, *reflect*, *edge-value padding*, *zero padding* and *periodic* may be forced. `scale_min` and `scale_max` (powers of two) bound the fitted levels, at least three; `gamma` is a pseudo-fractional integration of the coefficients by $2^{\gamma j}$, every readout given in the integrated frame and shifted back ($\tau - \gamma q$, $h - \gamma$, $c_1 - \gamma$); `dither` perturbs quantized data by half a least significant bit; `p` is the p-leaders' order. The window of the (θ, ω)-leader follows `wse_rule`: the paper's $\theta(j) = j + j^\beta$, $\omega(j') = j'$ in the resolution index (`wse_beta`, default $\frac{1}{4}$), or a fixed `wse_depth` of octaves below each level and `wse_width` of coarse cells either side (depth 1, width 1 lies inside 3λ). `with_leaders`, `with_p_leaders` and `with_wse` choose which quantities a run computes, with their masks, local

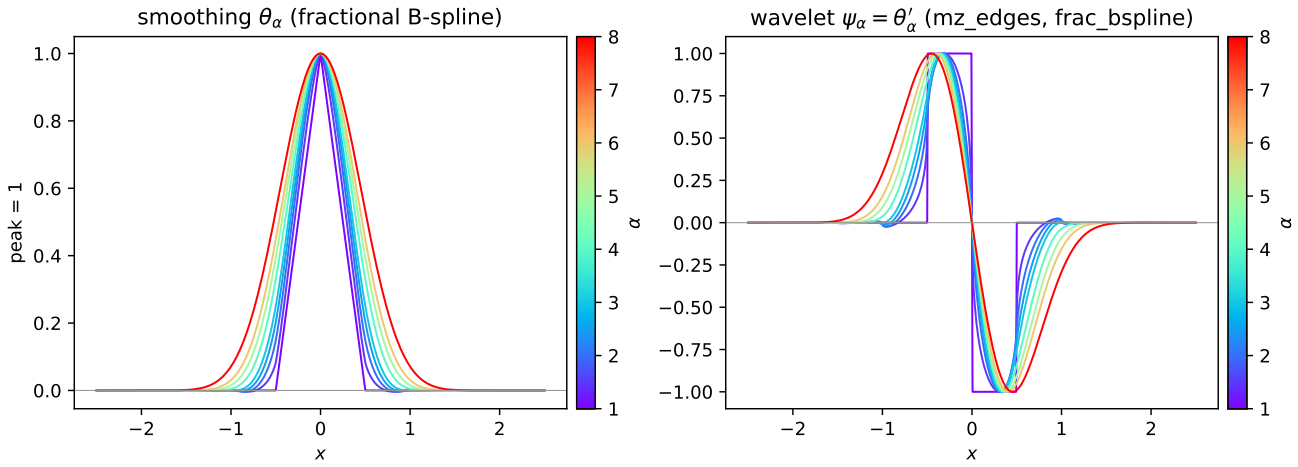


Figure 2: `mz_edges`, `frac_bspline`: the smoothing function $\theta_\alpha(x) = 2\beta^\alpha_*(2x)$, the *symmetric* fractional B-spline of degree α (Unser & Blu 1999, eqs. 10–11; $\hat\beta^\alpha_*(\omega) = |\sin(\omega/2)/(\omega/2)|^{\alpha+1}$), and the wavelet $\psi_\alpha = \theta'_\alpha$ as α varies — the analytic limit the filter cascade is pinned to (its maxima sit exactly where the classical $\alpha = 3$ path puts them: no causal phase). α is θ's Hölder degree; its approximation order is $\alpha + 1$. $\alpha = 1$: a triangle and a box pair; $\alpha = 3$: the paper's quadratic-spline wavelet. Only odd integer α gives a compactly supported classical B-spline; other orders decay algebraically, with small sign-changing tails at the knot spacing (the lobes near $x \approx \pm 1$ on the $\alpha \approx 1.5$–2.5 wavelets are these, not ringing), and as $\alpha \to 0$ the symmetric spline is *not* a box but develops logarithmic peaks and oscillating tails (Unser & Blu, fig. 2). Drawn from the exact time-domain series (their eq. 11), never by inverting the Fourier transform, so the $\alpha = 1$ jumps carry no Gibbs ringing. This is Mallat–Zhong's dyadic *derivative* wavelet built on a fractional spline — exactly one vanishing moment ($\hat\psi \sim i\omega$ at low frequency), since maxima-are-edges is the method — not one of Unser & Blu's fractional spline wavelet bases, which behave like fractional differentiators $|\omega|^{\alpha+1}$.

exponents and sample sets; the WSE, which costs about as much as the transform, is off by default, and the oscillation ratio needs the leaders and the WSE. The tool waits for Run whatever the auto-run setting, so the boxes are chosen before a run. A chain saved before the boxes reopens with all three ticked. `spin` (s grids per axis, 1 by default) runs the leaders on $s \times s$ phases of the dyadic grid, stepped by $2^J/s + 1$ px so every level's cells shift: the grids' per-level samples are pooled with one grid's cell counts and σ's (the shifted samples are not independent), and the h-map is the pixel-wise mean of the grids' maps. On a region the tool reads real data from the file around it: the bank's border in coarsest cells, or the WSE's reach at the coarsest level when it is ticked, plus one cell; the WSE's resolution index is then the file's, so a region's quantities are the whole field's there. Admissibility: leaders need $h_{\min} > 0$ (the slope of $\log_2 \max|d|$ against the level), p-leaders $\eta(p) > 0$ (the slope of $\log_2$ mean $|d|^p$), the (θ, ω)-leaders nothing; a method that fails is computed, stamped and named in the reading with the γ in steps of $\frac{1}{2}$ that would clear it. The tool fits nothing. It publishes the three quantities and the bare coefficients over their valid cells as per-level sample sets (the `Samples` contract: the scales, the field's cell count per level — so a structure function counts the dyadic grid's cells whatever the border takes, $\tau(0) = -2$ — and one set per quantity); the Structure functions and Log-cumulants tools read them with their own moment orders and scale ranges, the Large deviations tool with its own bins. The h-map is a local exponent per cell of the finest fitted level (the slope of $\log_2 \ell$ over that cell and its ancestors) with its R^2, painted on the native grid. `method` chooses which of the three the h-map shows; `show` draws the h-map or, on the Scale slider, $\log_2$ of a quantity per level or the oscillation ratio $\log_2(\ell/\ell^w)$ (0 at a cusp, large at a chirp when the window lies inside 3λ). The record carries the bank, the levels, γ, p, the window schedule, the edge used and the border, the registration, the counts and the admissibility.

Footprint preview. View → Edge Footprint shows the processing extent, the displayed level's dyadic grid, edge exclusions and a representative cell with its 3λ neighborhood, wavelet support and optional WSE window. A region includes its full halo and labels reflected file-edge support. Cycle spinning shows the base grid and reports the number of shifted grids. The preview reads geometry and sampled pixels without running the transform.

6.5 CWT leaders · `cwt_leaders` — Jaffard's leaders on the WTMM's continuous transform

The leader at a point and a scale a is the supremum of the lifted modulus over the scale-space cylinder: every finer scale, within the disc of radius $R(a) = k\,c\,a$, where c is the analyzing kernel's 99 % energy radius per unit scale (`energy`, measured once from the kernel; `radius_k` the multiple, 1 by default). It is the continuous form of the 3λ leader of the DWT pyramid (Jaffard 1994, Lemma 1; Jaffard, Lashermes & Abry 2006), positive everywhere, so negative moments are finite, and it covers the plane, not the maxima skeleton. The front end is the WTMM's own: the smoother (`wavelet` and its knobs), the ladder (`n_oct`, `n_voice`, one voice per octave by default, `a_min`), the lift `fracint_alpha` applied to each scale's modulus before the supremum, and the edge handling, so the transform stage is shared with WTMM 2-D · `wtmm2d`. `operator` chooses the analyzing kernel: the gradient of the smoother (the WTMM's modulus, the shared stage) or the smoother itself, which makes the Mexican smoothers the even wavelets g_2 and q-g_2 and the Gaussians zero-order mass kernels for nonnegative fields. `measure` chooses what is covered: every pixel (the leaders) or the WTMM's maxima only, Jaffard's repair of the wavelet-maxima method, one maximum per cell of radius $R(a)$, read from the shared extrema stage with the WTMM's `thresh`, `detector`, `tie_rule` and `interpolate`. `sup_from` names the finest octave admitted to the supremum, a check that fine-scale noise does not propagate up. `p` adds the p-leaders, the p-mean over the same cylinder (0 turns them off). No fit happens here: the sample sets (every valid pixel; `sample_stride` thins them to a lattice for very large fields, with the independent-disc count $N_{\text{eff}} = A/\pi R^2$ carried for the error bars) go to the Structure functions, Log-cumulants and Large deviations windows, and the h-map is each pixel's slope of $\log_2 L$ against $\log_2 a$ over the fit range (`fit_min`, `fit_max`; 0 is the ladder minus the scales whose disc is under 3 px), which also seeds the windows. Lazy outputs `leader` and `p_leader` show $\log_2$ of either at the scale shown, with its radius in the reading; `show` picks the raster drawn.

6.6 Directional scaling functions · `directional_scaling` — the directional formalism on the hyperbolic DWT

The directional thermodynamic formalism of Ben Slimane et al. 2019 on the hyperbolic (fully separable) wavelet coefficients of a band: the anisotropic scaling function $\eta_\alpha(p)$ on an α_1 grid of step $\frac{1}{8}$ ($\alpha_2 = 2 - \alpha_1$), read along the index families $(j_1, j_2) = ([j\alpha_1], [j\alpha_2])$ as the slowest family's envelope slope over the levels no family clips (at least three); $\eta(p, e_1) = \sup_\alpha \eta_\alpha(p)/\alpha_1$ with the maximizing α_1 as the anisotropy profile $\alpha_1^*(p)$, $\eta(p, e_2)$ by symmetry, $\eta(p)$ the diagonal, and the directional Legendre bound of their Theorem 6 capped at d. `wavelet`, `edge`, `scale_min`, `scale_max` and `gamma` are the Wavelet leaders' (the pyramid to $J = \log_2$ `scale_max`, the fit from $\log_2$ `scale_min`; under *periodic* no border is lost); the moment orders `p_min`, `p_max`, `p_step` are the formalism's $p > 0$. The readings are curves for the Curves window, $\eta(p, e_1)$, $\eta(p, e_2)$, $\eta(p)$ and $\alpha_1^*(p)$ against p, $L(H, e_1)$ and $L(H, e_2)$ against H; the two directional exponents $\eta(2, e)/2$ stand beside the WTMMM's H_x and H_y in the Directional roughness window. Anisotropy needs depth: at the finest levels $[j\alpha]$ cannot resolve a fractional α, and a run that keeps no anisotropy three unclipped levels says so. The record carries the bank, the edge used, the levels, γ, the orders, the levels per anisotropy and the three readings.

6.7 Wavelet skeleton · `wavelet_skeleton` — medial-axis (modulus-minima) extractor

Tang–You 2003 (the constructed compact-support wavelet κ: support radius = scale, maxima-pair separation = scale, width- and gray-invariant) + You et al. 2006 (modulus *minima* sit on the medial axis, scale-independently). A detector, not statistics: directional minima gated by a flanking ridge *pair* (each side within reach $\sim s$ must clear `edge_frac` of the peak modulus), valley contrast (`t_frac` of the flanking ridge — the paper's threshold T made local), and pair dominance (a pair under `pair_dom` of the strongest reachable ridge is someone else's slope). Border and nodata neighborhoods (one kernel support) are distrusted. On `input=gradient` the detector runs over $\|\nabla s\|$, where a diffused scarp (an *edge* in the signal) becomes a ribbon whose axis is the fault trace. Output is single-scale extrema (H-lines via `line_id`).

Parameter	Default	Range	Meaning

`s1` (s_1)	6	$[2, 64]$ px	stage-1 scale = kernel support radius; should cover the ribbon width
`s2` (s_2)	6	$[2, 64]$ px	refinement-stage scale
`t_frac` (T/ridge)	0.5	$[0.1, 0.9]$	keep minima below this fraction of their own flanking ridge
`edge_frac` (Edge/peak)	0.1	$[0.01, 0.6]$	flanking-pair significance vs the *peak* modulus; lower to admit faint features
`n_stages`	2	$[1, 5]$	WT passes (1 = detection only; each further pass re-runs on the skeleton image)
`input`	signal	signal / gradient	ribbons in the signal, or scarps via $\|\nabla s\|$
`wavelet`	tang_you	tang_you / gaussian / bspline2	the papers' own comparison set; theorems hold for κ only
`pair_dom` (Pair dom.)	0.5	$[0, 0.9]$	pair-dominance gate; 0 disables

6.8 CDF edges · `cdf_edges` — complex cross-diffusion edges

One complex cross-diffusion evolution (Gilboa small-θ; Barbeiro 2023 for the scale-space reading), snapshotted at the dyadic schedule $n = \sigma^2/(2\cos\theta)/\mathrm{d}t$: the Re channel evolves as the Gaussian-smoothed image and $\mathrm{Im}/\theta \approx t \cdot \mathrm{LoG}$ (machine-pinned), so one evolution *is* a Mexican-hat scale space. The nonlinear variant modulates diffusivity by the Im channel (Perona–Malik-style edge stopping through k). Final Re (`filtered`) and Im (`edge_channel`) rasters ride the result. Scales are effective Gaussian σ in px.

Edge readouts (*Edges*, `edge_mode`), each read from every snapshot:

- **Canny (Re)**, stored as `mz`: the gradient-modulus maxima of the Re channel. The gradient is the central difference; its direction is snapped to the nearest of the eight neighbor directions, and a pixel is kept where its modulus is at least that of both neighbors along it (neighbors wrap at the field's edges). This is Canny's non-maximum suppression with a single floor, $|\nabla\mathrm{Re}| > \texttt{floor} \cdot \max|\nabla\mathrm{Re}|$, and no hysteresis: one threshold and no edge tracking. The Re channel is already the Gaussian-smoothed image, so the smoothing is the diffusion's own. The same reading is the Mallat–Zhong edge definition, which makes `cdf_edges` directly comparable to `mz_edges` at matched scales; applying it to the diffusion is this app's own graft.
- **Marr (Im)**, stored as `marr`: the zero crossings of the Im channel ($\approx t \cdot \mathrm{LoG}$, the Marr–Hildreth edges), where a pixel and the next one along its row or column differ in sign, kept where the slope of Im exceeds `floor` times its maximum. This is the CDF authors' own edge concept: Barbeiro's v "in the role of edge detector".

Under the `follow` detector both readouts give way to xsmurf's contour following on the Re channel, and `edge_mode` and `interpolate` are inert.

Parameter	Default	Range	Meaning
`n_levels` (Levels J)	4	$[1, 8]$	dyadic sigmas $1, 2, \ldots, 2^{J-1}$ px; cost is quadratic in the top σ
`variant`	linear	linear / nonlinear	LCDF, or NCDF with edge-stopping
`k_edge` (k)	1.0	$[0.001, 1000]$	Perona–Malik edge threshold, image units (nonlinear only)
`theta` (θ)	0.1047	$[0.001, 0.5]$ rad	complex-diffusion angle (small-θ regime; $\pi/30$ default)
`dt`	0.2	$[0.001, 0.24]$	explicit-Euler step (stability-capped)

`edge_mode` (Edges)	mz	mz / marr	the edge readout: Canny (Re), stored as `mz`, or Marr (Im), stored as `marr` (above)
`detector`	nms	nms / follow	maxima detector
`show` (*view*)	edges	edges / filtered / edge_channel / re_scale / im_scale	the edge stack, the final Re or Im, or the Re or Im at the shown level (the Outputs rows; the *edges* row hides the maxima over any raster)
`interpolate`	off	bool	sub-pixel refinement of the maxima
`floor`	0.05	$[0, 0.5]$	detection floor (fraction of the per-snapshot maximum)

6.9 Perona–Malik edges · `pm_edges` — anisotropic diffusion edges

One Perona–Malik (1990) evolution — real, gradient-driven conduction $g(|\nabla I|)$, $\lambda \leq \frac{1}{4}$, adiabatic borders — snapshotted at the same nominal dyadic σ schedule as `cdf_edges`, with gradient-NMS maxima per snapshot. What it adds: past the flux peak $\phi(s) = s\,g(s)$ the diffusion runs locally *backward*, sharpening edges while the discrete maximum principle keeps it stable, so edges stay sharp and in place across the stack. Scales are nominal effective σ (the $K \to \infty$ heat-equation limit); `filtered` (the final diffused raster) rides the result.

Parameter	Default	Range	Meaning
`n_levels` (Levels J)	4	$[1, 8]$	dyadic nominal sigmas
`k_edge` (K)	1.0	$[0.001, 1000]$	edge threshold in image units: contrasts above K sharpen, below it blur
`g`	exp	exp / frac	conduction function (the paper's two)
`lam` (λ)	0.2	$[0.001, 0.25]$	explicit step (stability needs $\lambda \leq \frac{1}{4}$)
`show` (*view*)	edges	3 outputs	canvas shows the edge stack, the diffused raster (`filtered`), or the diffused raster at the shown level (`filtered_scale`); the Outputs rows, and the *edges* row hides the maxima over either
`detector`	nms	nms / follow	maxima detector
`interpolate`	off	bool	sub-pixel refinement of the maxima
`floor`	0.05	$[0, 0.5]$	detection floor (fraction of the per-snapshot maximum)

6.10 Lineal path · `lineal_path` — the two-phase lineal-path function

Family Microstructure. The tool thresholds the field into two phases (`threshold`: phase 1 lies above it) and computes the lineal-path function $L(r)$, the probability that a segment of r pixels dropped along a row or a column lies entirely in phase 1 (`direction`: horizontal, vertical or the mean of both), for $r = 0 \ldots$ `r_max`; $L(0)$ is the phase fraction and L never increases. It produces a Curve, the pair $(r, L(r))$, counted exactly from the phase-1 run lengths without periodic wrap.

6.11 Disorder function · `disorder_function` — disorder of a striped pattern

Family Measures. The tool computes the disorder function of a locally striped pattern, a Euclidean-invariant measure built from the local density of its second derivatives, in its field form (Gunaratne, Hoffman and Kouri 1998, PRE 57:5146, Eq. 2): the density $D(x) = |(\Delta + k_0^2)u|/(k_0^2\langle|u|\rangle)$ vanishes on perfect stripes of wavenumber k_0, and its moments give the mean disorder $\bar{\delta}(\beta) = (2-\beta)\langle D^\beta\rangle$ (dimensionless), the total $|\Omega|\,\bar{\delta}(\beta)$ in px^2 and the rooted mean $\bar{\delta}(\beta)^{1/\beta}$ (Hu et al. 2005), for $0 < \beta < 2$ at $\beta \in \{0.1, 0.25, 0.5, 1, 1.5, 1.9\}$. β is the moment order throughout; the disorder function's first paper (Gunaratne, Jones, Ouyang and Swinney 1995, "An invariant measure of disorder in patterns", PRL 75:3281) writes it μ. `disorder_form` chooses the field form (default) or the PRL's envelope form: the complex envelope $A(x)$ about the wavevector k_0 along the spectral peak is the analytic

signal from the Fourier half-plane $k \cdot k_0 > 0$, demodulated by $e^{-ik_0 \cdot x}$ (Jones and Gunaratne 1998, §4), and the density is $|\odot A|/(k_0^2\langle|A|\rangle)$ with $\odot = k_0 \cdot \nabla - (i/2)\nabla^2$ (PRL Eqs. 2–3); the outputs then carry the envelope's names, and $|A(x)|$ joins the rasters. `envelope` = *wavelet ridge* takes the envelope from Passot and Newell's wavelet ridge instead (Physica D 74:301, 1994, Appendix A): the 2-D analytic q-Morlet of `dynamix.core.analytic_cwt` — a Gaussian in k centered where $aR_\theta k = \omega$ (A.15), the Morlet at $q = 1$, the q-Gaussian envelope of Borges et al. 2004 through its closed 2-D transform otherwise — is applied on a log-polar bank of wavevectors around k_0 (half an octave to an octave either side at 4 voices, 12 orientations over the half-plane about k_0's direction), and at each pixel the fixed point $aR_\theta \nabla\Psi = \omega$ (A.12–A.13, A.17–A.18) is found by interpolation on the bank with Aitken's Δ^2, giving the local wavevector $K = \nabla\phi$, the amplitude $2|T|\,|\det N|^{1/2}/\sqrt{\alpha_1\alpha_2}$ with $N = R\,\mathrm{diag}(\alpha)R^T - ia^2H$ (A.19 as the determinant it comes from, H the Hessian of the ridge phase) and the phase $\Psi + \frac{1}{2}\arg\det N$ (A.8). `omega` is the carrier (default 3, their 2-D runs; the window's phase curvature grows as ω^2, and on a target $\omega = 6$ reads the total 3–6% high where 3 reads it within 1%), `anisotropy` the across/along stretch α_2/α_1 of A.16, `ridge_q` the q with its two widths (`q_pairing`, `q_beta`). `density` picks the phase-field form (default): $\odot A = e^{i\phi}\frac{1}{2}[\nabla\cdot(a^2K)/a + i(a(|K|^2 - k_0^2) - \nabla^2 a)]$ from a, K and $\nabla\cdot K$ alone, which needs no $\pm k$ partition and is the same under $K \to -K$; or the PRL operator on $A\,e^{i(\Psi - k_0\cdot x)}$, blind within one node of the partition's fold. Pixels where the ridge does not converge (defect cores, a local wavenumber outside the bank) are left out and counted; $|K(x)|$ joins the rasters; the record carries ω, α, q, β_q, the bank, the steps and the counts. The curves are Curve outputs against β; the densities are rasters, `show` picking the one on screen. Derivatives (`deriv`) are Fourier multipliers: the papers' Hermite DAF (the default; its order `daf_M` and width `daf_sigma` $= \sigma/\Delta$ are knobs, PRE 57 p. 5147's M = 12 and 4 by default), a derivative of Gaussian of width `sigma_d` px, or spectral derivatives, which need a periodized input — a Periodic extension's stamp, or `periodic` declaring the field periodic as it stands. The two Helmholtz terms are filtered alike and the normalizing $\langle|u|\rangle$ carries the same smoothing, so a perfect stripe reads exactly zero whatever the scheme; a non-periodic field is reflected past its edges by the kernel's reach first. `k0` is the minimum of $\bar{\delta}(1)$ over a bracket around the spectral peak (Gunaratne, Ratnaweera and Tennekone 1999; convex in $1/k_0^2$, so unique), the spectral peak itself, or a given wavelength `lambda0` in px. `harmonic_filter` (default *daf*): Hoffman et al. 2000's pipeline — the field made periodic by the self-tuned DAF extension of §4.2 (skipped when the input is periodic already; where it does not apply the field is reflected instead and the record says why), then the band $[0.7, 1.3]\,k_t$ around the spectral peak kept (Goldman et al. 2002), the derivatives then running on the periodic extension and every statistic over the real pixels alone; or a Gaussian radial band-pass at k_0 of relative width 0.5 (full width at half maximum); or none. `detrend` removes the mean (default) or a plane first. A density below 10^{-12} is a numerical zero. Negative moment orders are not in the default grids; when asked for, the density is clipped at `beta_floor` (0 takes 10^{-6} of the mean density), the floored pixels are counted and a warning says how much a tenfold higher floor would change the value. The record names the scheme, its reach, k_0 and how it was found, the filter, the normalization of each published curve and the floor.

6.12 Hessian measure (Nathan–Gunaratne) · `hessian_measure` — a surface without a planform

Family Measures. The tool computes the Euclidean-invariant measure of a surface that need not have a planform, from the local density of its second derivatives. (Nathan and Gunaratne 2005, PRE 71:035101, Eq. 1): $|\Lambda(x)| = |u_{xx}u_{yy} - u_{xy}^2|/\mathrm{Var}(u)$ and $\mu(\beta) = \langle|\Lambda|^\beta\rangle^{1/(4\beta)}$ in px^{-1} at $\beta \in \{0.1, 0.25, 0.5, 1, 2, 3, 3.4, 5\}$. β is the moment order. The curve is a Curve output against β; the density is the raster. Derivatives (`deriv`, `daf_M`, `daf_sigma`, `sigma_d`) and `periodic` are those of the disorder function (§6.11); so are `detrend` and the floor for negative moment orders. `hessian_norm` keeps the variance of PRE 2005 Eq. 1, the standard deviation being a diagnostic that breaks the inverse-length dimensions. The record names the scheme, its reach, the norm and the floor.

6.13 Wavelet ridge (Passot–Newell) · `wavelet_ridge` — amplitude and wavevector of stripes

Family Patterns. The tool reads a striped field's slowly varying amplitude and its wavevector field off the ridge of a 2-D analytic wavelet transform (Passot and Newell 1994, Physica D 74:301, Appendix A): at each pixel the scale and orientation where the transform's phase is stationary, $aR_\theta\nabla\Psi = \omega$ (A.12–A.13, A.17–A.18), found on a log-polar bank of wavevectors around the planform wavenumber k_0 (half an octave to an octave either side, 4 voices, 12 orientations over the half-plane) by interpolation with Aitken's Δ^2. The wavelet is the 2-D analytic

q-Morlet of `dynamix.core.analytic_cwt`: a Gaussian in k centered where $aR_\theta k = \omega$ (A.15), the Morlet at $q = 1$, the q-Gaussian envelope of Borges et al. 2004 otherwise (`ridge_q`, with its two widths `q_pairing` and `q_beta`); `omega` is the carrier (default 3, their 2-D runs), `anisotropy` the across/along stretch α_2/α_1 of A.16. The amplitude is $2|T|\,|\det N|^{1/2}/\sqrt{\alpha_1\alpha_2}$ with $N = R\,\mathrm{diag}(\alpha)R^T - ia^2H$ (A.19 as the determinant it comes from, H the Hessian of the ridge phase, smoothed over half the window) and the phase $\Psi + \frac{1}{2}\arg\det N$ (A.8). `k0` centers the search: the spectral peak, or a given wavelength `lambda0`. The ridge rasters are scale-free, chosen by `show`: $|A(x)|$ (default), $|K(x)|$, the roll direction $\vartheta(x)$ as a director in $[0, 180°)$, the ridge scale $a_r(x) = |\omega|/|K|$ and the phase $\varphi(x)$; pixels where the ridge does not converge (defect cores, a local wavenumber outside the bank) are blank and counted. The transform the ridge is read from sits on the Scale slider: over the ladder `a_min`, `n_oct`, `n_voice`, each level offers $|T|$ maximized over `orientations` wavelets centered on $|\kappa| = \omega/a$ (*modulus*), the orientation that wins and the phase there, computed on request and cached per level. Figure 12 draws the q-Morlet's real and imaginary parts and modulus in 3-D for three q. The scale bar labels each level with $\lambda^* = 2\pi a/\omega$, the carrier's wavelength, and draws the q-Morlet's section along its carrier. A non-periodic field is reflected by the widest wavelet's reach; a Periodic extension's stamp, or `periodic`, runs it periodic. The record carries ω, α, q, β_q, the bank, the steps, the counts, k_0 and the ladder.

7 Decompositions

Standalone transforms heading their own chain, displaying through the ordinary raster path. They are primary analyzers: dropped onto a layer headed by another analyzer, each forks a sibling layer (§2). Every output is computed once and cached; which one is on screen is a *view* choice. To keep an output as data of its own, fork it (§1).

Group (`tucker_HOOI_HOSVD`, `ssa2d`; *view*): which components form the reconstruction — `all` (the whole kept reconstruction, exactly), or a list such as `1-3, 5`. **recon** shows the sum of the group; **residual** shows the data minus that sum (data = recon + residual of the whole kept reconstruction). So removing just the first component is Group `1` with Show = residual. The layer row names it: `recon [1-3, 5] (88%)`, `data − [1] (42%)` (the group's share). An unreadable group shows the whole reconstruction and says so. The Components window picks the group by clicking thumbnails.

7.1 PCA · `pca` — principal components of a multi-band field

Principal component analysis over the bands of a multi-component field ($n_y, n_x, n_c \geq 2$) (covariance-trick PCA); shows one component as a raster.

Parameter	Default	Range	Meaning
`n_components` (Components)	6	$[1, 64]$	components computed
`component` (Show PC; *view*)	1	$[1, 64]$	the component shown
`standardize`	off	bool	scale each band to unit variance first (correlation PCA)

7.2 Tucker (HOOI / HOSVD) · `tucker_HOOI_HOSVD` — delay-embedded Tucker decomposition (Tucker-SSA)

A Tucker decomposition of a delay embedding of the field — Broomhead–King singular-spectrum analysis carried to 2-D and multiband data by a Tucker decomposition instead of one SVD. (It was first named `tucker_havok`; HAVOK's regression on the delay coordinates was never part of it.) Nothing is ever materialized: bases come from Grams of contiguous window slabs, and the reconstruction averages each pixel over the windows covering it.

- **Embedding.** `delay_2d` (default): *both* axes delayed — every $L \times L$ patch (L = Delays), tensor (delay$_y$, delay$_x$, y', x' [, band]), so no image axis plays "time". `delay`: a 1-D Hankel embedding along one axis (Tape axis) — for data where one axis really is time (e.g. a reconstructed tape); on an ordinary 2-D field it biases the result toward that axis. `none`: the array's own modes (rows, columns[, band]).

- **Method** — a two-state button. **HOSVD**: one pass, each mode's leading singular vectors. **HOOI**: that HOSVD refined by *HOOI sweeps* alternating passes, each mode re-derived with the others fixed (monotone in the kept core energy; the fast path equals a materialized HOOI to 10^{-8}).
- **Components** (at most 32, largest core-energy share first): under `delay_2d` one per kept (vertical, horizontal) delay-pattern pair (a, b); *Orientation* `combined` adds each pair to its twin (b, a) ($r(r+1)/2$ components for delay rank r). Tape: one per delay pattern; none: one per rows-mode vector. They sum to the reconstruction.
- Separable by construction: a pattern is a vertical pattern $\otimes$ a horizontal one, so an oblique structure spreads over several pairs. `ssa2d` is the rotation-invariant alternative.

Parameter	Default	Range	Meaning
`embed` (Embedding)	delay_2d	delay_2d / delay / none	see above
`axis` (Tape axis)	rows	rows / cols	the time axis of the 1-D tape (grayed out unless `delay`)
`n_delays` (Delays)	32	$[2, 512]$	embedding dimension L (patch $L \times L$ under delay_2d); the lag between delays is 1 px
`rank_delay` (Rank: delay)	4	$[0, 512]$	delay patterns kept (both delay modes under delay_2d); 0 = full
`rank_time`, `rank_space`	0	$[0, 10^5]$	ranks of the rows/cols (y'/x') modes; 0 = full
`rank_band` (Rank: band)	0	$[0, 64]$	rank of the band mode (multiband only); 0 = full
`method` (Method)	HOSVD	HOSVD / HOOI	the algorithm (a toggle button)
`sweeps` (HOOI sweeps)	2	$[1, 16]$	HOOI refinement passes (grayed out under HOSVD)
`show` (*view*)	recon	recon / residual / component	what the canvas shows
`component` (*view*)	1	$[1, 64]$	the component shown (Show = component)
`pairs` (Orientation; *view*)	separate	separate / combined	component numbering under delay_2d; also what a Group refers to
`group` (Group; *view*)	all	text	the components the recon sums (above)
`cluster` (*view*)	1	$[1, 64]$	the cluster shown (largest share first)
`cluster_distance` (*view*)	0.5	$[0.05, 0.95]$	where the clustering tree is cut

Superseded: `tucker_havok` stays registered (browser Dev group) so saved projects open and resolve exactly as before; its single *HOOI sweeps* knob chose the algorithm (0 = HOSVD).

7.3 2-D SSA · `ssa2d` — 2D singular-spectrum analysis

Golyandina & Usevich (2010). Every $L_r \times L_c$ window of the image is one column of the Hankel-block-Hankel matrix $\mathbf{W}$ ($L_r L_c \times K_r K_c$, $K = N - L + 1$); *one* SVD $\mathbf{W} = \sum_i \sqrt{\lambda_i}\, U_i V_i^{\mathsf{T}}$ gives the eigentriples. Reshaped, U_i is an **eigenarray** — an $L_r \times L_c$ pattern of any shape, not restricted to separable products, which makes 2D-SSA invariant to rotations of the image axes — and V_i a **factor array** ($K_r \times K_c$). Each rank-1 term, averaged back onto the image (every pixel over the windows covering it), is an **elementary component**; all of them sum to the image. Ranks (the paper's §4, pinned by tests): an oblique cosine $\cos 2\pi(\omega_x k + \omega_y l)$ is rank 2, a product of cosines rank 4, a sum of m 2-D exponentials rank m.

Solvers. None forms $\mathbf{W}$.

- *Lanczos* (the default), *PROPACK* and *randomized* use FFT products: $\mathbf{W}^\mathsf{T}u$ and $\mathbf{W}v$ are 2-D correlations of the image, done at its own size, whose kept parts never wrap (Korobeynikov 2010; Golyandina et al. 2015). They find only the k leading eigentriples — ARPACK's Lanczos on the implicit $\mathbf{W}\mathbf{W}^\mathsf{T}$; PROPACK's bidiagonalization of $\mathbf{W}$, with full reorthogonalization (its default partial one returns ghost copies of the leading triplet when the 32-bit FFT products are adjoint only to $\sim 10^{-7}$); or a randomized range finder whose power iterations are re-orthonormalized after every product (Halko et al. 2011; Lopes et al. 2024). Components are FFT convolutions of eigenarray and factor array. They agree with the dense route to float32 tolerance. Measured on a 2048^2 field with a 64×64 window and $k = 16$: Lanczos 6 s, randomized 25 s, PROPACK 42 s.
- *Dense*: the lag covariance $\mathbf{W}\mathbf{W}^\mathsf{T}$ ($(L_r L_c)^2$) accumulated from blocks of window rows, its full eigenbasis, then $\mathbf{W}^\mathsf{T}U$ in the same blocks; float64 BLAS, equal to an explicit SVD of $\mathbf{W}$ to 10^{-9}. Limited to windows of 1600 entries (40×40, about 4 s on a 256^2 field). An FFT solver hands over to it when the kept count reaches the window size.

The shares $\lambda_i / \operatorname{tr}(\mathbf{W}\mathbf{W}^\mathsf{T})$ are exact on every route (the trace is the sum of the squared pixels weighted by their window counts). Each eigenarray is signed so that its largest entry is positive. Large outputs are stored at the app's precision. Single-band fields only. NaN pixels are filled with the finite mean for the decomposition and re-masked in every output. Also returned: the **w-correlation** matrix (the paper's Def. 3.18) of the kept components — the grouping aid shown in the Components window.

Keeping and grouping (Lopes et al. 2024). *Keep: energy* treats `n_components` as the most to find and keeps the fewest whose cumulative share reaches the threshold (the row note says when it was not reached). This 2D-SSA is uncentered, so on a raw field the first eigentriple, the mean, often carries most of the energy on its own. *Show: cluster* shows one group of an automatic grouping: average-linkage clustering on $1 - |\text{w-correlation}|$, cut at the cluster distance, clusters ordered by share. The cluster knobs are view settings, so changing them never re-decomposes.

Edges. A pixel near the edge of the analyzed region is covered by fewer windows. On a region of interest the tool reads a margin of real data around it (`margin_windows` windows of $\max(L_r, L_c) - 1$ pixels; at one window every ROI pixel is fully covered) and crops back. On the whole field there is no data beyond the edge. Lopes et al.'s mirror extension is available (*Whole-field edges: mirror*), but off by default: in 2-D a reflected wave is a different wave (an oblique wave flips a frequency component), so the extension adds rank. On two test waves it raised the edge error 3–18×, where the real-data margin of an ROI lowered it by about 10%.

Parameter	Default	Range	Meaning
`rows_window`, `cols_window`	16	$[2, 512]$	window $L_r \times L_c$ ($L_r L_c \leq 1600$ for dense)
`n_components` (Components kept)	16	$[1, 64]$	leading eigentriples kept (the most, with Keep: energy)
`keep`	count	count / energy	keep `n_components`, or the fewest reaching the energy threshold
`energy_threshold` (Energy)	0.9	$[0.5, 0.9999]$	cumulative share to reach (Keep: energy only)
`solver`	lanczos	lanczos / propack / randomized / dense	the eigentriple route (above)
`power_iterations`	2	$[0, 10]$	randomized solver only
`margin_windows` (Edge margin)	1	$[0, 4]$	ROI margin in windows; also the mirror width
`edge_extension` (Whole-field edges)	none	none / mirror	mirror-extend the whole field (off: it adds rank in 2-D)
`show` (*view*)	recon	recon / residual / component / cluster	what the canvas shows
`component` (*view*)	1	$[1, 64]$	the elementary component shown

`group` (Group; *view*)	all	text	the components the recon sums (above)
`cluster` (*view*)	1	$[1, 64]$	the cluster shown (largest share first)
`cluster_distance` (*view*)	0.5	$[0.05, 0.95]$	where the clustering tree is cut

7.4 DWT · `dwt2d` — the periodized orthogonal pyramid

The field is mirror-extended once in each axis and rounded up to a multiple of 2^J, then decomposed with an orthogonal bank (symlets, Daubechies, coiflets, or `bl3`, the Battle–Lemarié order-3 bank of Pottier, Turiel & Garçon 2008; every bank's φ and ψ in Figure 9, its 2-D atoms in Figure 10) in periodization mode, the one mode whose synthesis is the analysis's adjoint. Every band is a per-level row reconstructed alone onto the native grid: the `approximation` at level j (the low-pass at scale 2^j px) and the `horizontal`, `vertical` and `diagonal` details; the details of every level plus the coarsest approximation sum back to the field exactly. Cycle spinning averages each row over random shifts (translation invariance at the price of that many transforms). Non-finite pixels are filled with the finite mean for the transform and restored on every row. The Transport walks the levels; the pyramid is the foundation of the wavelet-leader tools.

Parameter	Default	Range	Meaning
`wavelet`	sym7	the orthogonal banks	the pyramid's bank
`level`	0	$[0, 10]$	levels; 0 = the full depth $\lfloor \log_2 \min(2n_y, 2n_x) \rfloor - 1$, at most 8
`cycle_spin`	off		average every row over random shifts
`n_spins`	8	$[1, 64]$	the shifts averaged
`show` (*view*)	coarsest	coarsest / the four bands	the coarsest approximation, or the row drawn at the Transport's level

7.5 Interpolating wavelets (DD / DAF) · `interpolating_wavelets` — a lifting pyramid

A biorthogonal lifting transform built on an interpolating predictor. Each level splits rows and columns into even and odd samples, predicts the odd ones from the even ones and keeps the misfit as the detail, then updates the even ones by one half of the predictor's transpose (a local lifting design), with the midpoints normalized for DC. The predictor is either the four-point Deslauriers–Dubuc stencil $(-1, 9, 9, -1)/16$ or the Gaussian–Lagrange DAF shell of US6847737 (Eqs. 47–48), restricted to its polynomial stencil of odd order M with Gaussian width σ in coarse cells. This local construction is biorthogonal (the orthogonal banks are `dwt2d`'s), and the leaders read the DWT pyramid, so its coefficients stay out of them. The outputs are the coarsest approximation (`low`) and the horizontal, vertical and diagonal details at each level (per-level outputs, §1); `show` picks the displayed band. Figure 11 draws the atoms of both predictors in 1-D and in 3-D.

Parameter	Default	Range	Meaning
`bank` (Predictor)	dd4	dd4, gldaf	Deslauriers–Dubuc 4-point, or the Gaussian–Lagrange DAF shell
`order` (Shell order M)	3	$[1, 15]$, odd	under gldaf: the shell's polynomial order
`sigma` (Gaussian σ)	1	$[0.25, 5]$	under gldaf: the shell's Gaussian width in coarse cells
`levels` (Levels)	3	$[1, 10]$	the pyramid's depth
`edge` (Boundary)	reflect	reflect, periodic	the stencil beyond the edges
`show` (*view*)	low	low, horizontal, vertical, diagonal	the displayed band

8 Inpainting and fusion

One engine, three routes (Pottier, Turiel & Garçon 2008 for the cascade, the compressed-sensing notebooks for the cycle-spun sparsity). The unknown lives on the mirror-extended working image; the data terms are the target under its weights and every guide on its own grid (a shift on the same grid, a block average on a nested one, a warp on a foreign one — a guide may be resampled inside its measurement, the target never); the prior is the cycle-spun soft threshold of the detail bands (*sparsity*, FISTA with restarts), the paper's parent-from-children inference over the abacus of $\ln|\eta|$ blended by ρ (*cascade*, the paper's plain loop), or the first then the second (*both*). The known pixels are corrected after every step through the universal kernel and put back exactly at the end. Held-out blocks of clear pixels measure the error of each route; the $\ln|\eta|$ and $\ln|c|$ distributions inside the regions against the clear ones say whether the fill kept the cascade (the Histograms window).

8.1 Inpaint · `inpaint` — fill declared regions

Regions: the non-finite pixels (always), a threshold on a band of the stack (connected components over `mask_min_px`), polygons drawn in the Mask mode (stored in file pixels), or a sent mask raster; each region is a *gap* (every scale inferred) or a *veil* (shadow, haze: the fine-scale geometry is data under a gain or an offset, the amplitude and the coarse levels are inferred; experimental). Sends (the Sends dialog): a *value* prior (the same quantity: another day, another satellite, a climatology), a *structure* guide (another band, or a DEM shaded under the scene sun), *weights* (a coherence or error raster, for the target or for a send), a *mask*, a *veil seed* DEM. A foreign value or structure send closes the cascade routes. Result: the filled band (`raster_out`, equal to the data on known pixels), `fill_mask`, the rows `residual` and `heldout`, and the stamps `_route`, `_grid_relations`, `_seed`, `_iterations`, `_returned`, `_stop`, `_variance_error`, `_heldout`, `_eta_distance`, `_amp_distance`, `_terms_dropped`, `_veil`, `_fill_kind`.

Parameter	Default	Range	Meaning
`route`	sparsity	sparsity / cascade / both	the prior
`wavelet`	sym7	the orthogonal banks	the pyramid's bank
`level`	0	$[0, 10]$	0 = the route's depth (sparsity: PyWavelets' maximum, at most 8; cascade: the paper's full depth)
`lam_rel`	0.2	$[0, 10]$	λ relative to $\hat{\sigma}\sqrt{2\ln N}$ (sparsity, both)
`n_iter`	0	$[0, 5000]$	0 = the route's default (200, 20)
`tol`	10^{-3}	$[0, 1]$	sparsity stop: relative change inside the regions
`stop_var_err`	0.01	$[0, 1]$	cascade stop: variance error on known pixels
`seed`	0	$[0, 2^{31}]$	cycle-spin shifts, the held-out draw, the synthesis pass
`batch`	1	$[0.05, 1]$	the mini-batch of known pixels per iteration
`cascade_synthesis`	off		the coarse-to-fine synthesis pass (beyond the paper)
`target_band`		text	the band id on a stack (empty = the field)
`mask_source`	nodata	nodata / threshold / region / send / all	where the regions come from
`mask_band`, `mask_op`, `mask_value`, `mask_min_px`, `threshold_kind`			the threshold: band id (empty = the target), > or <, the value, the smallest component, gap or veil
`region_kind`	gap	gap / veil	the kind of a sent mask's regions

`veil_model`, `veil_levels`, `veil_passes`	gain, 3, 2		the veil: gain (positive data) or offset (log-domain), its fine levels J_v, the outer passes
`sun_azimuth`, `sun_elevation`	0, 45	degrees	the scene sun when the loader stamps none (ASTER does)
`prior_weight`	0.3	$[0, 10]$	λ_v of a value prior
`structure_weight`, `structure_levels`	0.1, 3		λ_s and J_s of a structure guide
`abacus`, `abacus_file`, `abacus_min_pairs`	self, , 1000		the $\ln\lvert\eta\rvert$ histograms: this field's clear pairs, or a corpus file
`correct`	on		the known-pixel correction after each step
`heldout_blocks`, `heldout_side`	8, 16		the held-out blocks (0 = none) and their side in px
`show` (*view*)	filled	filled / residual / heldout	the raster on the product layer

9 Microcanonical MFA

Per-pixel singularity exponents $h(x)$ from the model $T(x, r) \approx \alpha(x)\, r^{h(x)}$ (Turiel 2008), $D(h)$ by direct histogram — the formalism that is *not* canonical/Legendre. One tool per functional; both expose the Pont 2006 estimator pair: **regression** (per-pixel log–log slope over the scale grid; the prefactor cancels exactly; resolves both limbs of $D(h)$) and **punctual** (the finest scale alone, ensemble-mean normalized; sharpest localization, the most-singular-set extractor; cannot resolve the right limb; carries an $\alpha(x)/\langle\alpha\rangle$ bias). Both estimators share the same cross-scale no-support mask — they agree on *where* an exponent exists and differ in *how*. Under punctual, κ/`n_scales` shape only the mask and `r2_map` is honestly all-NaN. Figure 8 draws both routes' kernels; Figure 16 their q sweeps.

9.1 Microcanonical MFA, measure · `holder_measure` — gradient-measure route

T = positive unit-mass kernel over $\|\nabla s\|$ (finite differences first, then g_0-class smoothing). For measure-like / edge-sparse fields; slope $= h$. The scale knob r_1 is the kernel's *dilation scale* (for the Gaussian: its standard deviation) — positive kernels have no zero crossings, and their resolution floor is the pixel, except for the compactly supported q-Gaussian (below). Kernel trade-off (Turiel App. A): Gaussian resolves every h; Lorentzian truncates at $h < 2\beta - d$ but localizes sharpest; q-Gaussian interpolates; `frac_gaussian` fractionalizes the envelope, $e^{-\rho^n/2}$ (n=2 is the Gaussian exactly).

The q-Gaussian is Tsallis's $e_q^{-\rho^2/2r^2}$ with $e_q^x = [1 + (1-q)x]_+^{1/(1-q)}$ (Borges et al. 2004, eq. 9), for $-1 \le q \le 3$. Above $q = 1$ it has power-law tails (Student-t / Lorentzian: $q \equiv 1 + 1/\beta$ with ρ^2/r^2 rescaled); $q = 1$ is the Gaussian; below $q = 1$ it is *compactly supported*, radius $r\sqrt{2/(1-q)}$ — $q = 0$ is the Epanechnikov paraboloid, $q = -1$ a half-dome of radius r, and $q \to -\infty$ the Dirac δ. For $q \ge 2$ the 2-D kernel has no finite mass (the Lorentzian $\beta \le 1$ case), so the result depends on the field size; the q knob shows a warning there. The compact kernel (below $q = 1$) is the steepest decrease there is, so it carries a resolution floor: the r_1 knob warns when its support disc, of radius $r/\sqrt{\beta(1-q)}$ for width β, is less than 2 px across, i.e. below $r_1 = \sqrt{\beta(1-q)}$ (at most 1 px at the default width).

9.2 Microcanonical MFA, multiaffine · `holder_multiaffine` — wavelet-projection route

$T = |\psi_r * s|$ with a zero-mean wavelet applied to the *raw signal* (derivative order lives inside the kernel). For smooth / function-class fields (DEMs); slope $= \gamma$, with $h_{\text{measure}} = \gamma - 1$. Here r_1 is the *zero-crossing radius*: every wavelet in the menu is calibrated to change sign at radius r, so for a wavelet with one crossing the fig.-2 minimum-resolution rule reads $r_1 \ge 1$ (more crossings, or a compact support, raise it: *Resolution floor* below). The g-ladder is the vanishing-moment axis (order n measures only $\gamma < n$): `g1`/`g3` are the fractional family at n=1, 3; `g2` *is* the Ricker; `lorentzian_marr` is the Laplacian (Marr operator) of the Lorentzian envelope (true tail $\rho^{-2\beta-2}$, truncation $\gamma \ge 2\beta$); `frac_gaussian` makes the order a real knob ($\psi \propto {}_1F_1(\frac{n+2}{2}; 1; -\rho^2/2\sigma^2)$, n=2

reproduces the Ricker exactly): n is a fractional *differentiation* order ($\hat{\psi} \sim |k|^n$ at low frequency), so exact moments vanish only for the integer orders below n — a fractional n is not "n vanishing moments" — and $\gamma < n$ is measurable.

`q_mexican` is the q-Mexican hat of Borges et al. (2004, eq. 15): the second derivative of the q-Gaussian *raised to the power* $2-q$. In 2-D (the Laplacian) this gives

$$\psi_q(\rho) \;\propto\; \left[1-(2-q)\,b\rho^2\right]\left[e_q^{-b\rho^2}\right]^q, \qquad b=\frac{1}{(2-q)\,r^2},$$

so the zero crossing sits at $\rho = r$ for every q, and $q = 1$ is the Ricker exactly. The paper's $q < 3$ is the 1-D bound: in 2-D the Laplacian of $[e_q]^{2-q}$ vanishes identically at $q = 2$ and never changes sign above it, so the knob runs $-1 \le q \le 1.95$. For $q \ge 0$ the wavelet is bounded; for $q < 0$ it diverges at its cutoff (still integrable) and the knob shows a warning. Because $[e_q]^{2-q}$ is itself a q'-Gaussian with $q' = 1/(2-q)$, this is the family of Laplacians of q-Gaussians under a relabeled index: `q_mexican` at q equals `lorentzian_marr` at $\beta = (2-q)/(q-1)$ for $1 < q < 2$ (e.g. $q = 1.5 \equiv \beta = 1$), with tail $\rho^{-2/(q-1)}$ and truncation $\gamma \ge 2(2-q)/(q-1)$.

The q width β (both q-kernels) is the β of $e_q^{-\beta x^2}$, a toggle away from the paper's two choices.

- **Fixed** (default $\beta = \frac{1}{2}$): the unit-variance Gaussian per component at $q = 1$, in any dimension. The scale r keeps its meaning at $q = 1$: $\sigma = r$ for the q-Gaussian, the zero crossing at r for the q-Mexican hat. At other q the q-Mexican hat's zero crossing drifts to $r/\sqrt{2\beta(2-q)}$.
- **q-paired**: Borges et al.'s $\beta = 1/(3-q)$ holds the q-Gaussian's *escort* (q-)variance $\int x^2[e_q]^q / \int [e_q]^q$ at one. In d dimensions that variance is $1/(\beta(d+2-dq))$ per component, so the pairing is $\beta = 1/(d+2-dq)$: $1/(3-q)$ in 1-D and $\beta(q) = 1/(2(2-q))$ here.
- The q-Mexican hat's zero crossing sits exactly at that escort RMS radius, so the pairing also pins it (at r, for every q: the tool's classic calibration). The two settings agree at $q = 1$.
- The pairing exists exactly where the 2-D q-Mexican hat is admissible, $q < 2$: the escort variance and the zero mean both need $d + 2 - dq > 0$. A q-paired q-Gaussian therefore refuses $q \ge 2$.
- In real space the normalization is done numerically at every scale, so the paper's closed-form constants are not used. (In 2-D they happen to be elementary: $A_q = \sqrt{\beta(3-q)(1+q)/\pi}$ for the q-Mexican hat, $B_q = \sqrt{\beta(3-q)/\pi}$ for the modulated q-Gaussian. The flat Jacobian $\rho\,d\rho = du/2\beta$ turns every integral into a Beta function with integer arguments, where 1-D keeps $\Gamma(\frac{1}{2})$-type ratios.)

Resolution floor on r_1. A discretized wavelet must separate its positive and negative parts, so its zero crossings set the minimum attainable resolution, and more crossings cost resolution (Turiel et al. 2008, fig. 2 and §4.2.1); Turiel (2009) puts a Mexican hat's inner minimum scale at "several pixels". Neither gives a number. The tools read the rule as: every sign lobe at least 2 px across at r_1, the central disc counted by its diameter. The edge of a compact support is not a sign change, but it closes the last lobe. The r_1 knob warns below the floor and never refuses.

- One crossing at r (`g1`, `g2`, `lorentzian_marr`, `frac_gaussian` for $n \le 2$): the floor is $r_1 = 1$, as before.
- More crossings: `g3` 1.32 px (its ring is $1.52\,r$ wide); `frac_gaussian` at $n = 4$ and 5, 1.41 and 1.46 px.
- `q_mexican`: its crossing sits at $\rho_0 = r/\sqrt{2\beta(2-q)}$ (r when q-paired) and, below $q = 1$, its support ends at $\rho_0\sqrt{(2-q)/(1-q)}$. The negative ring between them narrows as q falls, and below $q \approx 0.875$ (q-paired) it is the narrowest lobe: floor 2.73 px at $q = 0.5$ and 4.42 px at $q = 0.1$. A fixed width β is a pure dilation of the wavelet, so it rescales the floor by $\sqrt{2\beta(2-q)}$.
- `holder_measure`: only the compact q-Gaussian has a floor (above).

Wavelet space: real or Fourier (both tools). *Real* (the default, Turiel's construction) samples each kernel on the pixel grid and normalizes it there. *Fourier* evaluates the same continuous kernel's closed-form 2-D transform at $r\mathbf{k}$: a Matérn function of k for a q-Gaussian with $q > 1$ (and the Lorentzian, $q = 1 + 1/\beta$), a Jahnke–Emde function for $q < 1$, $k^n e^{-k^2/4u_0}$ for the fractional family, evaluated stably at every q. The kernel is then band-limited rather than sampled.

- Measure route: unit mass exactly. Multiaffine route: zero mean exactly, and the kernel is the L^1-dilated continuous wavelet, so T equals the real construction's times the mother wavelet's L^1 norm (a constant that cancels in every slope) wherever both are exact.
- They differ where sampling matters: at the smallest scales, for compact kernels (the Fourier one rings slightly), and for heavy tails, which real space cuts at the edge of the padded grid (about 10^{-4} at the default padding for a $\rho^{-6.7}$ tail).
- Fourier is refused on the measure route for `frac_gaussian` (no closed-form transform), and for a q-Gaussian at $q \geq 2$ or a Lorentzian with $\beta \leq 1$ (no finite mass to normalize by).

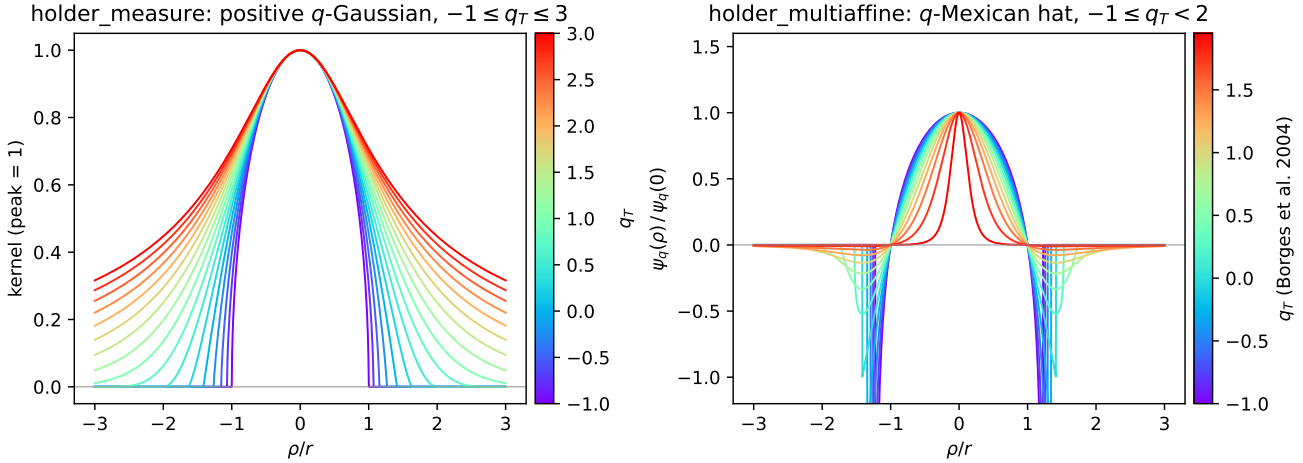


Figure 3: The q family. Left: `holder_measure`'s positive q-Gaussian, $-1 \leq q \leq 3$ (compact below $q = 1$). Right: `holder_multiaffine`'s q-Mexican hat (Borges et al. 2004, in 2-D), $-1 \leq q < 2$; every curve crosses zero at $\rho = r$, and for $q < 0$ the negative lobe diverges at the cutoff. Profiles scaled to 1 at the center.

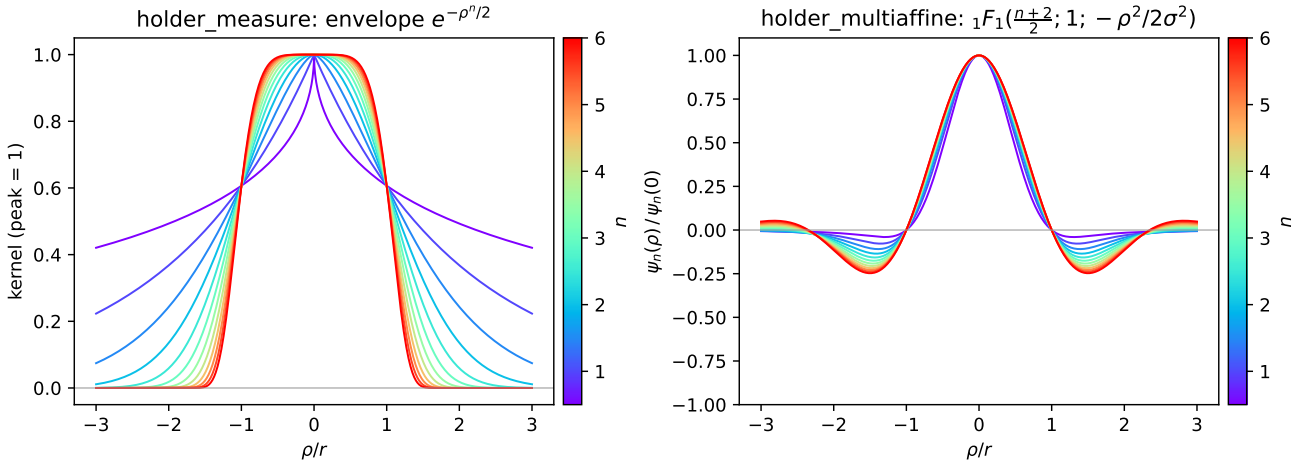


Figure 4: The fractional Gaussians. Left: `holder_measure`'s envelope $e^{-\rho^n/2}$ ($n < 2$ fattens the tail, $n > 2$ flattens the top; every curve passes $e^{-1/2}$ at $\rho = r$). Right: `holder_multiaffine`'s order-n wavelet ${}_1F_1(\frac{n+2}{2}; 1; -\rho^2/2\sigma^2)$, σ set per order so the first zero crossing sits at $\rho = r$.

The same r_1 value is not numerically comparable across the two tools, and γ values are not comparable across wavelets.

Parameter	Default	Range	Meaning (both holder tools)
`estimator`	regression	regression / punctual	Pont estimator pair (see above)

`r_min` (r_1)	1.0	$[0.1, 32]$ px	finest scale (dilation scale / zero-crossing radius per tool); warns below the resolution floor
`kappa` ($\kappa = r_2/r_1$)	8	$[1.1, 200]$	scale-range width; the knob that changes the physics
`n_scales`	6	$[2, 64]$	samples in the range (nearly irrelevant vs κ; keep small)
`wavelet`	(per tool)	method-true menu	see tool text
`q_tsallis` (Tsallis q, q_T)	1.5	$[-1, 3]$ / $[-1, 1.95]$	q-family shape, measure / multiaffine (q_gaussian / q_mexican only); warns at measure $q \geq 2$, multiaffine $q < 0$
`beta` (β)	1.0	$[0.1, 8]$	Lorentzian tail exponent (Lorentzian family only)
`frac_n` (n)	2.0	$[0.05, 8]$	fractional order (frac_gaussian only)
`q_pairing` (q width)	fixed	fixed / q-paired	the q-kernel's width β: fixed, or $\beta(q) = 1/(2(2-q))$ (below); grayed out unless q_gaussian / q_mexican
`q_beta` (q β)	0.5	$[0.05, 5]$	the fixed width of $e_q^{-\beta\rho^2/r^2}$ (grayed out while q-paired)
`construction` (Wavelet space)	real	real / fourier	sample each kernel on the grid, or build it from its closed-form transform (above)

9.3 Band reconstruction: the `band` and `band_mask` outputs

Both Microcanonical MFA tools reconstruct the field from a singularity band. The band $h \in [h_{\text{lo}}, h_{\text{hi}})$ of the cached h map, cleared of islands outside the size range, is the mask; the reconstruction is the gradient-restricted Fourier inversion of the analyzed values on that mask (Turiel eq. 65, symmetric-extension border fix). Two output rows carry them, *band reconstruction* (`band`) and *band mask* (`band_mask`), and `Show` picks what the canvas draws: the h map, the band or its mask. The band row reads its PSNR against the data and the share of pixels in the band, the mask row the share.

The band knobs are view-only and sit in the right panel's *Reconstruction* section. Only the two outputs read them: scrubbing the band recomputes the reconstruction from the cached h map, and the analysis key leaves them out. A saved project gains them at their defaults. Sweeping the band and watching the reconstruction is the workflow. *Reconstruct…* (the Band dialog) previews the band live; on a whole-field layer headed by a Microcanonical MFA tool, a band row of a stack included, it writes the band into that step's knobs and shows the band row. On an ROI layer, and on a layer headed by `holder_map`, it adds a `band_recon_measure` or `band_recon_multiaffine` child, which runs the same $h(x)$ engine with the band as keyed knobs and shows the reconstruction as its raster.

Parameter	Default	Range	Meaning (band knobs; engine knobs above)
`show` (*view*)	h_map	3 outputs	what the canvas draws: `h_map`, or the `band` reconstruction or its `band_mask` (the Outputs rows)
`h_lo` / `h_hi` (*view*)	-2 / 0	$[-6, 6]$	the band $h_{\text{lo}} \leq h < h_{\text{hi}}$, in the estimator's own frame
`min_island` / `max_island` (*view*)	0 / 0	px	drop mask components outside this size range before inversion
`connectivity` (*view*)	8	8 / 4	island connectivity (8 keeps diagonal ribbons whole)

Superseded: `band_recon_measure` and `band_recon_multiaffine` stay registered in the browser's Dev group, so a project that holds one opens and resolves as before; at the same knobs their reconstruction equals the `band` output bit for bit. `holder_map` and `band_recon` (the pre-split conflated tools, estimator × wavelet crossed) remain

registered for saved projects and live in the Dev group too; their physics is identical to the split tools at the old defaults.

10 Topology

10.1 Chain topology · `chain_topology` (Transform)

Builds the incidence graph over the WTMM maxima-line skeleton (no parameters). It applies after an analysis that carries an extrema stack and V-chains (the WTMM tools), passes that analysis through and adds the topology, which downstream topology filters query.

10.2 Min V-chains per H-line · `min_vchains` (Filter)

Keeps H-line points that at least `min_vchains` maxima lines (V-chains) thread. It applies after Chain topology. Default UI opens at 1 (show everything computed; turn up to prune).

11 Selection: extrema

Act on the extrema of the level the Transport shows (§1), after any tool that produces an extrema stack (the applicability table, §2). In mode `sup_all_scales`, `hline_modulus` also reads the V-chains threading each line, and `hline_holder` reads the per-point α the Mallat–Zhong analysis stamps.

Device.param	Default	Range	Meaning
`orientation_wedge.centre`	0	deg	keep extrema whose gradient strike falls in the wedge
`orientation_wedge.half_width`	90	$[0, 90]$ deg	wedge half-width (90 = no-op)
`orientation_wedge.north`	grid	grid/true/magnetic	the NorthRef of `centre`; frame errors are multi-degree errors
`modulus_threshold.frac`	0	$[0, 1]$	per-point: keep $\|W\| \geq$ frac $\times$ scale peak
`hline_length.min_len/max_len`	1 / 0	points	per-line point count range; kills orphan confetti and stitching seams; -1 orphans pass
`hline_modulus.frac`	0	$[0, 1]$	per-*line* modulus keep-fraction
`hline_modulus.mode`	sup	sup / sup_all_scales / mean / segments	line statistic: peak at this scale; peak over the threading chains (needs chains); mean; or per-segment between modulus minima
`hline_holder.min_alpha/max_alpha`	-3 / 3	$[-3, 3]$	keep lines whose Hölder α lies in the range (`mz_edges`: the per-line α of its analysis; a layer without α passes)
`hline_holder.keep_unfitted`	on	bool	keep points whose chain could not be fitted

12 Selection: V-chains

Act on cross-scale chains, the WTMM maxima lines (V-chains). They apply after an analysis that produces V-chains (the applicability table, §2); the other analyzers stamp `chains: []`, and their cross-scale linking is reserved work.

Device.param	Default	Range	Meaning
`chain_holder.cutoff`	-3	$[-3, 3]$	keep chains with Hölder $\geq$ cutoff

`chain_holder.estimator`	ols	ols / max	per-chain h: log–log OLS slope, or the max-modulus scaling read
`chain_modulus.threshold`	-99	$[-99, 32]$	keep chains whose max $\log_2 \lvert W \rvert \geq$ threshold
`chain_length.min_len`	2	$[2, 64]$	minimum scales spanned

12.1 Seam classifier · `chain_classify`

Classifies chains by their Hölder exponent and geometry into step-like ($h \approx 0$: fault-scarp class) vs delta-like ($h \approx -1$: thin-ridge class) lineaments, with a straightness gate; flagged or excluded per `action`, excluded chains optionally drawn as ghosts.

Parameter	Default	Range	Meaning
`estimator`	ols	ols / max	per-chain h estimator
`fit_a_min`	2.0	$[0.01, 64]$	fit floor: scales below this are excluded from the h fit
`h_step_lo` / `h_step_hi`	-0.15 / 0.15	$[-3, 3]$	the step-class h window
`h_delta_lo` / `h_delta_hi`	-1.2 / -0.8	$[-3, 3]$	the delta-class h window
`straightness_min`	0.85	$[0, 1]$	end-to-end / arc-length straightness gate
`action`	flag	flag / exclude	tag matches, or drop non-matches
`show_ghosts`	on	bool	draw excluded chains desaturated

13 Groups, points, Dev

`group_paint` / **Group filter** · `group_filter`: the arrangement view's commit transaction — `group_paint` stamps committed group tags onto chain copies (`spec_json`/`signature` are written by the commit flow, not hand-edited); `group_filter` keeps or drops one named group.

Back-project points · `backproject`: registers a point catalog onto a raster layer's grid (the point-layer transform). `target` names the raster layer; the `_target_*` params are the resolved grid geometry, written by the shell — not hand-edited.

`derived_vectors`: a derivative dataset's frozen extrema and maxima lines, read from its own file (§1). Placed by the shell (browser Dev group); its one parameter (the file) is hidden.

`stub_wavelet` / `stub_holder` / `stub_wedge`: parameter-schema test fixtures (the browser's Dev group); no analysis role. They declare no contracts, so every step after them applies.

Superseded tools stay registered in the Dev group, so a saved project that names one opens and resolves: `holder_map` and `band_recon` (the pre-split Hölder tools), `band_recon_measure` and `band_recon_multiaffine` (§9.3), `tucker_havok`, and `scale_select` (§1).

14 q-generalized wavelets: built and open

Borges et al. (2004) generalize the Mexican hat, the Morlet wavelet (a q-Gaussian-modulated plane wave, their eq. 23) and the trigonometric functions (an even/odd pair of q-trigonometric wavelets, their §4), and note that "other wavelets, as well as scaling functions, may be generalized along the lines here introduced, for instance, the Meyer wavelet or the sinc function." The appendix (§16) draws every one that is built.

- **Built: q-g1 and q-g3 for `wtmm2d`** (`q_gaussian`, `q_mexican`). WTMM's analyzing wavelets stay *derivatives of a smoothing function* ($\psi = \partial\theta$), so that maxima mark edges; the filters are built in the Fourier domain from the *2-D* transform of the q-Gaussian, a Hankel transform with Bessel order $1/(q-1) - d/2$, one half lower than the paper's 1-D order, evaluated stably at every q (the Matérn and Jahnke–Emde functions of *Wavelet space: Fourier*, §9).
  - q-g1: $\partial\theta_q$ with θ_q the unit-mass q-Gaussian, so the L^1 normalization is unchanged ($\hat{\theta}_q(0) = 1$); it exists for $q < 2$.

  - q-g3: the power-$(2-q)$ recipe makes the Laplacian's envelope a q'-Gaussian with $q' = 1/(2-q)$, whose mass is finite only for $q < 3/2$. For $3/2 \leq q < 2$ the wavelet is still finite (k^2 cancels the divergence at $k = 0$), but $\hat{\theta}(0) = 1$ can no longer fix its scale, so it needs another normalization there, such as unit L^1 norm.

- **Built: the q-Meyer smoother** (`q_meyer`), Meyer's radial taper with Borges's q-cosine, matched to the Gaussian pair by its analyzer's band peak (Figures 19 and 20).

- **Built: the 2-D analytic q-Morlet** of `wavelet_ridge` (`dynamix.core.analytic_cwt`, §6.13), the Morlet at $q = 1$; its 1-D form reads a chirp's two exponents off the modulus ridge (Figure 22).

- **Open:** the q-Morlet inside the WTMM (it needs a complex-wavelet WTMM), the q-trigonometric pair (oriented and directional analysis) and the q-sinc.

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Appendix: result contracts

- The named data contracts with the result keys each covers, and what each tool consumes and produces, are tabled in §2.
- **Extrema schema** (all edge/skeleton analyzers): per-scale {`x, y, mod, arg, line_id`}; `line_id` groups H-lines, −1 marks isolated points. `scales` is the matching scale array (px).
- **h-map contract** (holder tools): `h_map` displays in place of the field through the ordinary raster path; `r2_map` rides along (all-NaN under punctual). The `band` and `band_mask` outputs are lazy rasters with keys of their own (§9.3). `raster_out` (`band_recon_measure`, `band_recon_multiaffine`) displays the reconstruction; `h_map`/`band_mask` ride for the dialogs.
- **cdf extras**: `filtered` (final Re) and `edge_channel` (final Im) rasters.
- The vector (3-D) view draws one extrema layer: the level the layer shows (§1).
- **Decomposition extras**: `ssa2d` — `ssa_recon`, `ssa_residual`, `ssa_components` (n, n_y, n_x), `ssa_eigen_share`, `ssa_eigenarrays` (n, L_r, L_c), `ssa_factor_arrays` (n, K_r, K_c), `ssa_w_correlation` (n, n). `tucker_HOOI_HOSVD` — `tucker_recon`, `tucker_residual`, `tucker_components` / `tucker_combined_components` with their energies, core, factors, energy per mode; on a multiband field also `tucker_recon_bands` / `tucker_residual_bands` (n_y, n_x, n_c) (the display shows band 1). A view step writes `raster_out` and, for a named view, `_view_note` (what the layer row appends).
- **Inpainting extras**: `inpaint` — `fill_mask` (region ids), `_estimate` (the estimate before the final step, float32), `_heldout_mask`, `_hists` (the per-level $\ln|\eta|$ and $\ln|c|$ histograms), `_abacus`, `_warnings`, `_filled_fraction`; `dwt2d` — `scales` (2^j px), `_dwt_levels`. A `fill` clone's provenance carries `fill`; an analyzer result after it carries `_fill`.
- **Derivative file**: the RasterField npz keys (`schema`, `name`, `values`, `x_axis`, `y_axis`, `units`, `frame_meta`, `provenance`) plus, when it carries vectors, `vec_e_*` (extrema, CSR-packed), `vec_c_*` (maxima lines) and `vec_scales`. A vector-only derivative's raster is all-NaN on the same grid.

The wavelet ridge scalebar previews current q and width controls without running the analysis. Its solid curve is the real Morlet section; dashed curves bound its window (Gaussian at q=1, the corresponding q-envelope otherwise). The bracket continues to mark the carrier wavelength.

Export provenance uses the landed computation identity and its captured numerical context. A changed recipe or source tree must be reconciled before a record can be written; a provenance refusal is reported alongside the data export. Derived raster records use the computed output and level. Marked levels remain export annotations. Cache hit/miss counters stay outside the scientific record.

15 Fitting controls

The preparation tools have their own sections: `fractional_integration` (§4.5), `dyadic_interpolation` (§4.6), `daf_smoothing` (§4.7) and the interpolating wavelets (§7.5, §4.8).

Directional scaling's `alpha_step` sets the anisotropy grid resolution. The Gamma estimator uses increasing-resolution indices and excludes unresolved coarse scaling crossovers; its finite fit remains an estimate. The legacy estimator remains selectable. Large deviations expose `h_min` and `h_max` for the histogram grid; adaptive tilted bins are labeled a heuristic and can miss nonconcave regions. Fixed epsilon sweeps are independent. The generalized Gaussian tool exposes its two moment grids and takes their finite-grid Legendre infimum rather than treating tilted means as a spectrum.

The prototype transform's `normalise` and `n_octaves` remain prototype controls, not scientific DAF settings. Preparation controls belong to their own tools. Inspector Apply actions reuse cached upstream coefficients when only moments, chaining or fit controls change.

16 The wavelet atlas

Every analyzing wavelet and smoothing function the app runs, drawn by `make_wavelet_figures.py` from the code the devices compute with (`core.smoothers`, `core.wavelet_profiles`, `core.hyperbolic_preview`, `core.interpolating`, `core.daf`, `core.analytic_cwt`, `core.wavelet_skeleton`), so a figure cannot drift from what a tool computes. The 3-D views follow Arnéodo, Decoster and Roux (2000), Fig. 1. Figures 14–22 are the appendix of the author's note on tuneable wavelets (14 October 2026), redrawn from the current code. Sections are scaled for comparison; the app itself normalizes every kernel to unit L^1 mass at every scale.

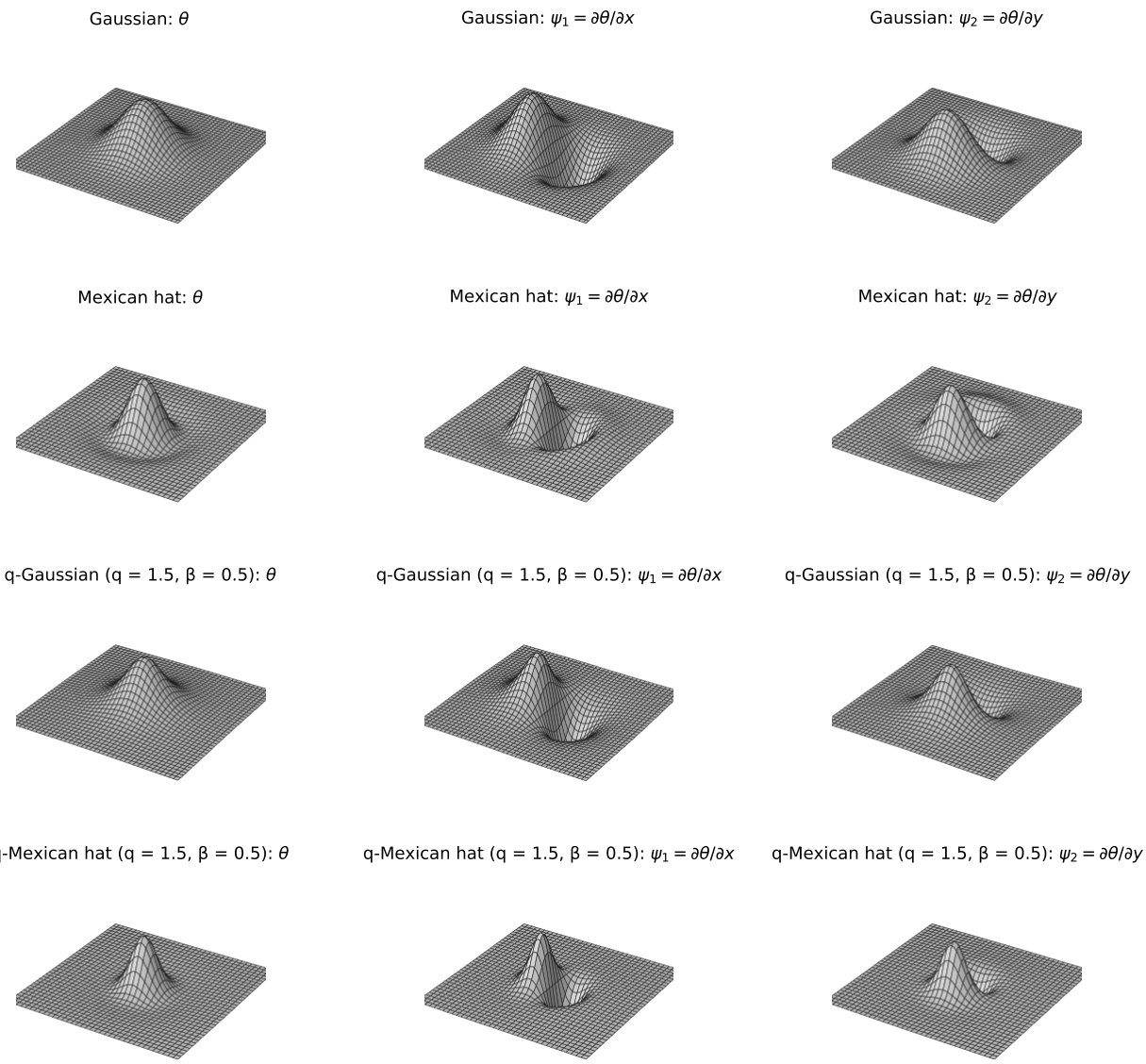


Figure 5: The WTMM smoothers in 3-D (continued in Figure 6): for each, the smoothing function θ and the two analyzing wavelets $\psi_1 = \partial\theta/\partial x$ and $\psi_2 = \partial\theta/\partial y$, at the default settings (q-Gaussian and q-Mexican hat $q = 1.5$, $\beta_q = \frac{1}{2}$), on $|u| \leq 4$–$5\,\sigma$. Each surface is inverted from $\hat{\theta}$ and the derivative filters the WTMM transform builds, so it is the kernel a run convolves with. The Gaussian pair is Arnéodo's first-order wavelet (one vanishing moment), the Mexican pair his third-order one (three).

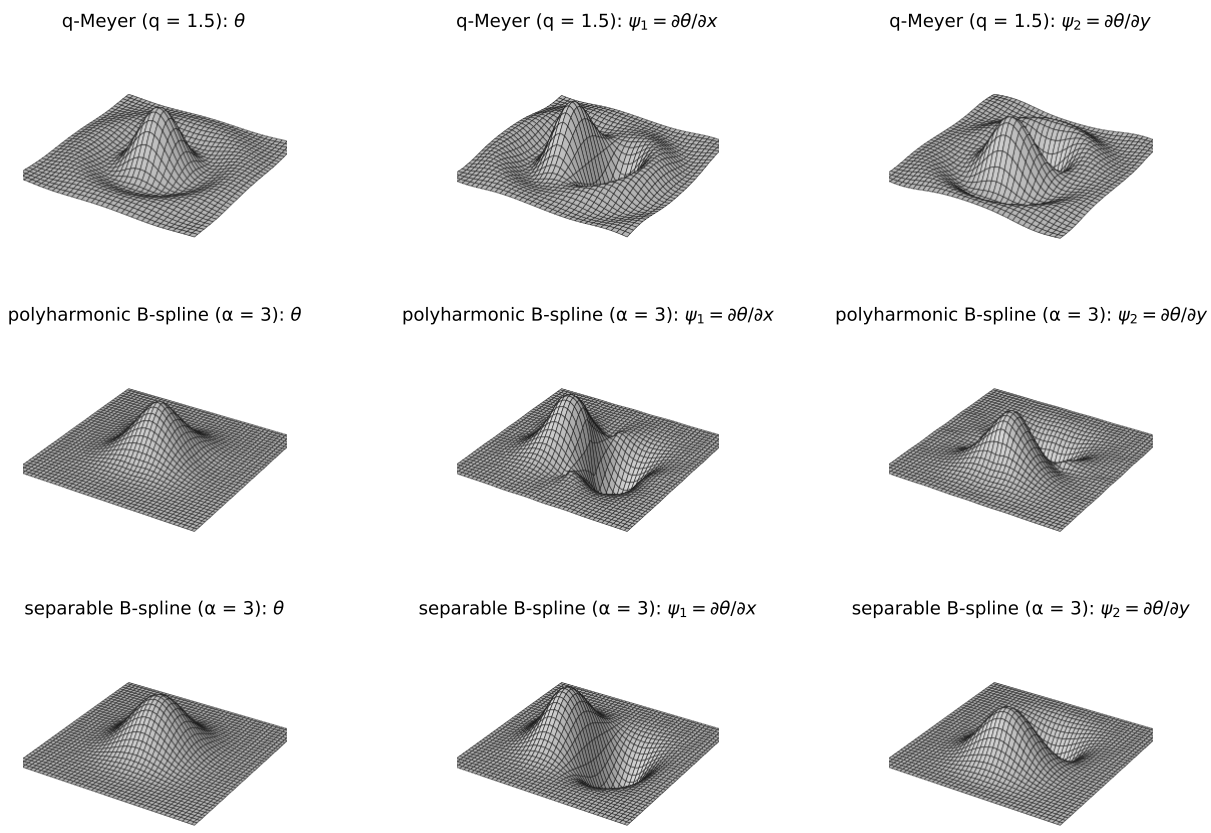


Figure 6: The WTMM smoothers in 3-D, continued: the q-Meyer ($q = 1.5$, inverted from $\hat{\theta}$ and the transform's filters on $|u| \leq 9\,\sigma$), the 2-D isotropic polyharmonic B-spline and the separable fractional B-spline ($\alpha = 3$), both evaluated in the space domain: the polyharmonic as its localized sum of Green's functions, the separable as the product of Unser & Blu's exact series.

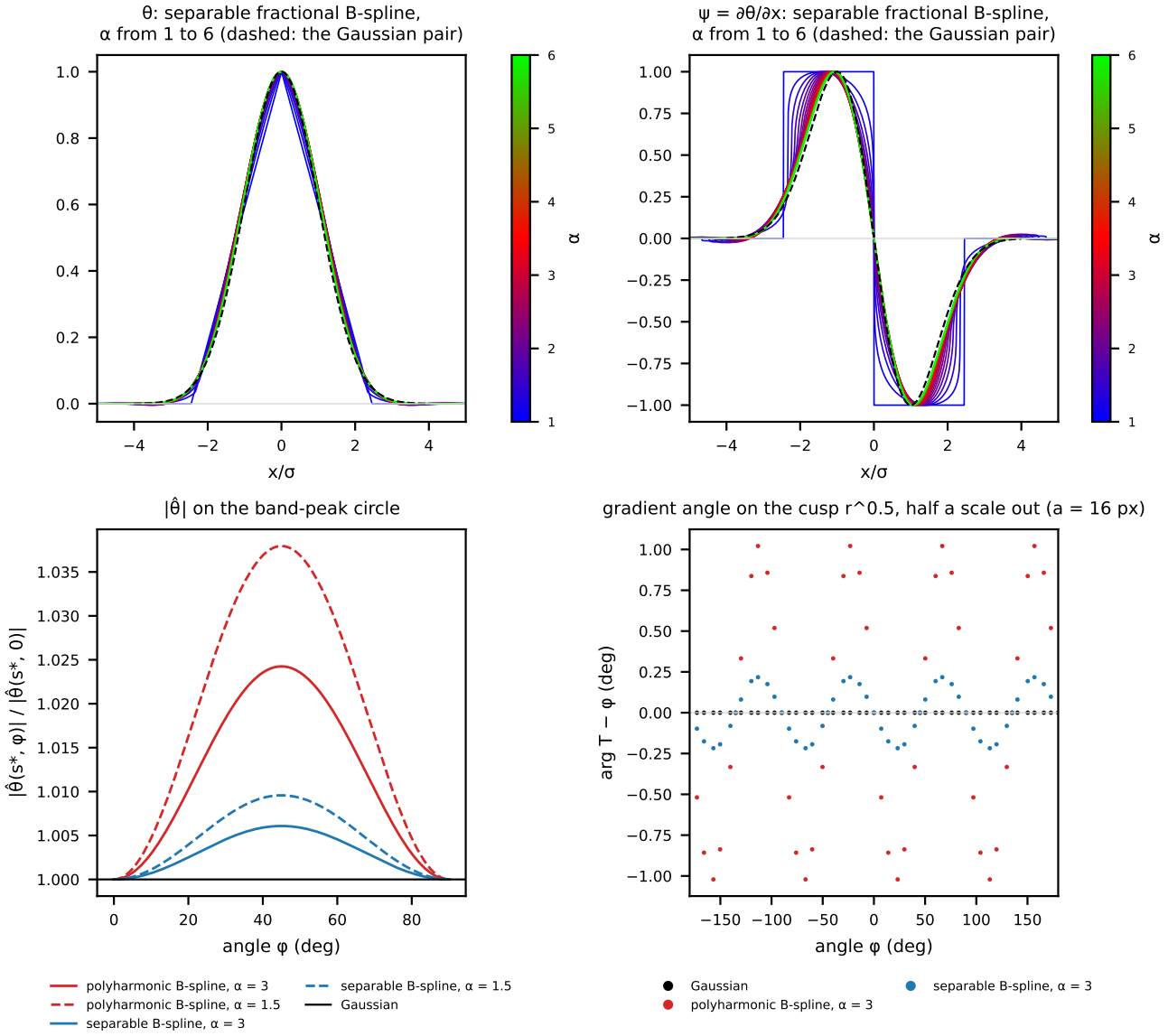


Figure 7: The two fractional B-splines. Top: the separable spline's θ and ψ sections at $a = 8\,\mathrm{px}$ as α runs from 1 (the box pair) to 6, from the exact series, the Gaussian pair dashed. Bottom left: $|\hat{\theta}|$ on the band-peak circle against the angle, relative to the axis, for both splines at $\alpha = 3$ (solid) and 1.5 (dashed); the Gaussian is flat at 1. Bottom right: the gradient argument's error on the isotropic cusp $r^{1/2}$, half a scale from its center ($a = 16\,\mathrm{px}$, $\alpha = 3$), through the WTMM's own derivative filters: a $\sin 4\varphi$ of about 1° for the polyharmonic spline and 0.2° for the separable one; the Gaussian's is zero.

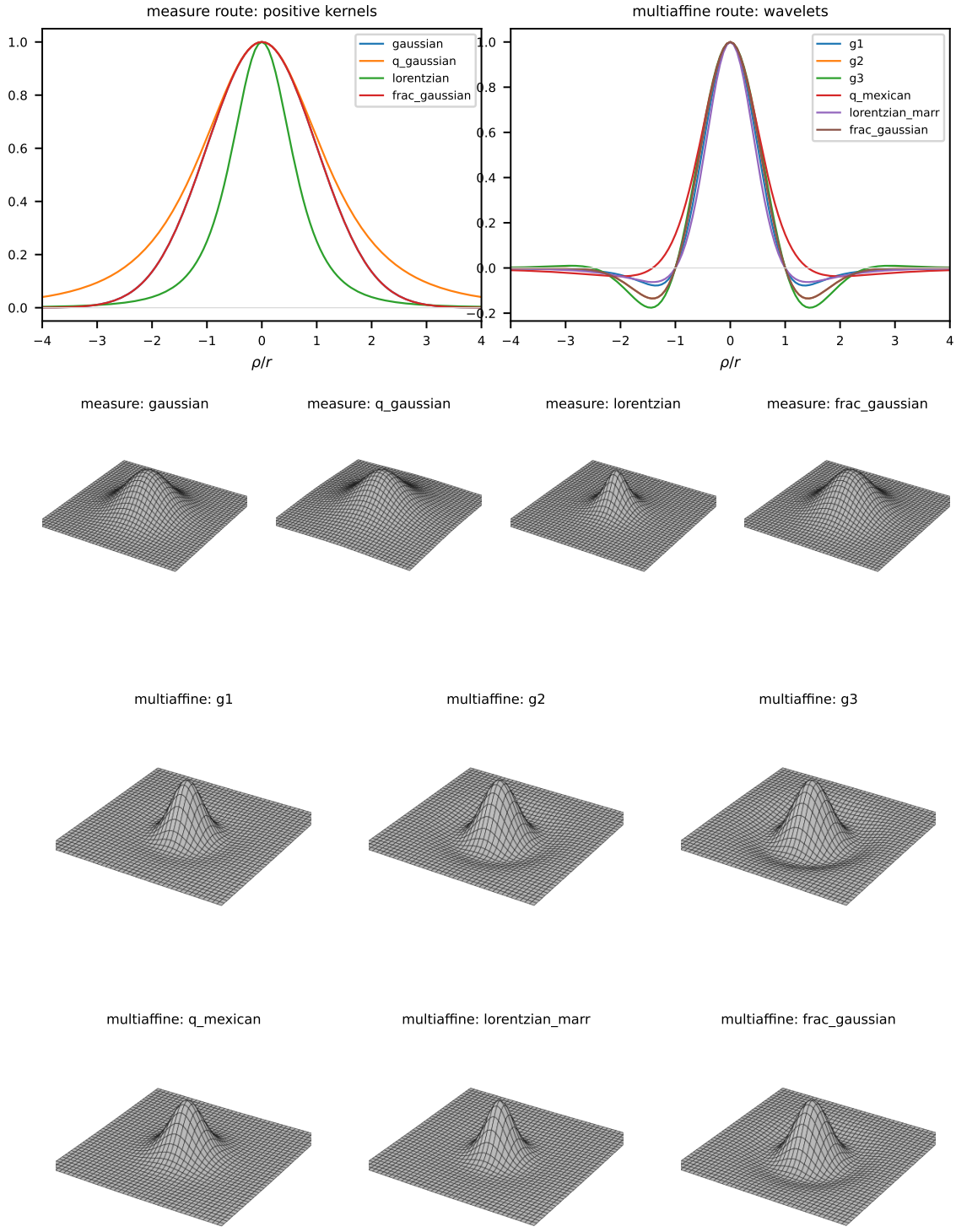


Figure 8: The microcanonical kernels at $r = 8\,\mathrm{px}$ (`holder_measure`, `holder_multiaffine`; $q = 1.5$, $n = 2$, and $\beta = 2$ for the Lorentzians, whose 2-D mass is infinite at $\beta \leq 1$). Top: sections of the measure route's positive kernels and the multiaffine route's wavelets against ρ/r. Below: their radial surfaces.

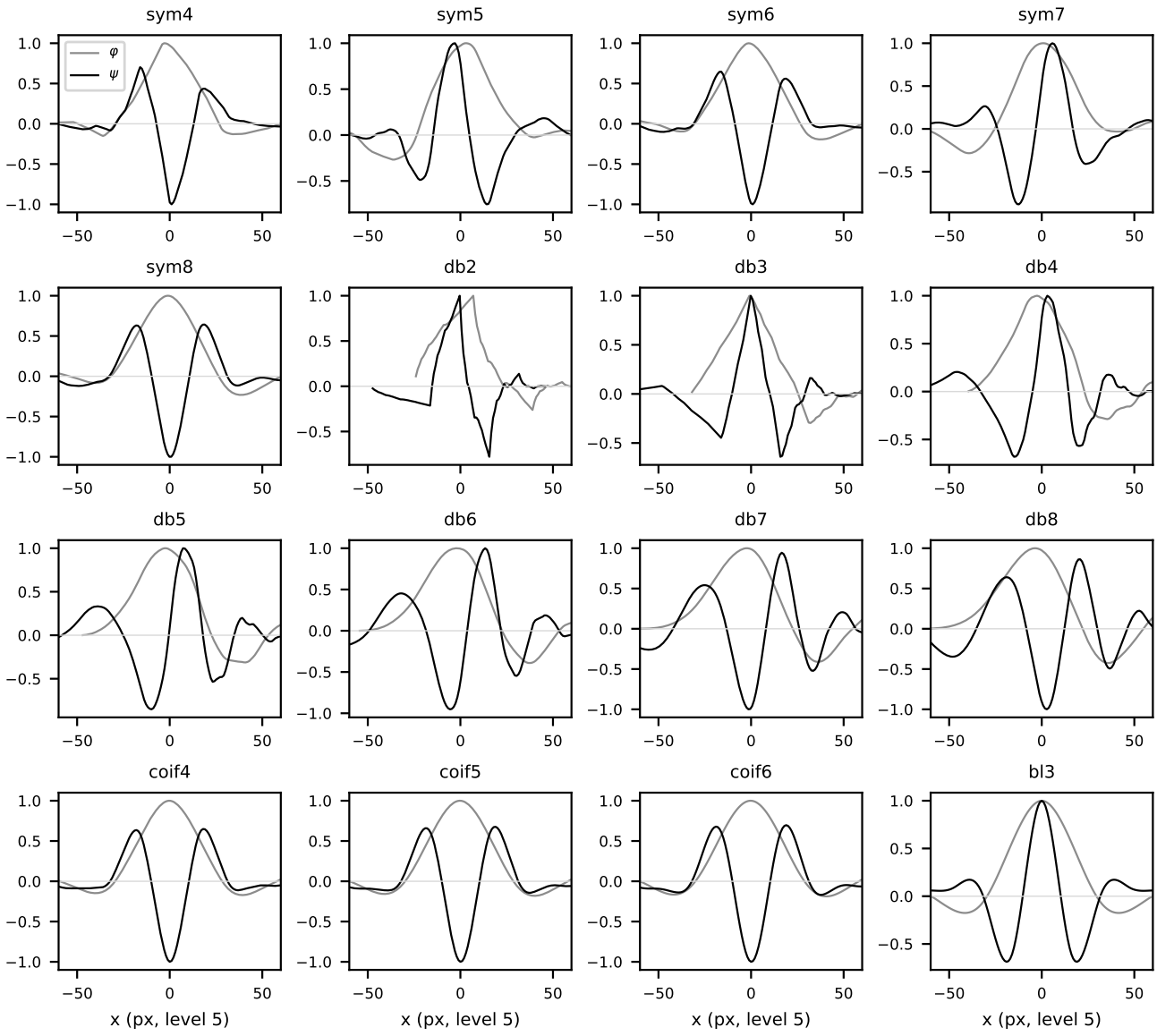


Figure 9: φ (gray) and ψ (black) of the sixteen orthogonal DWT banks at level 5, the cascades the pyramid applies, ψ as the analysis atom (the taps reversed), each centered on its energy centroid. `bl3` is the Battle–Lemarié order-3 bank, built by the app.

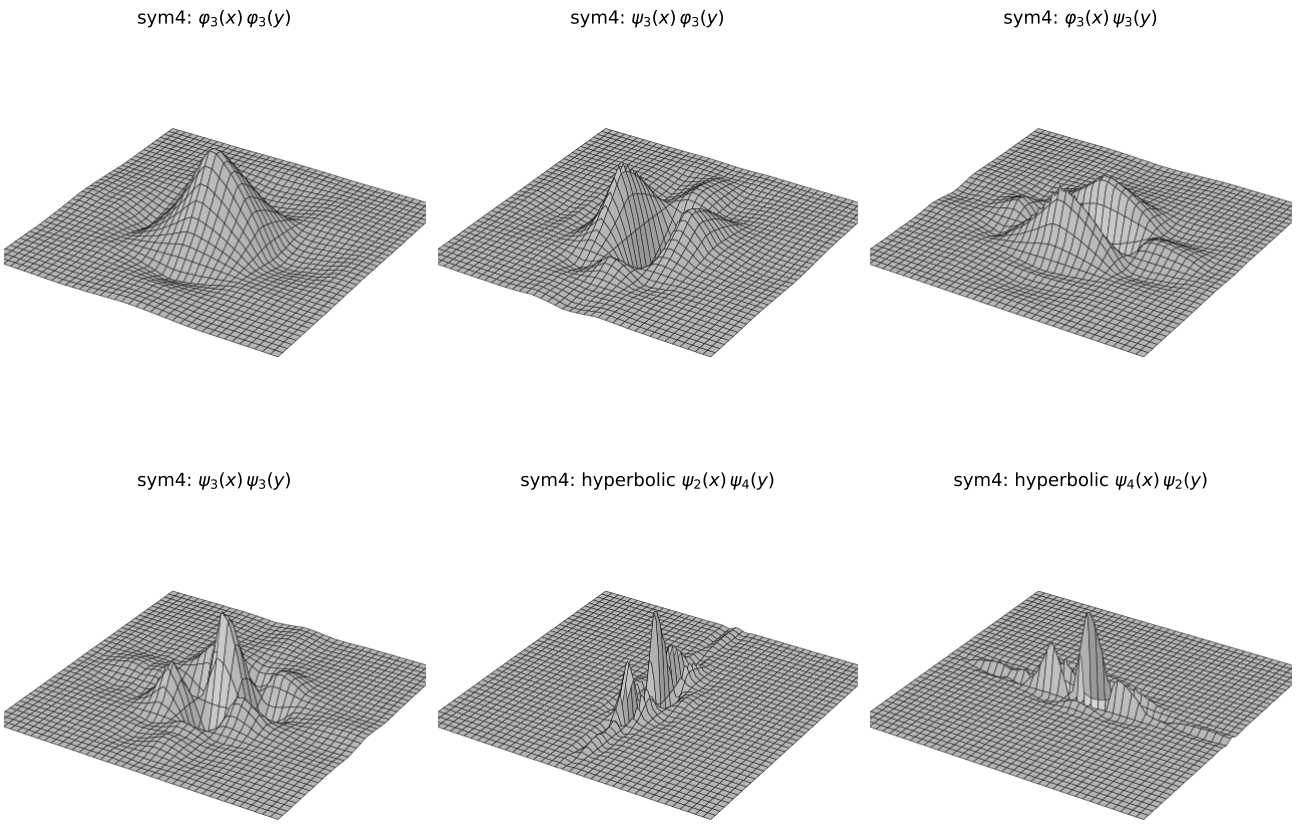


Figure 10: The 2-D atoms of `sym4` (the default bank of `wavelet_leaders` and `directional_scaling`) at level 3: the approximation $\varphi\varphi$ and the three separable details, then two hyperbolic atoms $\psi_{j_1}(x)\,\psi_{j_2}(y)$ with $j_1 \neq j_2$, the anisotropic members the directional tool fits; each on a square window as wide as its larger extent.

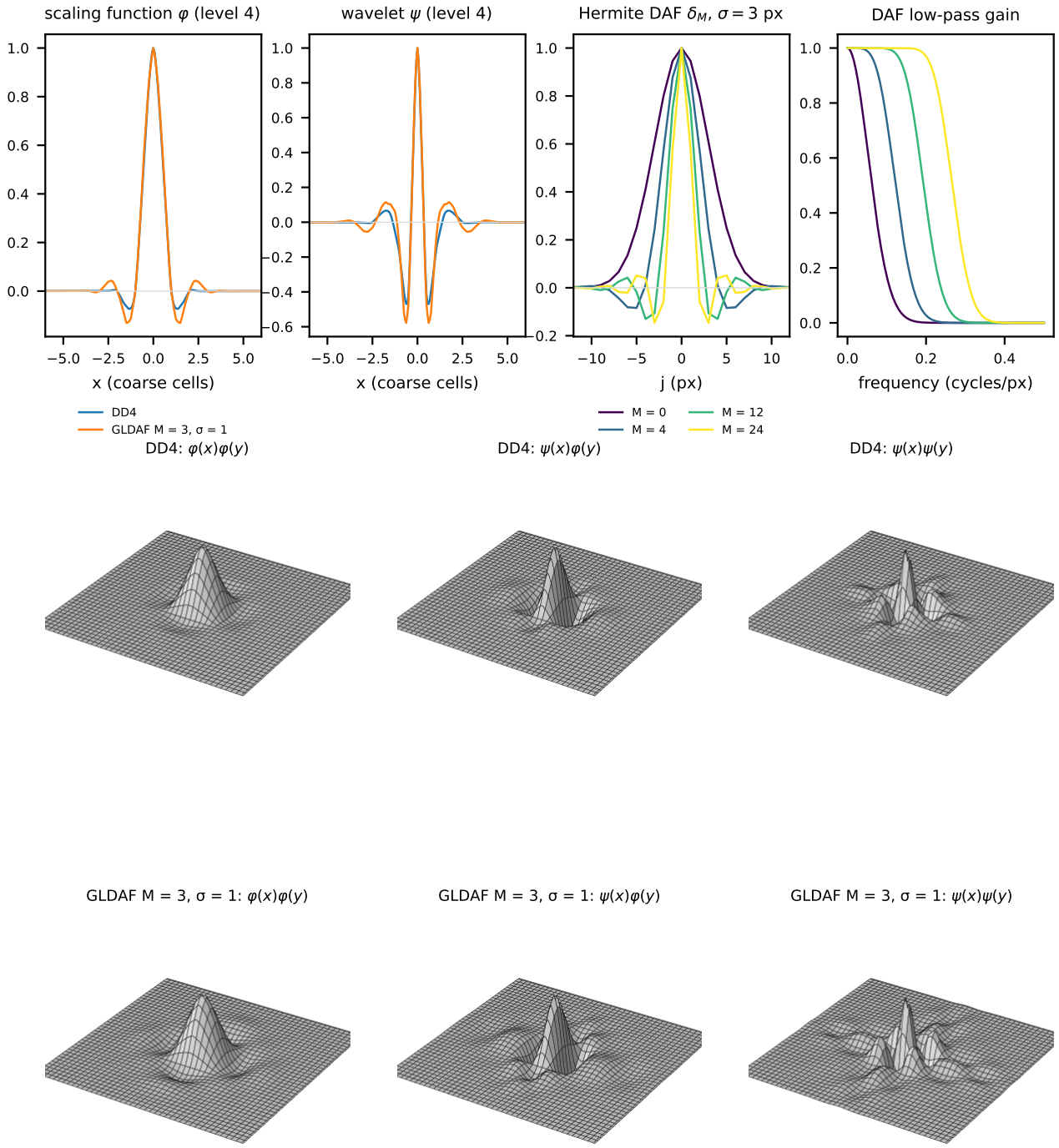


Figure 11: Top: the interpolating transform's scaling function and wavelet at level 4 for the two predictors at their defaults (Deslauriers–Dubuc and GLDAF with $M = 3$, $\sigma = 1$), as the central rows of the 2-D atoms: one coefficient at the coarsest level, reconstructed by the device's own inverse; then the Hermite DAF's taps and gain at $\sigma = 3\,\text{px}$ for $M = 0, 4, 12, 24$. Below: the 2-D atoms.

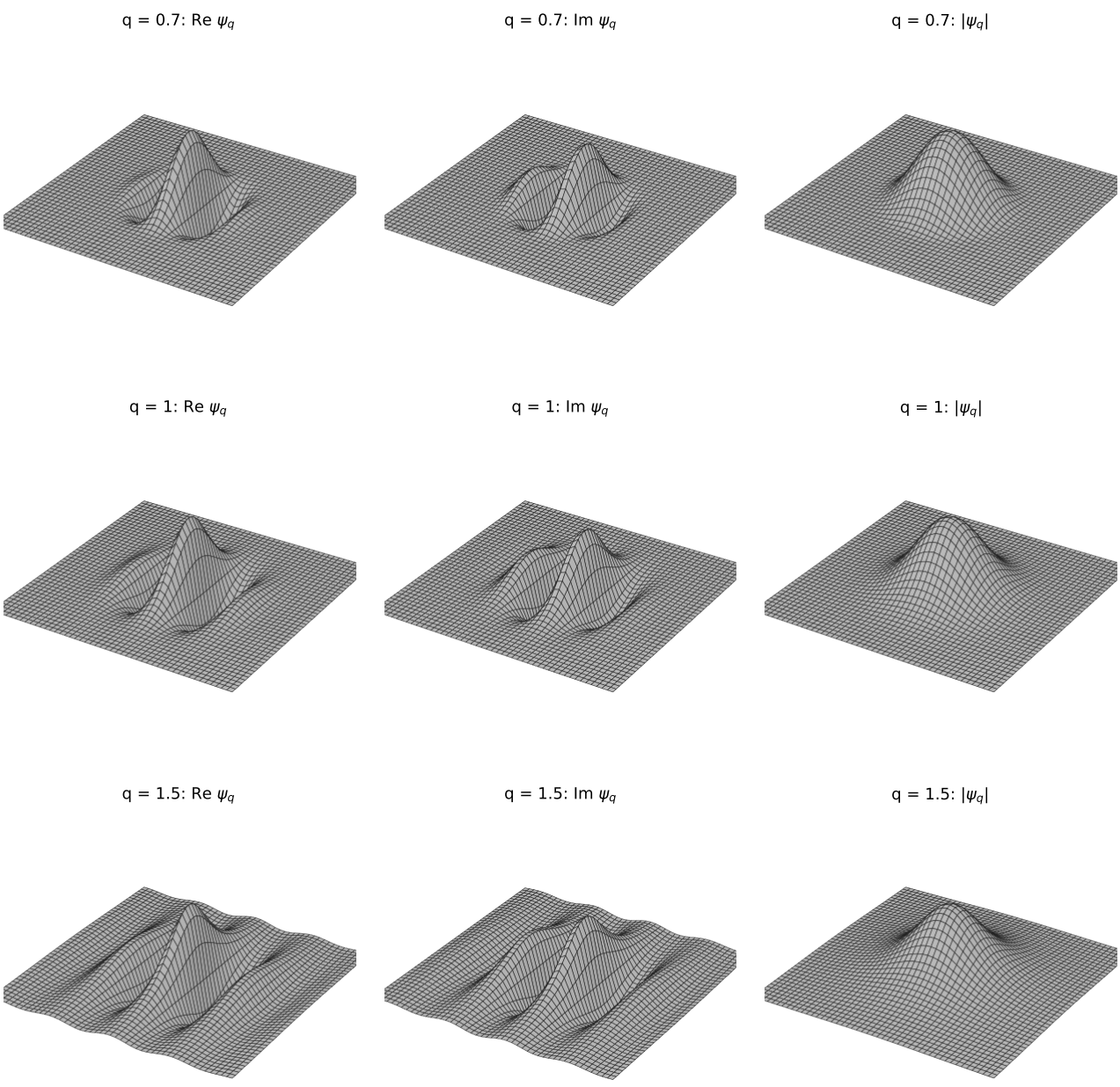


Figure 12: The 2-D analytic q-Morlet of `wavelet_ridge` at its default carrier $\omega = 3$, centered on $\kappa = 2\pi/12\,\mathrm{px}^{-1}$ ($a = 5.7\,\mathrm{px}$), with the zero-mean correction: real part, imaginary part and modulus for $q = 0.7$, 1 (the Morlet) and 1.5.

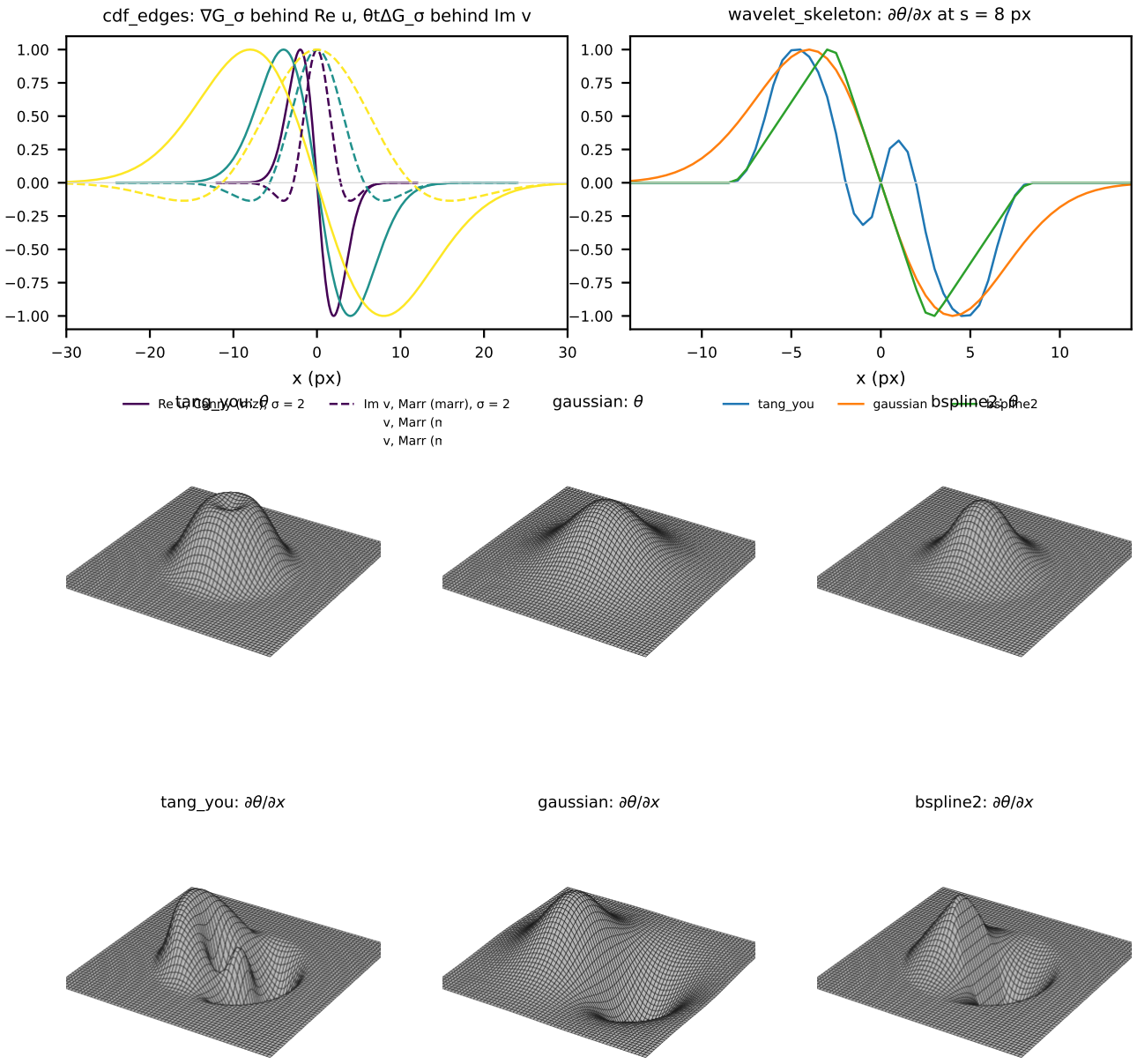


Figure 13: Top: `cdf_edges`' equivalent kernels of the linear evolution at small θ at $\sigma = 2, 4, 8$ px: the Gaussian derivative ∇G_σ behind the real part u, whose gradient maxima are the Canny edges (`edge_mode mz`), and the Mexican hat $\theta t\, \Delta G_\sigma$ behind the imaginary part v, whose zero crossings are the Marr–Hildreth edges (`marr`), and the three gradient wavelets of `wavelet_skeleton` at $s = 8$ px. Below: the skeleton's smoothing kernels θ and their x-gradients in 3-D at $s = 20$ px.

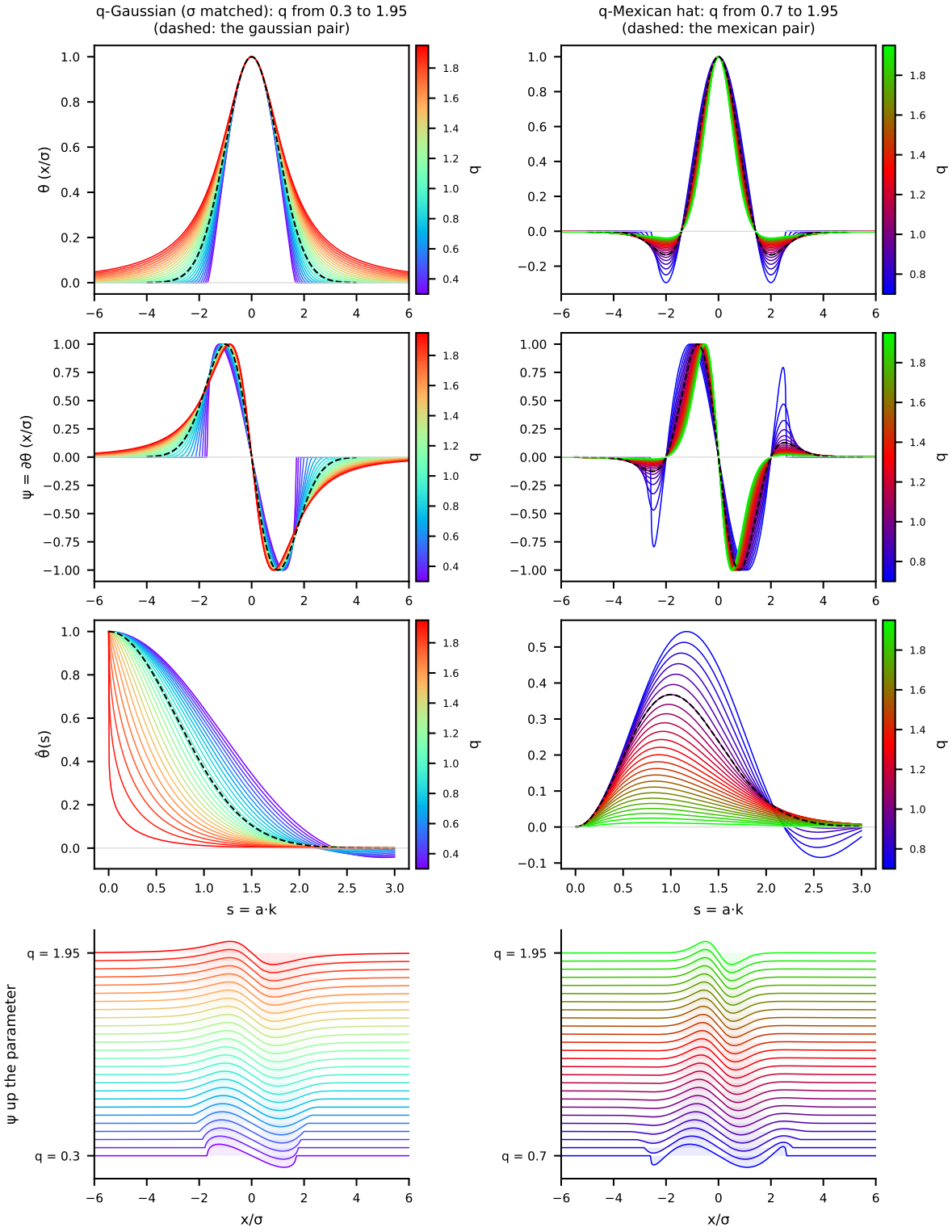


Figure 14: A1. WTMM smoothing functions at nominal scale $a = 8\,\mathrm{px}$: the smoother θ, its gradient $\psi = \partial\theta$ (the analyzer) and $\hat{\theta}(s)$, for the q-Gaussian (σ matched) and the q-Mexican hat ($-\Delta$ of the q-Gaussian; its analyzer the q-g3). 26 values per sweep as hairlines on a color ramp; the bottom row stacks the sections up the parameter.

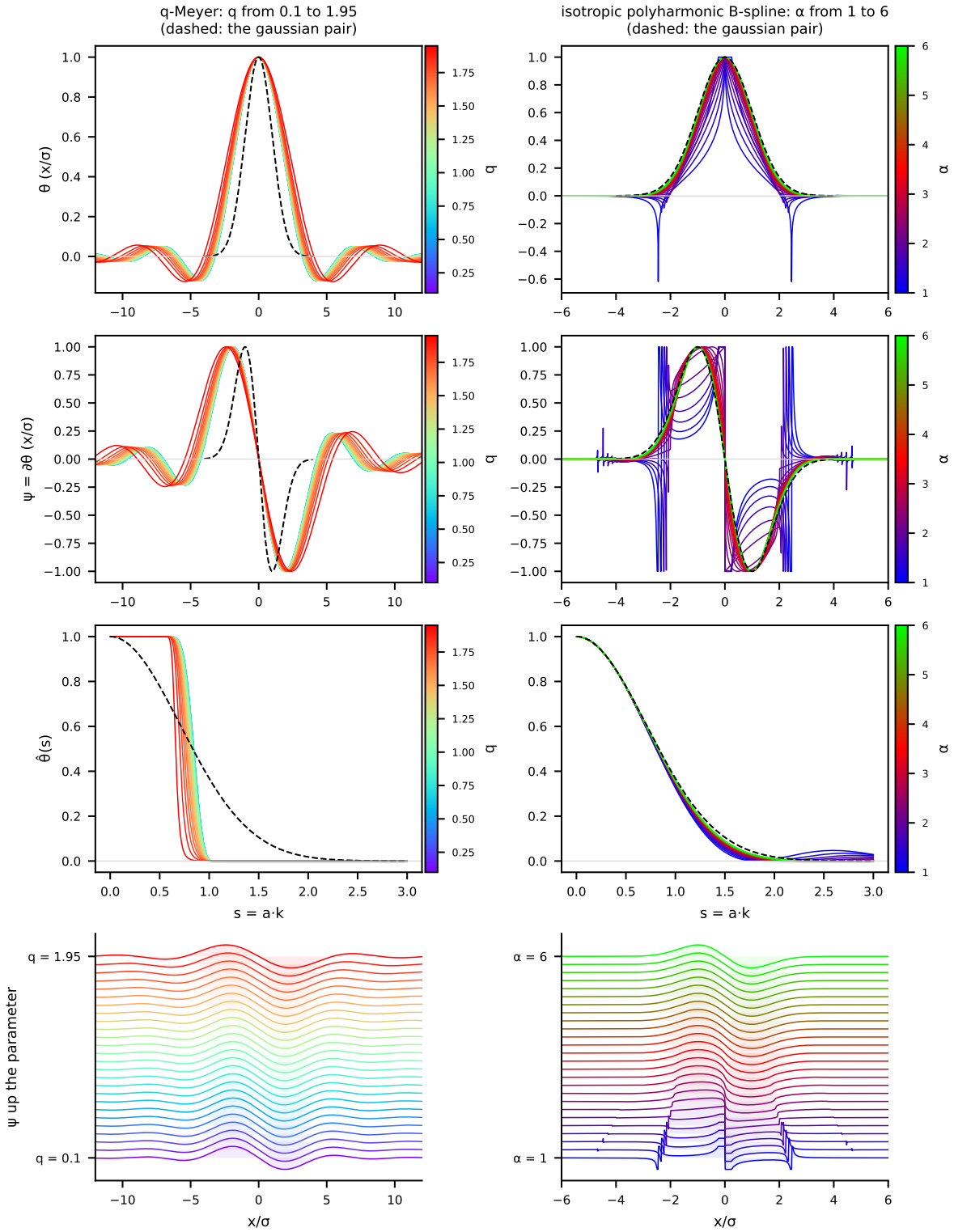


Figure 15: A1, continued: the q-Meyer (Meyer's radial taper with Borges's q-cosine, drawn over the $\pm 12\,\sigma$ of its section, where it still rings) and the 2-D isotropic polyharmonic B-spline (order $\alpha + 1$, evaluated in the space domain; its singularities at the knots below $\alpha = 3$ are clipped). Its ψ is not continuous in 2-D for $\alpha \leq 2$ (the knob's soft floor).

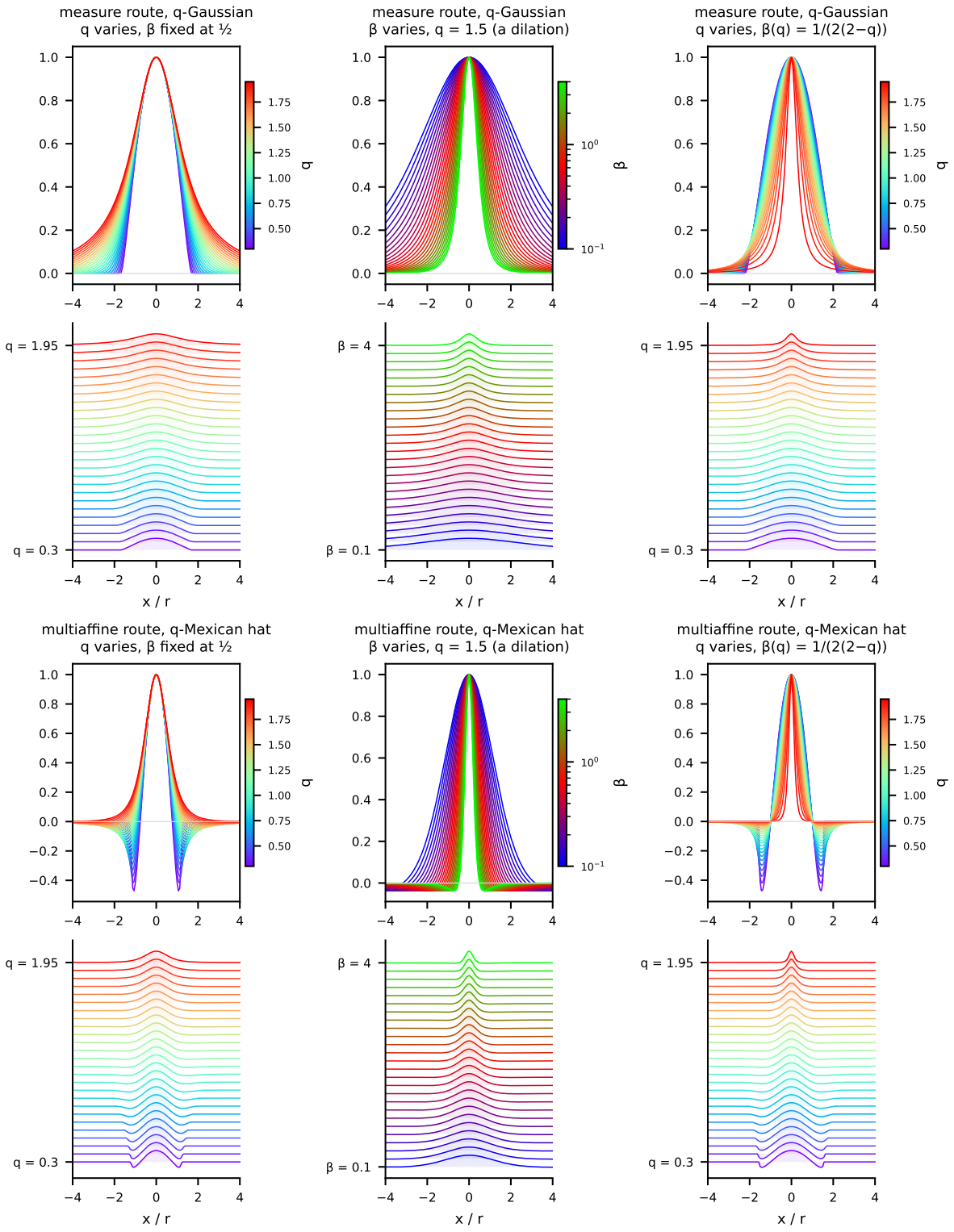


Figure 16: A2. Turiel's kernels (the microcanonical routes) at $r = 8\,\mathrm{px}$: the measure route's q-Gaussian and the multiaffine route's q-Mexican hat under three sweeps: q with β fixed at $\frac{1}{2}$ (the zero crossing moves), β with q fixed (a pure dilation), and q with β responding to q, $\beta(q) = 1/(2(2-q))$, which holds the escort variance so the zero crossing stays at r.

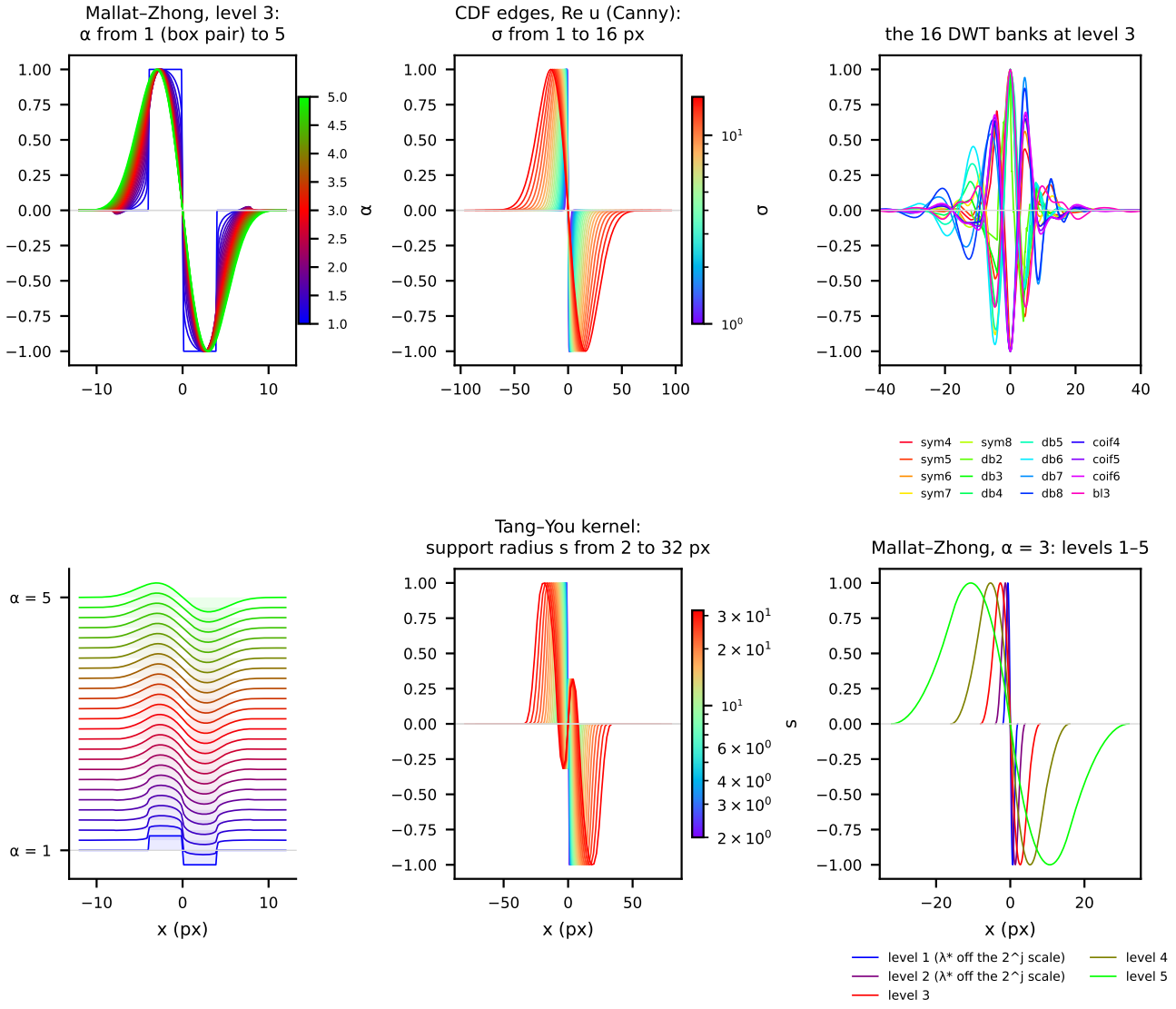


Figure 17: A3. Dyadic banks and edge detectors: Mallat–Zhong edges across the spline order α ($\alpha = 1$ the box pair, $\alpha = 3$ the quadratic spline) and across levels, the CDF edge detector's real part (the Canny channel) across σ, the Tang–You skeleton kernel across its support radius, and the sixteen DWT banks at level 3 (`bl3` added since the note).

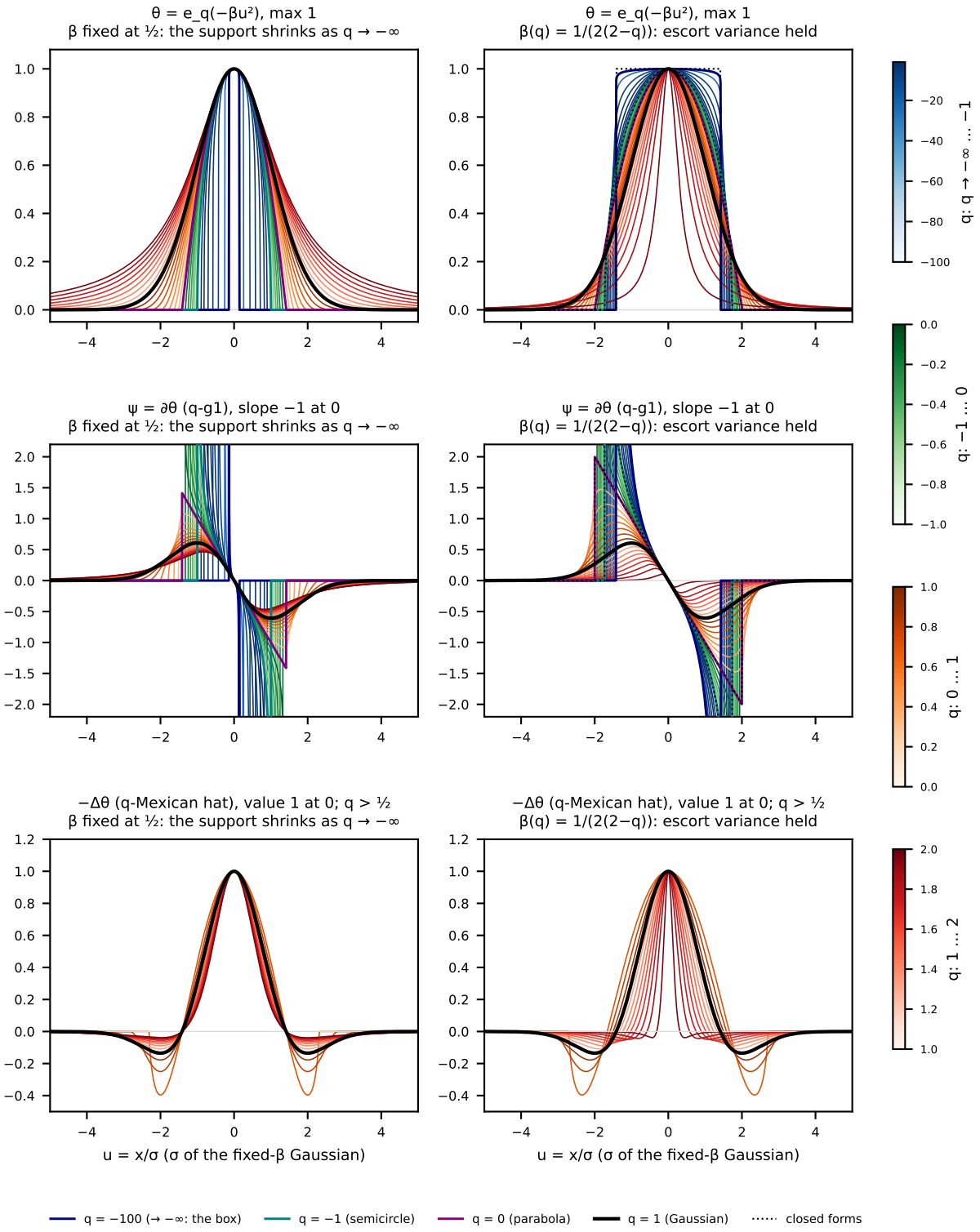


Figure 18: A4. The q family over its whole range from the app's own e_q: with β responding to q the kernels run box ($q \to -\infty$) → semicircle ($q = -1$) → parabola ($q = 0$) → Gaussian ($q = 1$) → heavy tails ($q \to 2$) at one escort width; the closed forms (dotted) coincide with the software's curves to 10^{-4}. Rows: θ, the WTMM analyzer ψ (drawn with slope -1 at the origin; its edge diverges for $q < 0$), and $-\Delta\theta$ for $q > \frac{1}{2}$.

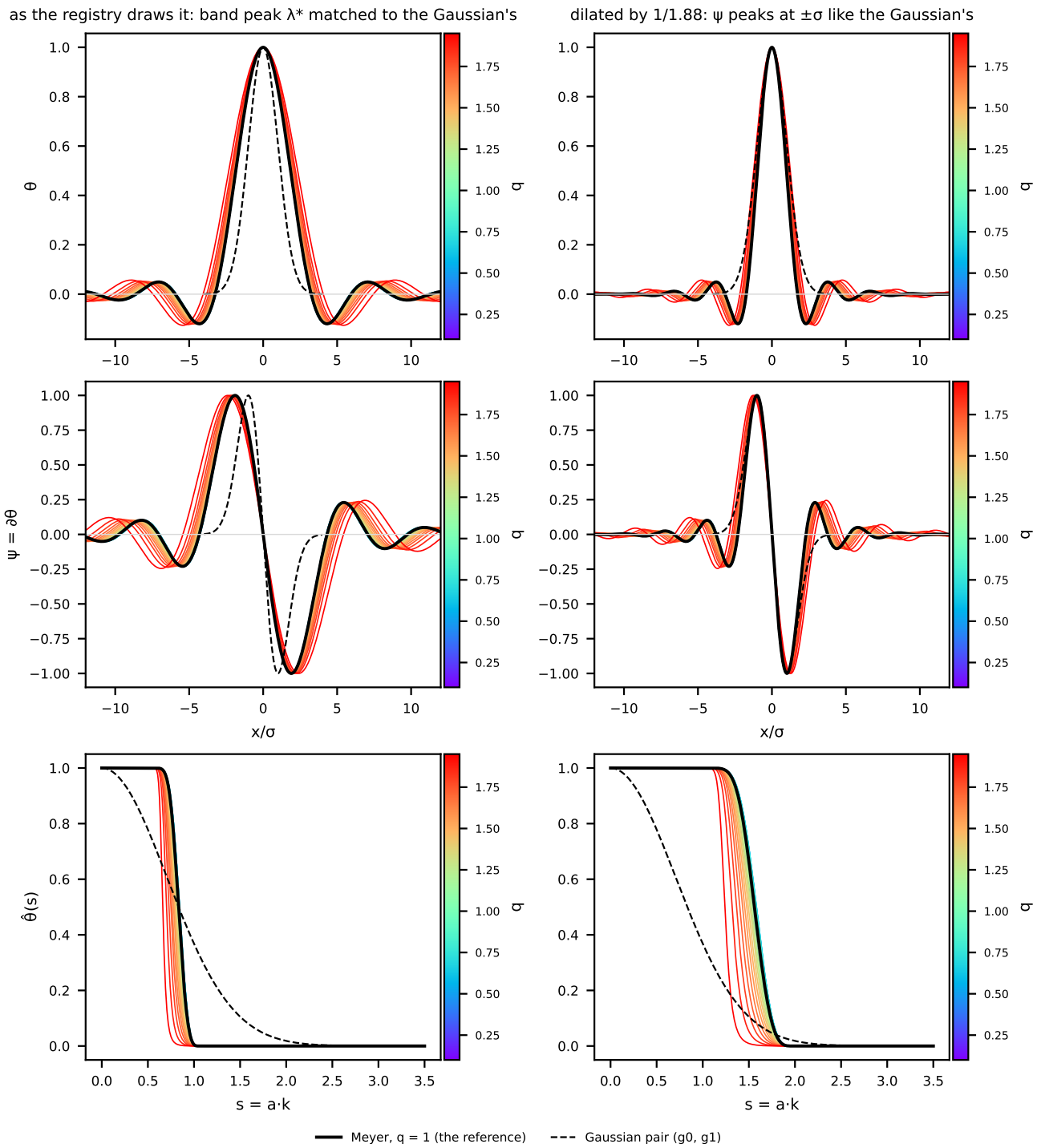


Figure 19: A5. The q-Meyer two ways. Left, as the registry draws it: the analyzer's band peak λ^* matched to the Gaussian's ($\lambda^* = \sqrt{2}\,a$), so the real-space lobe is 1.9× wider and rings. Right, dilated to the Gaussian's real-space lobe for visualization (λ^* then $0.75\,a$). The plain Meyer ($q = 1$) is the thick reference in both.

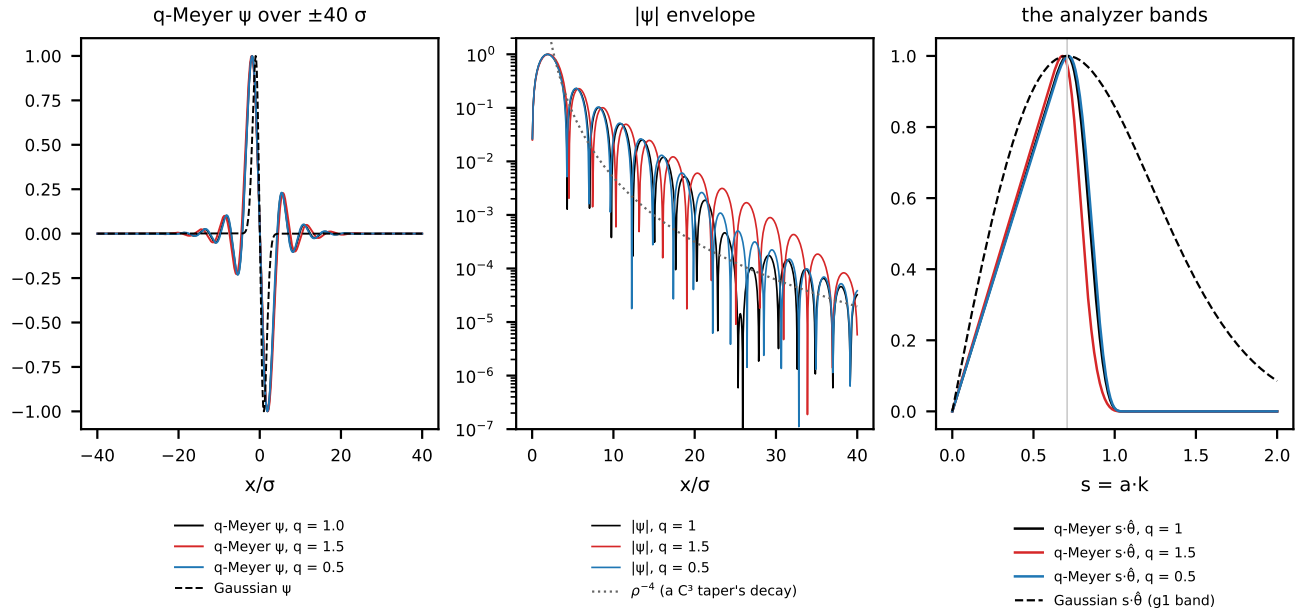


Figure 20: A6. Why a bandlimited analyzer looks wide: the q-Meyer ψ over $\pm 40\,\sigma$ rings with the period of its cut-off ($4.2\,\sigma$) and decays as ρ^{-4} (a C^3 taper); both analyzer bands $s\hat{\theta}$ peak at $s^* = 1/\sqrt{2}$, but the Meyer's band is a narrow triangle against the Gaussian's broad hump.

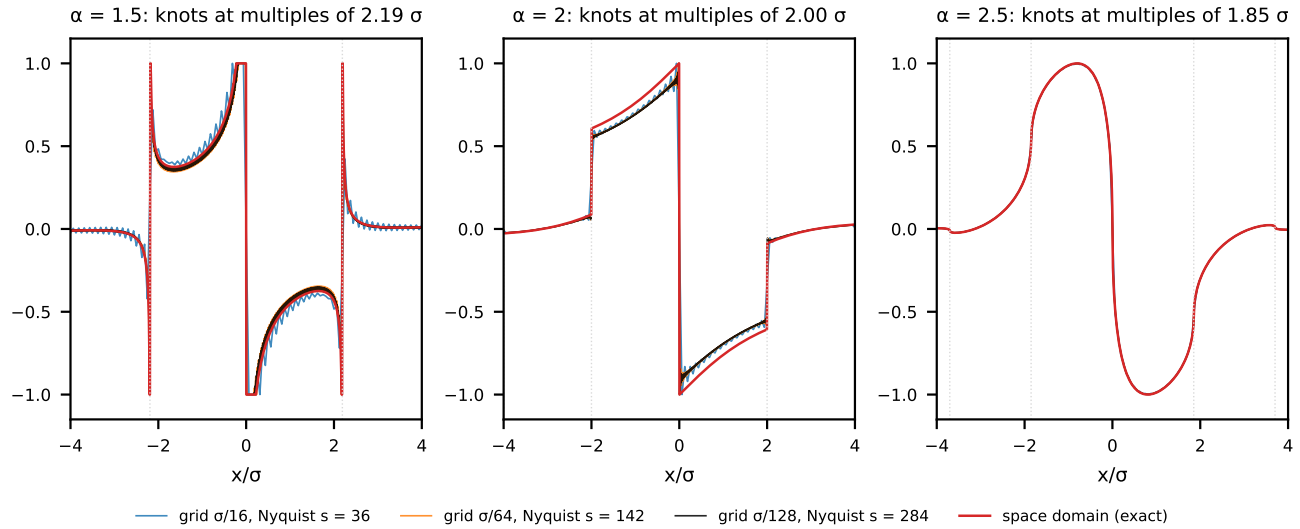


Figure 21: A7. The isotropic polyharmonic B-spline below $\alpha = 3$ on three inverse-FFT grids, against its space-domain evaluation (red), all scaled by the main lobe and clipped at the knots: the zig-zag between the knots is Gibbs truncation of the display grid (it shrinks as the grid refines and converges on the red curve); the spikes and jumps at the knots are the function itself, a localized Green's function of $(-\Delta)^{\gamma/2}$ whose gradient goes as $\rho^{\gamma-3}$ at every knot: divergent for $\alpha < 2$, a jump at $\alpha = 2$, a cusp for $2 < \alpha < 3$.

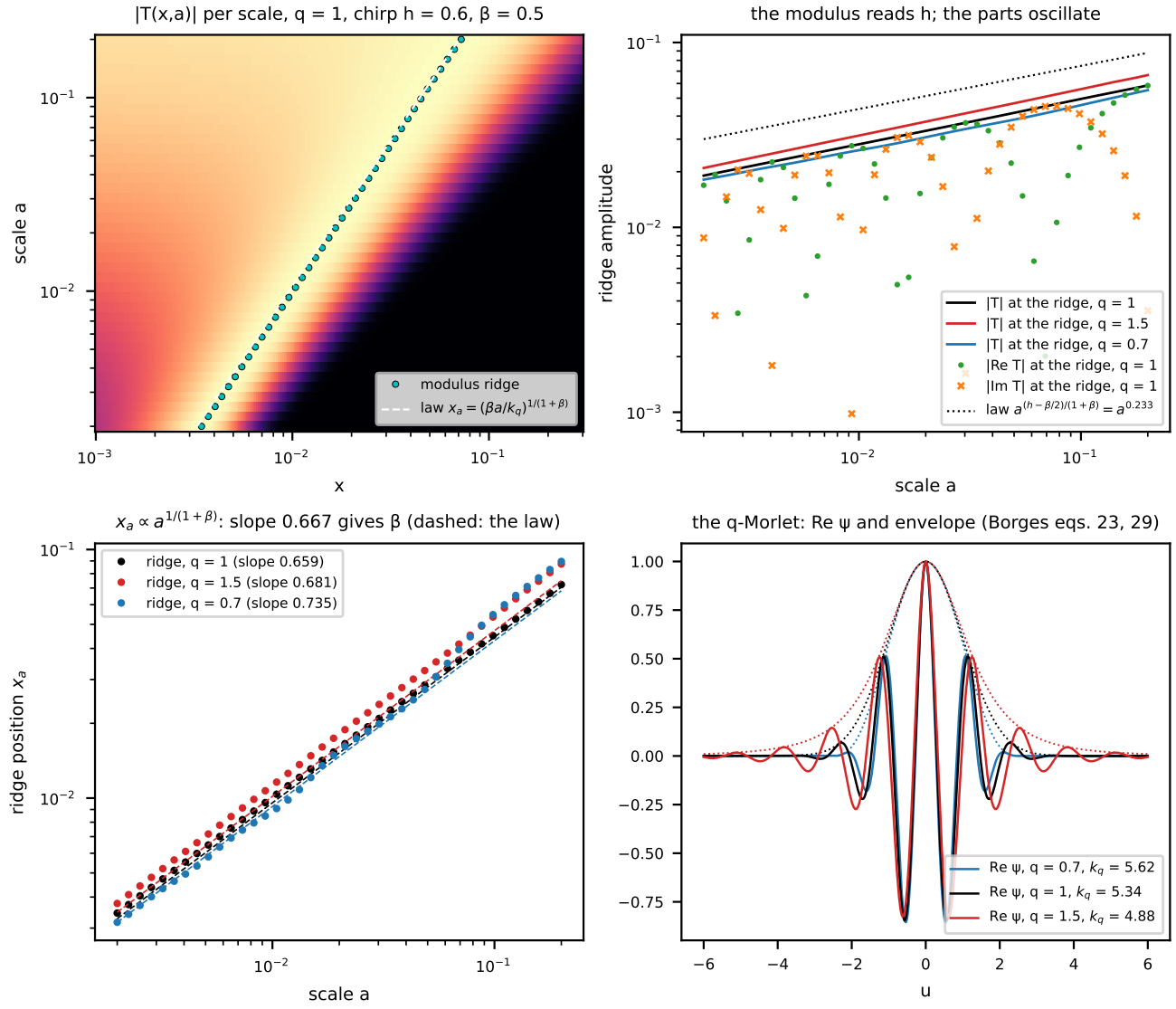


Figure 22: A8. A single oscillating singularity $x_+^h \sin(x^{-\beta})$ ($h = 0.6$, $\beta = 0.5$) under Borges's q-Morlet (the modulated q-Gaussian, the carrier k_q of their eq. 29, exact zero-mean correction), 1-D, L^1: the modulus ridge sits on $x_a = (\beta a/k_q)^{1/(1+\beta)}$, which gives β, and its amplitude follows $a^{(h-\beta/2)/(1+\beta)}$, which gives h, cleanly for $q = 1$, 1.5 and 0.7; the real and imaginary parts at the ridge oscillate with the carrier's phase and read neither.